Corrective Overoptimism

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Abstract

Booms in infrastructure investment often coincide with investor exuberance and losses, yet the infrastructure remains long after these episodes. I show that investor overoptimism can correct for a general class of market failure: agents with diagnostic expectations over-extrapolate good news and conduct irreversible investments, pushing what was once a market failure toward the efficient level. This mechanism – corrective overoptimism – is embedded in a dynamic spatial model with endogenous infrastructure and estimated on data from England and Wales, 1700–1881, spanning the canal and railway booms. Corrective overoptimism can be a potent channel for growth.

JEL Classification: D84, E22, E32, N73, O18, O33, R12, R42.

Keywords: Corrective overoptimism, diagnostic expectations, infrastructure investment, spatial development, technology diffusion, Industrial Revolution.

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> Wherever you are in the country, you are just a short distance from the great arterial system. At any moment, you could integrate with this great life-force traversing England; become an essential part of the nation or retreat to the suburban domesticity that you call home.
>
> — Simon Foxell, *Mapping England*

The great arterial expansions of the modern economy—the railways, the electric grid, and the fiber-optic network—each coincided with dramatic boom–bust cycles: the Railway Mania of 1846, the crash of 1929, and the Dot-Com Bubble of 2000.[1] When expectations reversed, investors suffered ruinous losses. Yet the physical works outlasted the fever: from each episode emerged enduring connective infrastructure that reshaped the landscape and the economy. Janeway (2018) and Morck (2022) argue that this pattern is no accident, and that exuberance-driven buildouts have been central to the growth process. I formalize this argument and quantify its importance.

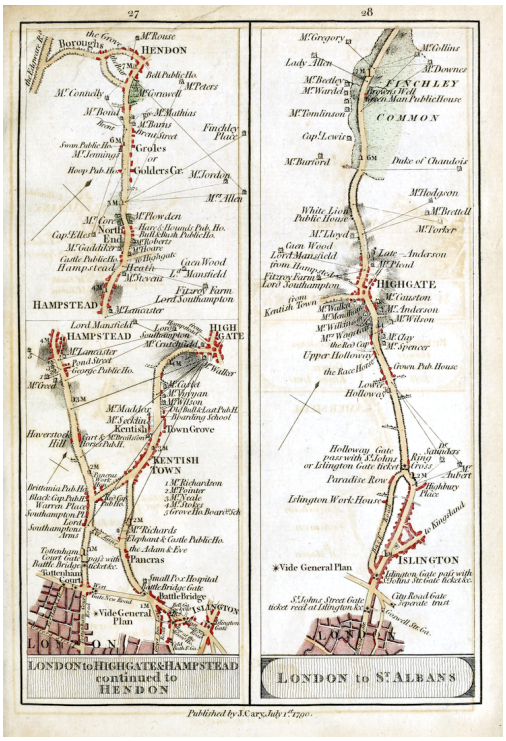
The core idea is simple. A monopolist who must sink capital into infrastructure before demand is realized underinvests, because market power drives a wedge between the private return and the social return. If that monopolist over-extrapolates good news – forming beliefs that are too optimistic after positive signals – the resulting over-investment can close the gap. And because the capital is irreversible, the correction survives even after beliefs revert (Section III.F). I call this mechanism *corrective overoptimism*.

The idea that financial exuberance can be productive has a long history (Caballero et al., 2006; Eatwell, 2004; Keynes, 1936; Perez, 2002). A large literature shows that bubbles can relieve financial constraints, deploying capital to productive firms and crowding in investment for innovation (see, e.g., Guerron-Quintana et al., 2023; Hirano and Yanagawa, 2016; Martin and Ventura, 2012; Miao and Wang, 2014; Nanda and Rhodes-Kropf, 2017). My contribution is to show that behavioral bias can correct not just for financial frictions, but for a broad class of market failures rooted in market power and externalities (Lipsey and Lancaster, 1956). Where the "big push" literature in development and spatial economics studies how coordinated state policy can overcome these externality and complementarity wedges (Garg, 2025; Kline and Moretti, 2014; Moneke, 2024; Murphy et al., 1989; Rosenstein-Rodan, 1943), the mechanism I identify operates through decentralized private investment even without funding constraints or state planning: what matters is that an agent invests above their rational choice in a market where the marginal social benefit of that investment exceeds its marginal private benefit – a monopoly markup as in Section I, or spillovers as in the full model – and that the resulting investment is sunk.

I model the bias using diagnostic expectations (DE), introduced by Bordalo et al. (2018) as a formalization of the representativeness heuristic of Kahneman and Tversky (1972). Under DE, agents overweight the likelihood of states that have become more likely in light of recent news. A growing literature has shown DE to improve our understanding of expectations and investment (Bordalo et al., 2020; Gennaioli et al., 2016), asset prices and credit cycles (Bordalo et al., 2026, 2019; Maxted, 2023), and business cycle dynamics (Bianchi et al., 2024; Elekdag et al., 2025; L'Huillier et al., 2023).[2] I exploit the interaction between the overreactive property of DE and irreversible investment. When a diagnostic infrastructure monopolist receives good news about

[1] See, for example, Kindleberger (1978), White (1990), Shiller (2000), Pástor and Veronesi (2009), Odlyzko (2010), Greenstein (2015), and Caoui and Steck (2025).

[2] In complementary work, Flynn and Sastry (2025) and Flynn et al. (2026) study contagious narratives and persistence learning over the business cycle, while Bizzarri and d'Arienzo (2024) analyze the social value of information overreaction in abstract coordination settings.



Great North Road out of London, 1790, John Cary. Courtesy of Duke University via HathiTrust.

future demand, it overestimates the return to building and invests beyond the rational level. Once the capital is sunk, the economy retains the network.

The simplest form of market failure is a static monopoly, and Section I shows how corrective overoptimism closes this static gap. Across space and time, externalities and spillovers take richer forms (Bresnahan and Trajtenberg, 1995; Lucas, 1988; Romer, 1990). Turnpike roads and railways lowered the cost of moving goods, mail, and people, while telegraphs and fiber networks sped the exchange of ideas across distant towns (Berkes et al., 2026; Cinnirella et al., 2024; Hanlon, 2025; Mokyr, 2005). Each connective revolution let strangers stand on one another's shoulders, yet no parish turnpike trust or county railway company could claim the returns its links created for invention across the realm. Because knowledge compounds once shared, a transient wave of optimism that leaves behind permanent roads and rails can set in motion a lasting tide of growth. Section II documents this mechanism in England and Wales over 1700–1881—where local projections and shift-share instruments confirm that regional construction surges reversed below trend and compressed subsequent railway operating yields (Arnold and McCartney, 2011; Campbell, 2012; Jordà, 2005; Odlyzko, 2010)—before Section III develops the model and Section IV quantifies its contribution to growth.

A distinguished literature has studied the relationship between transport infrastructure and economic outcomes. Fogel (1964) and Fishlow (1965) asked what the American economy would have looked like without the railway. Donaldson and Hornbeck (2016) and Donaldson (2018) brought this question into a market-access framework.[3] Hornbeck and Rotemberg (2024) showed, by accounting for input distortions (Baqaee and Farhi, 2020; Hsieh and Klenow, 2009; Restuccia

[3]Redding and Turner (2015) survey the market-access literature, and Ahlfeldt et al. (2015), Hodgson (2018), Heblich et al. (2020), Nagy (2023), and Balboni (2025) quantify how transport networks shape urban density, firm location, and spatial persistence.

and Rogerson, 2008), that the aggregate productivity gains from American railways were far larger than the earlier estimates of Fogel (1964) and Donaldson and Hornbeck (2016) suggested, because infrastructure has outsized effects in distorted economies. Fajgelbaum and Schaal (2020), Allen and Arkolakis (2022), and Allen et al. (2025) characterize optimal infrastructure networks in static environments, while Brancaccio et al. (2020), Bordeu (2025), and Jiang (2026) study decentralized transport provision. In the dynamic spatial literature, Kleinman et al. (2023) incorporate forward-looking capital accumulation and migration into a dynamic spatial general equilibrium, and Allen and Donaldson (2020) develop a theory of path-dependent spatial economies through agglomeration.

The counterfactual tradition running from Fogel to Donaldson and Hornbeck has produced sharp answers to the question of what infrastructure is worth once built. A complementary question is whether the right amount gets built in the first place, and whether the psychological biases of the investors who finance it might, against the usual intuition, improve the outcome. I take up this question. Doing so requires a model in which the network emerges from forward-looking investment decisions that are subject to systematic bias, and a structural approach to evaluate counterfactuals we cannot observe.

I build on the spatial growth framework of Desmet and Rossi-Hansberg (2014) and Trew (2020), adding four extensions before introducing diagnostic expectations. First, endogenous idea diffusion (Cai et al., 2022; Nelson and Phelps, 1966) links route capacity to technology catch-up beyond static trade-cost reductions (Proposition 3). Second, local manufacturing operating wedges (Desmet et al., 2020; Hornbeck and Rotemberg, 2024) and competitive county carriage services (Keibek, 2017; Shaw-Taylor and Wrigley, 2014) capture allocative efficiency gains as transport shifts labor from agriculture into distorted industry. Third, forward-looking county infrastructure companies build route capacity under parliamentary acts while competitive carriers operate freight (Bogart, 2014; Ward, 1974); their dynamic Markov perfect game reduces to a Tobin marginal-q rule (Hayashi, 1982) with a unique stage equilibrium under own-cost diagonal dominance (Proposition 4) and a unique Markov perfect equilibrium within a class of county values under a contraction condition (Proposition 7). Fourth, the firm and worker problems extend to diagnostic expectations while preserving identical within-period choices (Propositions 1 and 2), with the conditional dated equilibrium consistent across ledgers and, on a quote domain satisfying three regularity conditions, existing and unique (Propositions 5 and 6).

Diagnostic expectations counteract decentralized underinvestment, and the effect compounds over time. Because a permanent increase in route capacity raises the adoption rate in every subsequent period (Proposition 3, equation (42)), early buildout accelerates long-run catch-up. A county company's marginal-q values only land rents inside its jurisdiction, omitting three wider social returns: through-trade price reductions, labor reallocation into wedge-distorted manufacturing, and faster spatial knowledge diffusion. Because diagnostic expectations shift perceived next-period productivity by $\theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$ in the direction of current news (equation (12)) while completion intensities are nonnegative and route capacity does not depreciate (equations (62) and (54)), boom-time construction is permanent, whereas bad news can at most halt new building.

The structural estimates reject rational expectations in favor of diagnostic extrapolation. Taking the model to 1710–1881 data on output, relative industrial prices, occupational shares, spatial sorting, and multimode transport capacity, the Simulated Method of Moments estimates $\widehat{\theta} = 0.739$ (SE $= 0.090$; $t = 8.21$, $p < 0.0001$), matching the $\theta \approx 0.74$–0.91 range documented for credit spreads, earnings forecasts, and firm investment (Bordalo et al., 2018, 2026, 2019). Re-estimating under rational expectations ($\theta = 0$) matches secular output and employment trends, but cannot jointly generate the 1830–1850 railway boom, the post-1850 slowdown, and the negative serial correlation and spatial reversals of construction. Out of sample, the diagnostic model tracks the untargeted post-1820 acceleration of aggregate real wages (3.11 times the 1710 level by 1881 versus

3.28 in the data), whereas the rational model stalls at 1.72 (Allen, 2009). Counterfactual simulations show that diagnostic overoptimism was a potent source of growth. Setting $\theta = 0$ at the diagnostic estimates while holding initial stocks, project opportunities, and primitive shocks fixed (R | DE) shows that diagnostic expectations expanded the 1881 British transport network by +43.79 percent (+54.47 percent in the Midlands–North corridor), raised real output per worker by +20.96 percent, and lifted worker real wages by +12.97 percent. Larger route capacity shifted −12.9 percentage points of labor out of agriculture, raised manufacturing frontier productivity by +10.90 percent, and lowered the relative price of manufactures by −13.79 percent (Table 7 and Figure 9). Decomposing the +20.96 percent output gain across the three spillover channels attributes 25.9 percent to direct transit savings, 29.5 percent to labor reallocation into wedge-distorted manufacturing, and 44.7 percent to faster spatial knowledge diffusion.

The remainder of the paper proceeds in four steps. Section I illustrates corrective overoptimism in a two-period monopoly model. Section II sets out the historical landscape and reduced-form evidence from England and Wales between 1700 and 1881. Section III develops the dynamic spatial growth model, with proofs in Appendix A. Section IV estimates the model and evaluates the counterfactual economy under rational expectations, and Section V offers concluding remarks.

I A Simple Example

Diagnostic expectations (DE) can overcome market failure. Consider a two-period model of a regional economy that produces wheat and exports it through a monopoly railway.

Demand. Exporters' willingness to pay for shipping depends on the global wheat price Z_t. With shipment volume Q_t and demand slope $b > 0$, inverse demand for railway transport is

$$P_t(Q_t) = Z_t - bQ_t, \qquad b > 0. \tag{1}$$

The global wheat price follows an AR(1) process $Z_{t+1} = \rho Z_t + \varepsilon_{t+1}$ with persistence $\rho \in (0, 1)$ and mean-zero innovations ε_{t+1}, so that $\mathbb{E}_t[Z_{t+1}] = \rho Z_t$.

Technology. To transport wheat at date $t+1$, the monopolist must build capacity K_{t+1} at date t and sells it forward to exporters at the market-clearing price $P_{t+1}(K_{t+1})$, so that shipments equal capacity, $Q_{t+1} = K_{t+1}$. Each unit of capacity costs an upfront investment of $c > 0$ to build, with $\rho Z_t > c$ so that capacity is worth building, and there is no discounting between dates. This investment is sunk and irreversible. Operating costs are zero, and $\varepsilon_{t+1} \geq -c$ keeps the clearing price nonnegative at the capacities considered below.[4]

A The First-Best Baseline

Expected social surplus from capacity K_{t+1} is $\mathbb{E}_t[S_{t+1}(K_{t+1})] = \mathbb{E}_t\left[\int_0^{K_{t+1}} (Z_{t+1} - bq)\, dq\right] - cK_{t+1} = (\mathbb{E}_t[Z_{t+1}] - c)K_{t+1} - \frac{1}{2}bK_{t+1}^2$. To maximize it, the social planner builds capacity until the expected price exporters are willing to pay for the last unit of transport equals the cost of building it (c):

$$\mathbb{E}_t\left[P_{t+1}(K_{t+1}^{\mathrm{SP}})\right] = c \quad \Longrightarrow \quad K_{t+1}^{\mathrm{SP}} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{b}, \tag{2}$$

by substituting inverse demand $P = Z - bK$.

[4] Forward sale is what makes the first-order conditions below exact. If the owner could instead leave capacity idle ex post, it would ship $\min\{K_{t+1}, Z_{t+1}/(2b)\}$ and the closed forms below would be lower bounds on optimal capacity. Online Appendix S.A analyzes ex-post quantity competition and route entry.

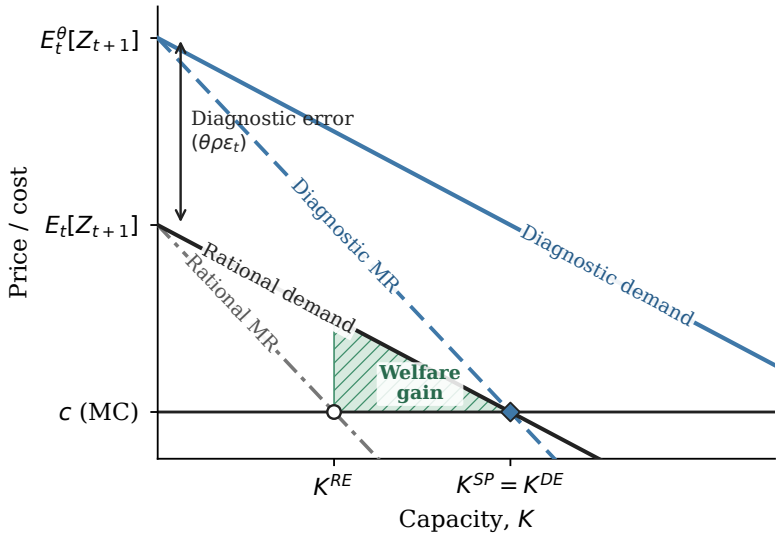


Figure 1: Railways for Wheat Exports

Notes: Linear inverse shipping demand $P = Z - bQ$ and marginal revenue $\text{MR} = Z - 2bQ$. Equating expected marginal revenue rather than marginal social benefit to unit capacity cost c leads rational expectations ($\theta = 0$) to build half the socially efficient capacity ($K^{\text{RE}} = \frac{1}{2}K^{\text{SP}}$), whereas diagnostic overreaction to positive news ($\theta > 0$, $\varepsilon_t > 0$) shifts perceived demand outward and expands sunk capacity toward K^{SP}.

B The Monopolist's Choice: Rationality vs. Overreaction

The future wheat price Z_{t+1} is the sole source of uncertainty. To capture behavioral bias, as in Bordalo et al. (2018), agents' beliefs are subject to diagnostic expectations:

$$\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta \underbrace{\left(\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}]\right)}_{\text{news}}, \tag{3}$$

where $\theta \geq 0$ is the diagnosticity parameter. When $\theta = 0$, we are back in the world of rational expectations. When $\theta > 0$, the railway owner over-updates their belief of the future wheat price based on good and bad news. Under an AR(1) process, $\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t$.

The profit-maximizing monopolist equates expected marginal revenue $\text{MR}_t = Z_t - 2bQ_t$ to cost. Setting $\mathbb{E}_t^\theta[\text{MR}_{t+1}] = c$:

$$K_{t+1}^{\text{DE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - c}{2b} = \frac{\mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t - c}{2b}. \tag{4}$$

Under rational expectations ($\theta = 0$), the monopolist underinvests:

$$K_{t+1}^{\text{RE}} = \frac{1}{2}K_{t+1}^{\text{SP}}. \tag{5}$$

However, diagnostic expectations provide a way out of this failure. Suppose $\theta > 0$ and that at time t a positive shock $\varepsilon_t > 0$ arrives: a failed harvest in Europe drives up the global commodity price for wheat. Then

$$K_{t+1}^{\text{DE}} - K_{t+1}^{\text{RE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - \mathbb{E}_t[Z_{t+1}]}{2b} = \frac{\theta\rho\varepsilon_t}{2b} > 0. \tag{6}$$

So the diagnostic investor builds more capital than the rational investor (Figure 1). In particular, if

$$\frac{\theta\rho\varepsilon_t}{2b} = \frac{1}{2}K_{t+1}^{\mathrm{SP}} \quad \Longrightarrow \quad \theta\rho\varepsilon_t = \rho Z_t - c, \tag{7}$$

then we attain the social optimum K_{t+1}^{SP} (and any positive news satisfying $0 < \theta\rho\varepsilon_t < 2(\rho Z_t - c)$ raises expected welfare above K_{t+1}^{RE}).[5] The example is static beyond $t+1$; in the dynamic model of Section III, where completion intensities are nonnegative and route capacity does not depreciate (equations (62) and (54)), capacity built during a boom remains in place after beliefs revert.

II Historical Setting and Motivating Evidence

Having illustrated the basic mechanism, I now bring corrective overoptimism to historical data from England and Wales between 1710 and 1881. Three features make the British Industrial Revolution an ideal laboratory. First, granular spatial and occupational records survive from the early eighteenth century (Broadberry et al., 2014; Crafts and Harley, 1992; Shaw-Taylor and Wrigley, 2014). Second, the north–south orientation of England and Wales allows the spatial investment problem to be represented on a one-dimensional corridor while preserving the geography of London, the Midlands, and the northern coalfields (Trew, 2020). Third, Britain experienced successive waves of private transport mania – in turnpikes, canals, and railways – under a decentralized Parliamentary regime (Arnold and McCartney, 2011; Bogart, 2014; Campbell, 2012; Odlyzko, 2010; Ward, 1974).

A Spatial Structural Transformation and Distribution Labor

The decline of agriculture during the Industrial Revolution was not absorbed by manufacturing alone; much of the labor released from the land flowed into a growing transport and distribution sector. To measure this transition on a consistent geography, I place county occupational series on fixed 1831 ancient-county boundaries across twelve benchmark dates from 1710 to 1881, combining Keibek (2017) with the CAMPOP national benchmarks (Shaw-Taylor et al., 2019), London registration-county series (Field and Shaw-Taylor, 2019), and the county population weights of Wrigley (2011) (Online Appendix S.B). Occupations follow the Primary–Secondary–Tertiary (PST) classification of Wrigley (2010): primary covers agriculture, forestry, and fishing; secondary groups mining with manufacturing and construction because coalfield geography parallels industrial production (Allen, 2009; Clark and Jacks, 2007); and tertiary comprises transport, communications, dealers, and sellers, isolating the distribution and haulage activities required to move goods across space.

Figures 2 and 3 highlight two features of this transition. At the national level, structural transformation proceeded from a large eighteenth-century industrial base: secondary employment reached half of included employment by 1817, while between 1851 and 1881 the primary share fell from 31 to 19 percent, the secondary share rose from 56 to 62 percent, and the tertiary share expanded from 13 to 18 percent as growing trade volumes required more distribution services (Herrendorf et al., 2012; Shaw-Taylor and Wrigley, 2014). At the regional level, primary employment remained concentrated in the South and East and secondary industry sorted toward the Midlands and Northern coalfields, whereas distribution and transport employment expanded in every county

[5] Expected social surplus relative to rational monopoly is $\mathbb{E}_t[S_{t+1}(K_{t+1}^{\mathrm{DE}}) - S_{t+1}(K_{t+1}^{\mathrm{RE}})] = \frac{\theta\rho\varepsilon_t}{8b}[2(\rho Z_t - c) - \theta\rho\varepsilon_t]$, which at $\theta\rho\varepsilon_t = \rho Z_t - c$ equals $b(K_{t+1}^{\mathrm{SP}})^2/8$, the shaded triangle in Figure 1. For any market failure where marginal social benefit exceeds marginal private benefit at K_{t+1}^{RE} (Lipsey and Lancaster, 1956), the envelope theorem implies that a small diagnostic expansion has zero first-order private loss to the investor while generating a positive first-order social gain.

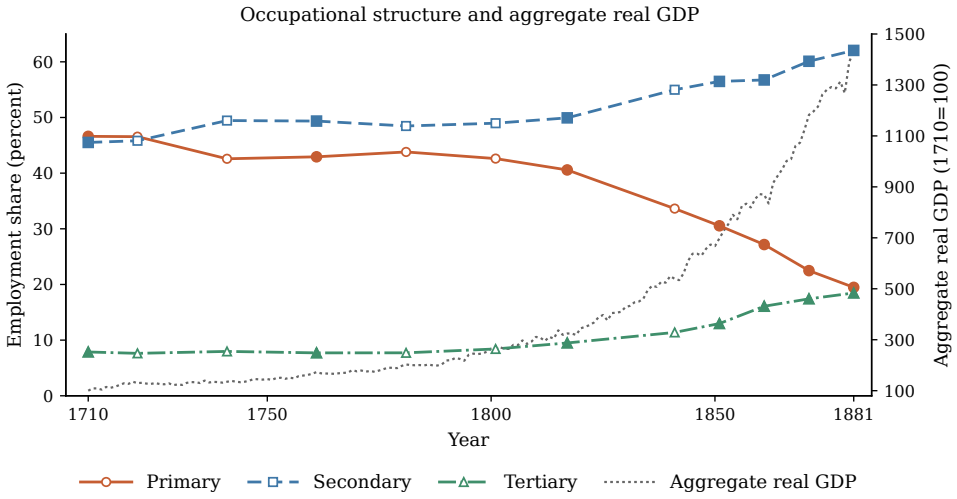


Figure 2: Occupational Structure and Aggregate Real GDP

Notes: Adult male employment shares across primary (excluding mining), secondary (manufacturing, mining, construction), and tertiary (dealers, sellers, transport, communications) sectors: filled markers denote CAMPOP benchmarks (Shaw-Taylor et al., 2019) and open markers denote aligned Keibek (2017) estimates. Right axis: Great Britain real GDP (1710 = 100; Broadberry et al. 2022; Feinstein 1972).

while remaining thickest around London and the northern industrial hubs (Bogart et al., 2022; Heblich et al., 2020). These facts matter because they are the moments targeted in estimation: a quantitative assessment of corrective overoptimism requires that the underlying economy first match the growth takeoff and the structural transformation across all three sectors.

B Decentralized Infrastructure Provision, Carriage, and Local Distortions

Unlike the state-planned trunk systems of continental Europe (Milward and Saul, 1973; Smith, 1990), British turnpikes, canals, and railways were financed by private regional investors under individual Acts of Parliament, and their completed links reshaped market access and idea diffusion (Berkes et al., 2026; Bogart, 2014; Mokyr, 2005; Trew, 2020). Across all three eras, statutory organization separated durable route provision from competitive carriage. Turnpike trusts, canal companies, and early railway firms raised local capital to build route capacity and collect tolls within authorized corridors, while independent carriers hired operating labor and vehicles to move freight (Hadfield, 1981; Turnbull, 1987; Ward, 1974).

Network expansion also interacted with persistent local distortions in manufacturing. Desmet, Greif and Parente (2020) document how interregional competition eroded craft resistance to mechanization across regions, and Hornbeck and Rotemberg (2024) show that transport improvements generate first-order productivity gains when they reallocate inputs toward wedge-distorted manufacturing producers. Figure 4 plots the cumulative multimode transport stock and annual physical additions over 1710–1881, showing the clustered construction waves of the 1790s canal mania and the 1830s–1840s railway booms. This institutional structure – local private finance, Parliamentary coordination of competing schemes, and separation between durable route capacity and competitive carriage – motivates the county-level investment game in Section III.

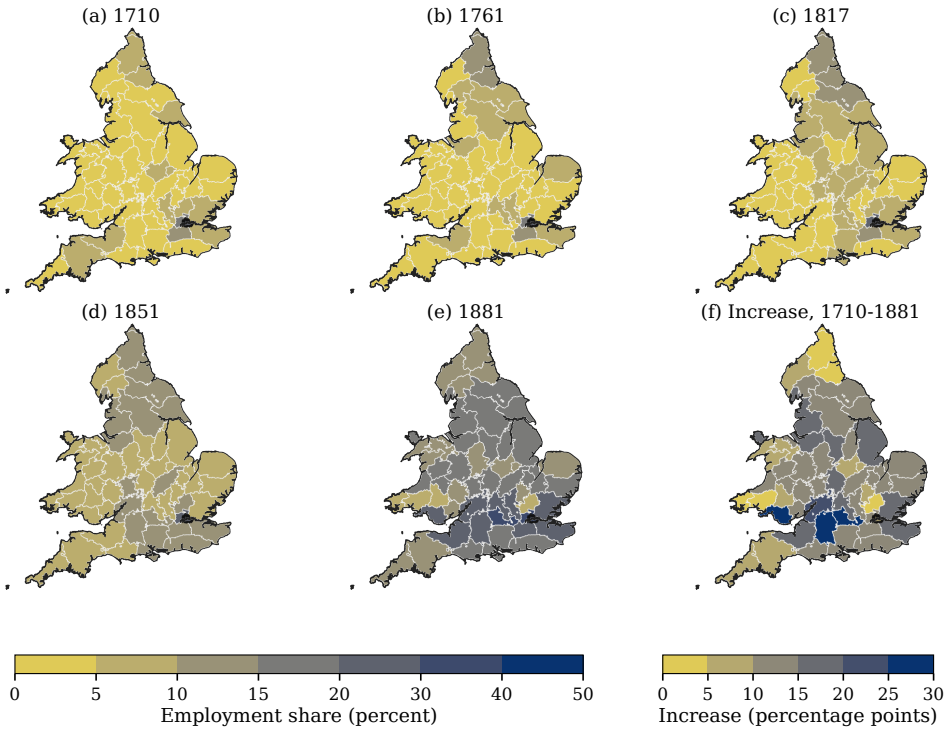


Figure 3: Spatial Expansion of Distribution and Transport Employment, 1710–1881

Notes: Panels (a)–(e) plot county tertiary employment shares at five benchmark dates; Panel (f) plots the 1710–1881 percentage-point increase. County shares through 1851 combine Keibek (2017) and Shaw-Taylor et al. (2019) on 1831 ancient-county boundaries, and 1881 projects the reconstructed latitude-band surface (Online Appendix S.B); the black outline marks London (Field and Shaw-Taylor, 2019).

C Boom–Bust Reversals and Yield Compression in Infrastructure

Each major wave of British transport investment surged after favorable news and ended in financial disappointment. During the 1830s and the 1845–1848 Railway Mania, company prospectuses extrapolated from early trunk successes to advertise 10–15 percent dividends, yet realized equity returns averaged under 5 percent and half of Mania-authorized mileage went unbuilt (Odlyzko, 2010). Campbell (2012) shows that weekly railway share prices overreacted to short-run dividend news, while Arnold and McCartney (2011) and Shaen (1847) document recurrent cost overruns and post-opening yield shortfalls across early canal and railway companies.

Local projections of future construction and operating returns on current investment surges test whether building tracked fundamentals or overshot during booms (Bordalo et al., 2018, 2020; Jordà, 2005). Let $r^I_{b,t} = (T_{b,t+1} - T_{b,t})/T_{b,t}$ denote the annual growth rate of transport capacity in latitude band b and $c^I_{b,t} = \mathrm{HP}_{6.25}[\log(10^{-3} + r^I_{b,t})]$ its Ravn and Uhlig (2002) cyclical component. Across horizons $h \in \{1, \ldots, 5\}$, I project future regional capacity additions $c^I_{b,t+h}$ on current additions $c^I_{b,t}$, and aggregate British railway gross operating yields R_{t+h}/K_{t+h} (working receipts over paid-up capital from Mitchell 1988) on log paid-up capital growth $\Delta \ln K_t$:

$$c^I_{b,t+h} = \alpha_{bh} + \delta_{th} + \rho_h c^I_{b,t} + u_{b,t+h}, \qquad \frac{R_{t+h}}{K_{t+h}} = \alpha^R_h + \beta^R_h \Delta \ln K_t + u^R_{t+h}. \tag{8}$$

Under rational construction smoothing, a building surge at date t predicts positive future additions above trend ($\rho_h > 0$). Under boom-time overshooting, excess capacity depresses subsequent

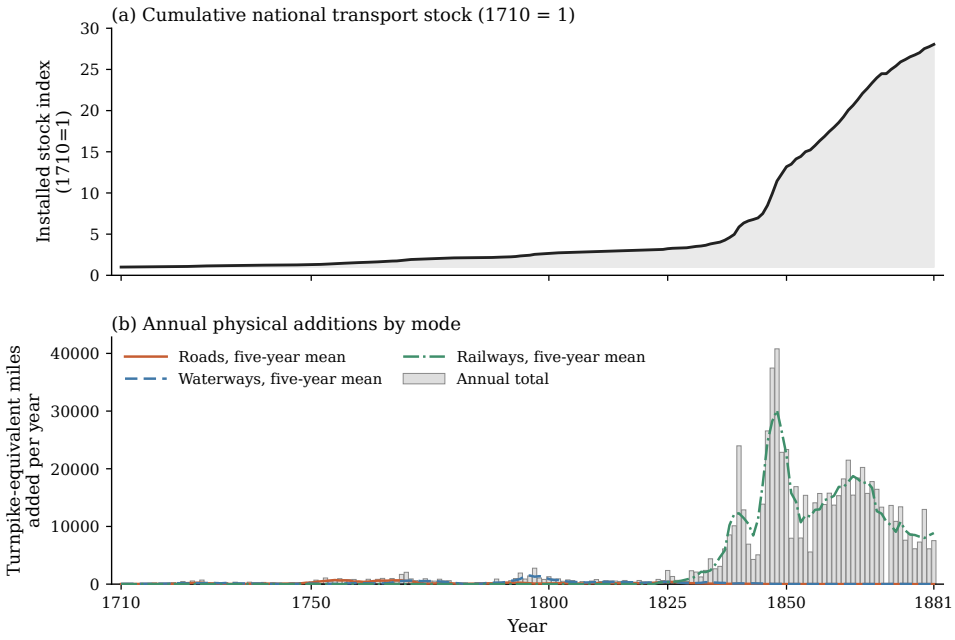


Figure 4: Transport Stock and Installed Additions, 1710–1881

Notes: Turnpikes, waterways, and railways aggregated into turnpike-equivalent miles (road+10×waterway+ 50 × railway). Top panel: cumulative national stock (1710 = 1). Bottom panel: annual physical additions (bars) and five-year moving averages by mode (lines).

additions below trend ($\rho_h < 0$) and compresses subsequent operating yields ($\beta_h^R < 0$).[6]

Table 1 documents sharp post-boom reversals in both physical construction and realized operating yields. In Panel A, a one-unit cyclical surge in regional capacity additions is followed by below-trend additions over the next three years ($\rho_1 = -0.221$, SE $= 0.019$; $\rho_2 = -0.215$, SE $= 0.020$; $\rho_3 = -0.121$, SE $= 0.020$). Instrumenting regional building with predetermined coalfield and 1710 manufacturing exposure interacted with leave-one-out regional addition cycles (first-stage $F > 172$) yields matching reversals ($\rho_1^{\text{IV}} = -0.273$, SE $= 0.054$; $\rho_2^{\text{IV}} = -0.202$, SE $= 0.057$), confirming that fundamental-driven booms overshoot and contract.[7] In Panel B, railway paid-up capital surges $\Delta \ln K_t$ predict lower subsequent gross operating yields across all five horizons ($\beta_1^R = -13.00$ to $\beta_5^R = -11.15$, $R^2 = 0.40$–0.52), matching the 34 percent shortfall between prospectus and realized net traffic yields in Shaen (1847)'s parliamentary audit.

These construction reversals ($\rho_h < 0$), yield compressions ($\beta_h^R < 0$), and prospectus shortfalls rule out four rational alternatives. First, under multi-period time-to-build and convex adjustment costs (Hayashi, 1982; Kalouptsidi, 2014; Kydland and Prescott, 1982), gestation lags create short-run momentum ($\rho_1 = +0.067$ in the railway era and $\rho_1^{\text{1s}} = +0.159$ under one-sided detrending; Online Appendix Table S.B.1), but additions return to trend upon completion rather than falling below trend ($\rho_2, \rho_3 < 0$), and known lags cannot explain why $\Delta \ln K_t \in \mathcal{F}_t$ predicts five years of yield compression or prospectus shortfalls (Shaen, 1847). Second, under Bayesian learning and

[6] Estimating ρ_h on the future level $c^I_{b,t+h}$ places no $-c^I_{b,t}$ term on the left-hand side, avoiding mechanical differencing (Jordà, 2005; López-Salido et al., 2017).

[7] Online Appendix S.B (Table S.B.1) shows that these reversals hold under two-way clustering by latitude band and year, leave-corridor-out ($|k - b| \geq 3$) shift-share instruments, one-sided backward-looking detrending, untransformed level and inverse-hyperbolic-sine addition rates, unfiltered cumulative changes, and separate pre-railway (1710–1829) and railway (1830–1880) subsamples.

Table 1: Local Projections of Infrastructure Investment and Railway Yields

	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
Panel A. Dependent variable: Regional investment cycle $c^I_{b,t+h}$ (25 latitude bands, 1710–1880)					
Investment cycle $c^I_{b,t}$ (TWFE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.019)	(0.020)	(0.020)	(0.020)	(0.019)
Investment cycle $c^I_{b,t}$ (2SLS)	−0.273	−0.202	−0.084	−0.088	−0.004
	(0.054)	(0.057)	(0.054)	(0.056)	(0.028)
Band and year fixed effects	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic (2SLS)	175.6	174.8	174.4	173.2	172.3
Observations	4,250	4,225	4,200	4,175	4,150
Panel B. Dependent variable: Railway gross operating yield R_{t+h}/K_{t+h} (percent, 1843–1881)					
Log paid-up capital growth $\Delta \ln K_t$ (OLS)	−13.00	−13.71	−13.65	−12.49	−11.15
	(2.23)	(2.06)	(2.31)	(2.37)	(2.19)
R^2	0.46	0.52	0.51	0.45	0.40
Observations	38	37	36	35	34

Notes: Local projections (8) at horizons $h = 1, \dots, 5$ years (Jordà, 2005). Panel A estimates regional regressions across 25 latitude bands (1710–1880) by TWFE and by 2SLS using shift-share instruments $Z_{b,t} = (e^{\text{coal}}_b \bar{c}^{I,\text{coal}}_{-b,t},\ \pi_{M,b,1710}\bar{c}^{I,M}_{-b,t},\ (0.5 + e^{\text{coal}}_b)\bar{c}^{I}_{-b,t})'$ that interact predetermined GIS coalfield exposure e^{coal}_b (Bogart et al., 2022) and 1710 manufacturing shares $\pi_{M,b,1710}$ with leave-one-out regional addition cycles; band-clustered standard errors are in parentheses. Panel B regresses British railway gross working receipts over paid-up capital R_{t+h}/K_{t+h} (percent) on annual log paid-up capital growth $\Delta \ln K_t$ over 1843–1881 (Mitchell, 1988), with Newey–West HAC standard errors in parentheses.

real options (Arrow, 1968; Bernanke, 1983; Dixit and Pindyck, 1994; Jovanovic, 1982; Pástor and Veronesi, 2009), uncertainty delays and smooths entry while martingale beliefs imply $\rho_h > 0$ and unpredictable return errors. Third, cyclical additions load on current and lagged output ($b_0 = 2.95$, $b_1 = 1.58$) but do not predict next-year output ($b_{\text{lead}} = -0.005$, SE $= 0.004$), showing that investors extrapolated recent industrial expansions rather than foreseeing future fundamentals (Cochrane, 1991). Fourth, year fixed effects δ_{th} absorb national discount-rate movements, and Mania prospectuses advertised 10–15 percent dividends—above, not below, historical norms (Campbell, 2012; Odlyzko, 2010).

III The Model

A The Spatial Economy

Time is discrete, $t \in \{0, 1, 2, \dots\}$. Building on the continuous-space geography of Lucas and Rossi-Hansberg (2002) and Rossi-Hansberg (2005) and its extension to spatial growth and endogenous transport by Desmet and Rossi-Hansberg (2014) and Trew (2020), the economy occupies the unit interval $\ell \in [0, 1]$, partitioned into J county jurisdictions $\mathcal{C}_j = [c_{j-1}, c_j]$, with output, employment, and route infrastructure measured as densities per unit of land. Three sectors operate across space: agriculture (A) and manufacturing (M), which produce the two consumption goods, and transport and distribution services (T). Competitive land leasing assigns each site $\ell \in [0, 1]$ to a single goods sector ($e_A(\ell, t) + e_M(\ell, t) = 1$, $e_i(\ell, t) \in \{0, 1\}$), while competitive transport carriers operate alongside goods producers within each county.

A continuum of worker households of mass $\bar{L}_t$ supplies one unit of labor and holds equal shares in the national portfolio of land leases, producer net profits, carrier capacity rents, and factor-wedge rebates. Under constant participation, England and Wales population P^{EW}_t determines the

household mass (normalized to 100 in 1710):

$$\bar{L}_t = 100 \frac{P_t^{EW}}{P_{1710}^{EW}}, \tag{9}$$

with each household receiving $1/\bar{L}_t$ of aggregate nonlabor income. In each county $j \in \{1, \ldots, J\}$, a parliamentary-authorized infrastructure company chooses completion intensity x_{jt} to maximize expected discounted land rents inside $\mathcal{C}_j$ net of manufactured construction expenditure.

As in dynamic spatial general equilibrium frameworks (Kleinman et al., 2023), each period t unfolds in two stages conditional on inherited knowledge $A_{it}(\ell)$, route capacity $T_t(\ell)$, and aggregate productivity gaps $\mathbf{z}_t$, separating forward-looking state transitions from within-period spot market clearing:

1. *Pre-discovery commitments:* County companies commit $\mathbf{x}_t \in \mathcal{X}_t$; goods entrants announce pure sector $e_i(\ell, t)$, research probability $\phi_i(\ell, t)$, innovation bill Ψ_i, and hired production labor $L_i(\ell, t)$ at a fixed local contract wage $w(\ell, t)$; carriers commit operating labor $L_T(\ell, t)$ at $w(\ell, t)$; and households choose residence density $n(\ell, t)$.

2. *Post-discovery spot clearing and transitions:* Local innovation gains $G_{it}(\ell, \omega)$ realize; output utilization $Y_i(\ell, t, \omega)$, county tolls $\tau_j(t, \omega)$, spot prices $p_i(\ell, t, \omega)$, trade flows $H_i(\ell, t, \omega)$, and consumption $c_i(\ell, t, \omega)$ clear markets statewise; and discoveries and completed projects update $A_{i,t+1}(\ell)$ and $T_{t+1}(\ell)$.

Table 2 summarizes the environment; Subsections III.B–III.G derive closed-form rules that reduce the equilibrium to a coupled spatial operator and a dynamic county marginal-q game.

B State and Beliefs

At the start of date t, the persistent spatial state consists of bounded, positive, measurable knowledge profiles $A_{At}(\ell)$ and $A_{Mt}(\ell)$ and installed route capacity $T_t(\ell)$, observed alongside aggregate sectoral productivity gaps $\mathbf{z}_t = (z_{At}, z_{Mt}, z_{Tt})'$ and the predetermined demographic, coalfield energy, manufacturing-wedge, and statutory project schedules $\mathcal{E}_t$, all known at the start of date t. Only the active sector at ℓ conducts research there, whereas both sectors adopt discoveries from the spatial frontier; infrastructure built at date t enters operating capacity at $t+1$ and accelerates knowledge adoption from $t+1$ onward.

Aggregate productivity gaps follow a stationary first-order Markov process on a compact symmetric box $\mathcal{B}$:

$$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma}), \tag{10}$$

where $\mathrm{TN}_{\mathcal{B}}$ is a multivariate Gaussian law truncated to $\mathcal{B}$ ($\pm c$ stationary untruncated standard deviations around zero, with $c = 3$ in the benchmark and $c = 4$ for robustness; Online Appendix S.C.1), $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$ with $|\rho_s| < 1$, and $\boldsymbol{\Sigma}$ is the positive definite innovation covariance matrix, with diagonal entries $\Sigma_{ss} = \sigma_s^2$ (Online Appendix S.C.1). The kernel has a unique stationary distribution π on $\mathcal{B}$, and effective productivity (19) is normalized by $\mathbb{E}_\pi[e^{z_i}]$ so that $e^{z_{it}}/\mathbb{E}_\pi[e^{z_i}]$ has unit stationary physical mean. Write $f(\mathbf{z}' \mid \mathbf{z})$ for this physical kernel and $M(\mathbf{m})$ for the mean of $\mathrm{TN}_{\mathcal{B}}(\mathbf{m}, \boldsymbol{\Sigma})$.

Investors and households form beliefs over future aggregate states under diagnostic expectations (Bordalo et al., 2018, 2020), which formalize Kahneman and Tversky (1972)'s representativeness heuristic (Bianchi et al., 2024; Bordalo et al., 2026; L'Huillier et al., 2023; Maxted, 2023). Relative to the no-news reference $\bar{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t] = M(\boldsymbol{\rho}\mathbf{z}_{t-1})$, observed news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$ tilts perceived

Table 2: Economic Environment

	Households
Lifetime utility	$W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s\geq t}} \mathbb{E}_t^\theta[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))]$
Period utility	$U(c_A, c_M) = (c_A - \gamma)^\eta c_M^{1-\eta}$
Budget	$p_A c_A + p_M c_M = w + b_t(\omega)$
Nonlabor income	$b_t = \bar{L}_t^{-1} \int_0^1 (R - p_M I + \Pi_F + \Pi_T + \Pi_\psi)\, d\ell$
	Production and Innovation
Bid rent	$R_i = \max_{\substack{0\leq\phi_i<1 \\ \bar{L}_i\geq 0}} \{\mathbb{E}_t^\theta[(p_i - r_i\tau_j)_+ Y_i] - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell)\}$
Production technology	$0 \leq Y_i \leq e_i Z_i G_i L_i^{\alpha_i}$, with equality whenever $p_i > r_i\tau_j$
Innovation expenditure	$\Psi_i(\phi_i; w, \ell) = w\left[f_i(\ell) + \dfrac{\chi_i}{1-\phi_i}\right]$ for $\phi_i > 0$, $\quad \Psi_i(0; w, \ell) = 0$
Research outcome	$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}$, $\quad g \geq 1$
Knowledge after innovation	$A_{it}^+ = A_{it}[1 + e_{it}(G_{it} - 1)]$
	Transport Services
Operating profits	$\max_{L_T\geq 0} \mathbb{E}_t^\theta[\tau_j Y_T - w L_T]$
Transport technology	$Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}$
County capacity and demand	$K_j = \int_{\mathcal{C}_j} Y_T\, d\ell$, $\quad D_j = \int_{\mathcal{C}_j} \sum_i r_i (Y_i + \lvert H_i\rvert)\, d\ell$
	County Infrastructure Companies
Prospectus value	$\Pi_{jt}^\theta(\mathbf{x}) = R_{jt}(\mathbf{x}) + \beta \mathbb{E}_t^\theta[V_{j,t+1}^\theta]$, $\quad V_{jt}^\theta = \max_{0\leq x_j\leq \bar{x}_{jt}} \Pi_{jt}^\theta(x_j, \mathbf{x}_{-j,t}^\theta)$
Net rents	$R_{jt} = \mathbb{E}_t^\theta \int_{\mathcal{C}_j} [R_t - p_{Mt} I_t(\ell; \mathbf{x}_t)]\, d\ell$
Installed capacity	$T_{t+1} = T_t + \sum_j x_{jt} \iota_{jt}$
	State Transitions
Productivity	$Z_{it}(\ell) = A_{it}(\ell) e^{z_{it}} / \mathbb{E}_\pi[e^{z_i}]$, $i \in \{A, M\}$; $\quad Z_{Tt} = A_T e^{z_{Tt}} / \mathbb{E}_\pi[e^{z_T}]$
Sectoral shocks	$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma})$
Available frontier	$F_{it}(\ell) = \sup_{r\in[0,1]} e^{-\delta\lvert\ell - r\rvert} A_{it}^+(r)$
Adoption	$A_{i,t+1} = A_{it}^+ + \left[1 - e^{-\lambda_0 (T_t/T_{t_0})^{1-\alpha_T}}\right](F_{it} - A_{it}^+)$
	Market Clearing
Residence	$\mathbb{E}_t^\theta[(w + b - \gamma p_A)/\mathcal{P}] = \bar{u}$
Goods	$\partial_\ell H_i = Y_i - n c_i - e_i \Psi_i / p_i - I_i - \kappa_0 e^{-T_t} \lvert H_i\rvert$
Labor and population	$n = \sum_i e_i L_i + L_T$, $\quad \int_0^1 n\, d\ell = \bar{L}_t$
Services	$D_j \leq K_j$, $\quad \tau_j \geq 0$, $\quad \tau_j(K_j - D_j) = 0$

Notes: Local and date arguments are suppressed; $i \in \{A, M\}$ unless stated otherwise, and j indexes the county. $e_i \in \{0, 1\}$ indicates the highest-bidding goods sector with land rent $R = \max_i R_i$; $f_A(\ell) = f_A$ and $f_M(\ell) = f_M + \xi(\ell)$; construction uses manufactured goods ($I_A = 0$, $I_M = I$). Borders are closed and the numeraire is $p_A(1, t) = 1$. Remaining definitions are in the text.

transition probabilities toward states made more likely by $\boldsymbol{\varepsilon}_t$, weighted by the representativeness ratio $\Lambda_t(\mathbf{z}') = f(\mathbf{z}' \mid \mathbf{z}_t)/f(\mathbf{z}' \mid \overline{\mathbf{z}}_t)$:

$$\mathbb{E}_t^\theta[X] = \frac{\mathbb{E}_t[X \Lambda_t(\mathbf{z}_{t+1})^\theta]}{\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]}, \qquad \theta \geq 0, \tag{11}$$

where $\theta = 0$ nests rational expectations. Because $\mathbf{z}_t$ is known at date t and private discoveries are independent of $\mathbf{z}_{t+1}$, $\Lambda_t(\mathbf{z}_{t+1})^\theta$ cancels from (11) for any within-period payoff conditional on $(\Omega_t, \mathbf{x}_t)$ (Proposition 2). Exponentiating the Gaussian kernel yields the diagnostic one-step law (Online Appendix S.C.1); inside continuation values, county companies carry their diagnostic forecast forward as the date-$(t+1)$ reference so that perceived future news is mean zero—a convention

on how the date-t company models its future reference rather than a consequence of (11):

$$\mathbf{z}_{t+1} \overset{\theta}{\sim} \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^{\theta}, \boldsymbol{\Sigma}), \qquad \mathbf{m}_t^{\theta} = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \overline{\mathbf{z}}_t), \tag{12}$$

$$\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^{\theta}), \qquad \boldsymbol{\varepsilon}_{t+1}^{\text{forecast}} = \mathbf{z}_{t+1} - M(\mathbf{m}_t^{\theta}), \tag{13}$$

whereas in physical time at date $t+1$, the observed reference resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$.

Equation (12) delivers both boom-time overreaction and post-boom disappointment. Relative to the no-news center $\boldsymbol{\rho}\overline{\mathbf{z}}_t$, the perceived location shifts by $(1+\theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$ instead of $\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$. Because the diagnostic location overshoots the physical one by $\mathbf{m}_t^{\theta} - \boldsymbol{\rho}\mathbf{z}_t = \theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, the physical expectation of the diagnostic forecast error, $\mathbb{E}_t[\mathbf{z}_{t+1}] - M(\mathbf{m}_t^{\theta}) = M(\boldsymbol{\rho}\mathbf{z}_t) - M(\mathbf{m}_t^{\theta})$, satisfies $(\theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t)'\boldsymbol{\Sigma}^{-1}[M(\boldsymbol{\rho}\mathbf{z}_t) - M(\mathbf{m}_t^{\theta})] < 0$ whenever $\theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t \neq \mathbf{0}$ by strict convexity of the log-partition function of $\mathrm{TN}_{\mathcal{B}}$ (and is negative componentwise when innovations are uncorrelated or truncation is slack). Subsection III.F links these properties to the construction reversals and yield compression of Section II through county marginal-q (65) and the clipped investment rule (62). The aggregate state is $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \overline{\mathbf{z}}_t, t)$ (Figure 5).

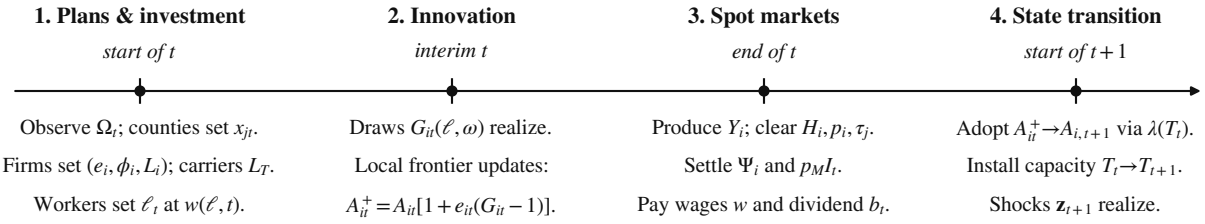


Figure 5: Within-Period Timing

Notes: County investment x_{jt}, producer plans (e_i, ϕ_i, L_i), carrier employment L_T, and worker residence ℓ_t at local contract wage $w(\ell, t)$ are committed before idiosyncratic discoveries $G_{it}(\ell, \omega)$ realize. Spot goods and service markets clear ex post, and completed capacity T_{t+1} and diffused knowledge $A_{i,t+1}$ enter at date $t+1$.

C Workers

Workers live where they work, supply one unit of labor, and hold equal shares in the national portfolio of land leases, goods producers, transport firms, and factor-wedge rebates, receiving common nonlabor income b_t in addition to the local contract wage $w(\ell, t)$. Goods are nonstorable, and households neither save nor borrow. Each period t, a worker commits residence $\ell_t \in [0,1]$ before private discoveries at the fixed contract wage $w(\ell, t)$ and chooses spot consumption $(c_A(\ell,t), c_M(\ell,t))$ ex post to solve

$$\begin{aligned} W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \; & \mathbb{E}_t^{\theta}\left[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))\right] \\ \text{s.t. } & p_A(\ell_s, s)c_A(s) + p_M(\ell_s, s)c_M(s) = w(\ell_s, s) + b_s, \quad c_A(s) \geq \gamma, \; c_M(s) \geq 0, \end{aligned} \tag{14}$$

with nonhomothetic Stone–Geary period utility (Fajgelbaum and Redding, 2022; Herrendorf et al., 2014; Kongsamut et al., 2001):

$$U(c_A(\ell,t), c_M(\ell,t)) = (c_A(\ell,t) - \gamma)^{\eta} c_M(\ell,t)^{1-\eta}, \qquad \eta \in (0,1), \quad \gamma > 0. \tag{15}$$

For $w + b_t \geq \gamma p_A$, spot optimization yields the Marshallian demands, marginal price index, and realized indirect utility:

$$\begin{aligned} c_A &= \gamma + \frac{\eta(w + b_t - \gamma p_A)}{p_A}, \qquad c_M = \frac{(1-\eta)(w + b_t - \gamma p_A)}{p_M}, \\ \mathcal{P}(p_A, p_M) &= \frac{p_A^{\eta} p_M^{1-\eta}}{\eta^{\eta}(1-\eta)^{1-\eta}}, \qquad u_t(\ell) = \frac{w(\ell,t) + b_t - \gamma p_A(\ell,t)}{\mathcal{P}(p_A(\ell,t), p_M(\ell,t))}. \end{aligned} \tag{16}$$

The subsistence floor $\gamma > 0$ generates Engel's law, shifting expenditure toward manufacturing as real income rises. The demands and $u_t(\ell)$ are defined where $w + b_t \geq \gamma p_A$; statewise feasibility at every site is part of the equilibrium definition (Definition 1) and is imposed as the nominal subsistence-slack clause of Condition 1 in Section III.G. Because residence is committed before private discoveries, workers evaluate locations by expected current indirect utility $\mathbb{E}_t^{\theta}[u_t(\ell)]$.

PROPOSITION 1: *If subsistence is feasible statewise at every site, expected indirect utility $\mathbb{E}_t^{\theta}[u_t(\ell)]$ is finite for every ℓ, and lifetime utility is finite, then, given the current price, wage and portfolio-income schedules, a worker's optimal dynamic residence choice maximizes expected current-period indirect utility.*[8] [Proof →]

Because annual relocation is costless and portfolio claims do not depend on residence, the continuation value of the worker's problem is invariant to current location, equalizing expected indirect utility across occupied sites; and because the contract wage is fixed before discoveries, it factors out of the expectation, so (17) inverts to the contract wage (18):

$$\mathbb{E}_t^{\theta}[u_t(\ell)] = \bar{u}_t \qquad \text{where } n(\ell,t) > 0 \quad (\mathbb{E}_t^{\theta}[u_t(\ell)] \leq \bar{u}_t \text{ elsewhere}), \tag{17}$$

$$w(\ell,t) = \frac{\bar{u}_t - \mathbb{E}_t^{\theta}[(b_t - \gamma p_A(\ell,t))/\mathcal{P}(p_A(\ell,t), p_M(\ell,t))]}{\mathbb{E}_t^{\theta}[1/\mathcal{P}(p_A(\ell,t), p_M(\ell,t))]}. \tag{18}$$

Because labor demand is positive at every site for any finite contract wage (carriers hire by (50) whenever $\mathbb{E}_t[\tau_{j(\ell)}] > 0$, and otherwise the winning goods sector hires by (30) at a positive expected net price), every site is occupied in equilibrium, (17) holds with equality on all of $[0,1]$, and (18) determines the contract wage at every site.

D Production, Research, and Land

Competitive entrants in sector $i \in \{A, M\}$ bid for one-period pure site leases at $\ell \in [0,1]$.[9] Effective productivity combines permanent local knowledge $A_i(\ell,t)$ (or constant carrier frontier $A_T > 0$) with the aggregate productivity factor, and realized output Y_i uses committed labor $L_i(\ell,t)$ under diminishing returns:

$$Z_i(\ell,t) = A_i(\ell,t) e^{z_{it}} / \mathbb{E}_{\pi}[e^{z_i}], \quad i \in \{A,M\}, \qquad Z_T(t) = A_T e^{z_{Tt}} / \mathbb{E}_{\pi}[e^{z_T}], \tag{19}$$

$$Y_i(\ell,t,\omega) = e_i(\ell,t) Z_i(\ell,t) G_i(\ell,t,\omega) L_i(\ell,t)^{\alpha_i}, \qquad i \in \{A,M\}, \quad 0 < \alpha_A < \alpha_M < 1, \tag{20}$$

where $\alpha_A < \alpha_M$ reflects the higher land intensity of agriculture relative to manufacturing (Valentinyi and Herrendorf, 2008). Entrants announce (e_i, ϕ_i, L_i) taking state-contingent spot prices $p_i(\ell,t,\omega)$, county tolls $\tau_{j(\ell)}(t,\omega)$, and the local contract wage $w(\ell,t)$ as given, earning the net unit price $p_i(\ell,t,\omega) - r_i \tau_{j(\ell)}(t,\omega)$ ex post.

[8] Proofs of the propositions are in Appendix A.

[9] Transport routes occupy narrow corridors and do not crowd out productive land.

Manufacturing firms face a predetermined local operating-labor wedge,

$$\psi_A(\ell) = 0, \qquad \psi_M(\ell) = \bar{\psi}_M \omega_M(\ell), \qquad \bar{\psi}_M \geq 0, \quad \omega_M(\ell) > 0, \tag{21}$$

paying $(1 + \psi_M(\ell))w(\ell, t)$ per hired production worker while agriculture is undistorted ($\psi_A = 0$) and wedge receipts are rebated lump-sum to households. Holding $\omega_M(\ell)$ fixed makes transport improvements a source of first-order allocative efficiency gains as labor reallocates toward distorted industrial producers (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024).

Before private discoveries, an entrant in sector $i \in \{A, M\}$ chooses a research success probability $\phi_i \in [0, 1)$. Building on the Pareto idea-flow representation of Kortum (1997) and Eaton and Kortum (2002), a successful draw yields a Pareto productivity multiplier with tail exponent $a_i > 1/(1 - \alpha_i)$:

$$\Pr(G_i = 1 \mid \phi_i) = 1 - \phi_i, \tag{22}$$

$$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}, \qquad g \geq 1, \quad a_i > 1. \tag{23}$$

Discoveries correlate across space. A success field $X^s_{it}(\ell)$ and a quality field $X^g_{it}(\ell)$ are stationary standard Gaussian Ornstein–Uhlenbeck processes on $[0, 1]$, independent of each other, across sectors and dates, and of aggregate productivity gaps, with correlation length $d_0 > 0$; with Φ the standard normal distribution function, an entrant choosing ϕ at ℓ draws

$$\begin{aligned} G_i(\phi; \ell, \omega) &= 1 + \mathbf{1}_{\{\Phi(X^s_{it}(\ell)) < \phi\}} \left[\Phi(X^g_{it}(\ell))^{-1/a_i} - 1\right], \\ \mathrm{Cov}(X^h_{it}(\ell), X^h_{it}(r)) &= e^{-|\ell - r|/d_0}, \qquad h \in \{s, g\}. \end{aligned} \tag{24}$$

Because $\Phi(X^h_{it}(\ell))$ is uniform at every site, (24) reproduces (22)–(23) at each ℓ, while neighboring sites draw correlated discoveries. All research intensities share one probability space: $G_i(\phi; \ell, \omega)$ is nondecreasing in ϕ, and $G_i(\phi_n; \ell, \omega) \to G_i(\phi; \ell, \omega)$ almost surely whenever $\phi_n \to \phi$.

Innovation requires a wage-indexed monetary expenditure committed before discovery and settled in the spot market by purchasing local own-sector output:

$$\Psi_i(\phi_i; w, \ell) = \begin{cases} w\left[f_i(\ell) + \dfrac{\chi_i}{1 - \phi_i}\right] & \text{if } \phi_i \in (0, 1), \\ 0 & \text{if } \phi_i = 0, \end{cases} \tag{25}$$

$$q^R_i(\omega) = \frac{\Psi_i(\phi_i; w, \ell)}{p_i(\omega)}, \tag{26}$$

where $f_A(\ell) = f_A \geq 0$ and $f_M(\ell) = f_M + \xi(\ell) \geq 0$ combines fixed participation cost $f_i \geq 0$ with predetermined manufacturing energy overhead $\xi(\ell) = \omega_E e(\ell) \geq 0$ ($\omega_E \geq 0$ scaling the standardized coalfield energy exposure schedule $e(\ell)$ of Allen 2009), and $\chi_i > 0$ governs research convexity. Gross marketed output Y_i includes research goods, using local distribution services $r_i Y_i$ once, while only the knowledge transition carries discoveries to future periods.

Firm optimization and innovation threshold.— Risk-neutral firms measure payoffs in the numeraire $p_A(1, t) = 1$ and hold neither future land claims nor patents. Taking state-contingent goods prices $p_i(\omega)$, county tolls $\tau_{j(\ell)}(\omega)$, and the local contract wage w as given, a producer with hired labor L_i and discovery $G_i(\omega)$ chooses spot output utilization to maximize operating surplus:

$$\max_{0 \leq Y \leq Z_i G_i L_i^{\alpha_i}} \left\{ (p_i - r_i \tau_{j(\ell)}) Y - (1 + \psi_i) w L_i \right\} = (p_i - r_i \tau_{j(\ell)})_+ Z_i G_i L_i^{\alpha_i} - (1 + \psi_i) w L_i, \tag{27}$$

where $[x]_+ = \max\{x, 0\}$, so full capacity $Y_i = e_i Z_i G_i L_i^{\alpha_i}$ operates whenever $p_i(\omega) > r_i \tau_{j(\ell)}(\omega)$. Expected net unit revenue per efficiency unit,

$$m_i(\phi_i; \ell) = \mathbb{E}_t^{\theta}\Big[\big(p_i(\ell, \omega) - r_i \tau_{j(\ell)}(\omega)\big)_+ G_i(\phi_i; \ell, \omega)\Big], \tag{28}$$

is affine in research intensity, $m_i(\phi_i; \ell) = m_i(0; \ell)[1 + \kappa_i(\ell)\phi_i]$, where $\kappa_i(\ell) = \mathbb{E}_t[(p_i - r_i\tau_{j(\ell)})_+(V_i - 1)]/m_i(0; \ell) \geq 0$ is the expected proportional revenue gain from a discovery (since success replaces the unit multiplier by V_i with probability ϕ_i independently of quotes at an atomistic site; Lemma A.1). The entrant maximizes expected land-bid surplus:

$$R_i(\ell) = \max_{\substack{0 \leq \phi_i < 1 \\ L_i \geq 0}} \Big\{ Z_i(\ell) m_i(\phi_i; \ell) L_i^{\alpha_i} - (1 + \psi_i(\ell)) w(\ell) L_i - \Psi_i(\phi_i; w(\ell), \ell) \Big\}. \tag{29}$$

First-order optimization in L_i yields the conditional hiring rule and scalar innovation problem:

$$L_i^*(\phi_i; w, \ell) = \left[\frac{\alpha_i Z_i(\ell) m_i(\phi_i; \ell)}{(1 + \psi_i(\ell)) w(\ell)} \right]^{\frac{1}{1-\alpha_i}}, \tag{30}$$

$$R_i(w, \ell) = \max_{\phi_i \in [0,1)} \Big\{ \Lambda_i(\ell) w(\ell)^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi_i; \ell)^{\frac{1}{1-\alpha_i}} - \Psi_i(\phi_i; w(\ell), \ell) \Big\}, \tag{31}$$

with $L_i^*(\phi_i; w, \ell) = 0$ if $m_i(\phi_i; \ell) = 0$ and $\Lambda_i(\ell) = (1 - \alpha_i) Z_i(\ell) [\alpha_i Z_i(\ell)/(1 + \psi_i(\ell))]^{\alpha_i/(1-\alpha_i)}$.

Convex research costs yield a reservation threshold for active innovation. On $\phi_i \in (0, 1)$, Ψ_i is strictly convex ($\partial^2_{\phi\phi} \Psi_i = 2w(\ell)\chi_i/(1 - \phi_i)^3 > 0$) and diverges as $\phi_i \uparrow 1$. By the envelope theorem, an interior research optimum $\tilde{\phi}_i(w, \ell) \in (0, 1)$ satisfies

$$Z_i(\ell) L_i^*(\tilde{\phi}_i; w, \ell)^{\alpha_i} \partial_\phi m_i(\tilde{\phi}_i; \ell) = \frac{w(\ell)\chi_i}{\big(1 - \tilde{\phi}_i(w, \ell)\big)^2}, \tag{32}$$

and for $\chi_i > \bar{\chi}_{it}$ the active objective is strictly concave on $(0, 1)$ with at most one interior maximizer. Dividing (31) by $w(\ell)$ with baseline profitability index $\pi_i^0(w, \ell) \equiv \Lambda_i(\ell)[m_i(0; \ell)/w(\ell)]^{1/(1-\alpha_i)}$ and discovery gain $\Gamma_i(\phi_i; \ell) \equiv [m_i(\phi_i; \ell)/m_i(0; \ell)]^{1/(1-\alpha_i)} - 1 = [1 + \kappa_i(\ell)\phi_i]^{1/(1-\alpha_i)} - 1$ gives

$$\frac{R_i(w, \ell)}{w(\ell)} = \pi_i^0(w, \ell) + \max\left\{ 0, \max_{\phi_i \in (0,1)} \left[\pi_i^0(w, \ell) \Gamma_i(\phi_i; \ell) - f_i(\ell) - \frac{\chi_i}{1 - \phi_i} \right] \right\}, \tag{33}$$

where (32) reads $\pi_i^0(w, \ell)\kappa_i(\ell)[1 + \kappa_i(\ell)\tilde{\phi}_i]^{\alpha_i/(1-\alpha_i)}/(1 - \alpha_i) = \chi_i/(1 - \tilde{\phi}_i)^2$. Because the bracketed gain in (33) is a supremum of affine functions of π_i^0 with nonnegative slopes Γ_i, equals $-f_i(\ell) - \chi_i < 0$ at $\pi_i^0 = 0$, and is unbounded above when $\kappa_i(\ell) > 0$, it has a unique zero crossing at a reservation threshold $\underline{\pi}_i(\ell) \in (0, \infty]$ (Appendix A.6.1): innovation is active ($\phi_i^*(w, \ell) = \tilde{\phi}_i(w, \ell) > 0$) if and only if $\pi_i^0(w, \ell) > \underline{\pi}_i(\ell)$. Transport capacity $T_t(\ell)$ raises export net prices and lowers county tolls while coalfield proximity lowers $\xi(\ell)$, pushing $\pi_M^0(w, \ell)$ above $\underline{\pi}_M(\ell)$ in industrializing counties; higher wages $w(\ell)$ weakly lower $\phi_i^*(w, \ell)$ and, wherever $m_i(0; \ell) > 0$, strictly lower $L_i^*(\phi_i^*(w, \ell); w, \ell)$.

Land competition and spatial sorting.— Competitive bidding assigns each site to the sector with the higher expected bid rent:

$$R(\ell) = \max\{R_A(\ell),\, R_M(\ell)\}, \qquad e_i(\ell) = \begin{cases} 1 & \text{if } R_i(\ell) = R(\ell), \\ 0 & \text{otherwise.} \end{cases} \tag{34}$$

Because $\alpha_A < \alpha_M$ and $\Lambda_i(\ell) \propto Z_i(\ell)^{1/(1-\alpha_i)}(1 + \psi_i(\ell))^{-\alpha_i/(1-\alpha_i)}$, the elasticity- and wedge-adjusted comparative-advantage profile $\frac{1}{1-\alpha_M} \log Z_M(\ell) - \frac{1}{1-\alpha_A} \log Z_A(\ell) - \frac{\alpha_M}{1-\alpha_M} \log(1 + \psi_M(\ell))$ governs the

sign of $R_A(w,\ell) - R_M(w,\ell)$. Under Condition 2 of Section III.G, which requires this profile to rise enough to dominate the spatial variation of wages, net prices, and research activity, $R_A(w,\ell) - R_M(w,\ell)$ changes sign once, at an interior boundary $b_{ct} \in (0,1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing; in the frictionless benchmark of Appendix A.6.3, where wages and prices are uniform across space and research is inactive, an increasing profile is by itself sufficient for at most one crossing. At the winning plan, local employment, worker participation, output, and wedge transfers satisfy

$$n(\ell) = \sum_{i \in \{A,M\}} e_i(\ell) L_i(\ell) + L_T(\ell), \tag{35}$$

$$\bar{u}_t = \mathbb{E}_t^\theta \left[\frac{w(\ell) + b_t - \gamma p_A(\ell)}{\mathcal{P}(p_A(\ell), p_M(\ell))} \right], \tag{36}$$

and realized output, innovation expenditure, and wedge transfers satisfy

$$Y_i = \mathbf{1}_{\{p_i > r_i \tau_{j(\ell)}\}} e_i Z_i G_i L_i^{\alpha_i}, \qquad \Psi_i = \Psi_i(\phi_i^*(\ell); w(\ell), \ell), \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i \tag{37}$$

(any utilization in $[0, e_i Z_i G_i L_i^{\alpha_i}]$ at $p_i = r_i \tau_{j(\ell)}$ is set by county dispatch).

PROPOSITION 2: *Fix the inherited physical state, the dated environment, and (where the conditional dated equilibrium is not unique) a common selection rule under which pre-discovery plans depend only on the inherited state, the dated environment, and the investment profile, and the post-discovery family depends in addition only on the date-t discoveries. Conditional on an investment profile, current household, firm, and carrier decisions, market prices, and the county current dividends (56) of Section III.F are invariant to θ. If two belief specifications select the same profile, the same private discoveries and sectoral shocks generate the same current allocation and next physical state.* [Proof →]

Because $\mathbb{E}_t^\theta$ in (11) reweights only future aggregate shocks $\mathbf{z}_{t+1}$ while leaving the conditional distribution of date-t private discoveries $G_{it}(\ell, \omega)$ unchanged, every within-period expectation conditional on $(\Omega_t, \mathbf{x}_t)$ reduces to rational expectations ($\mathbb{E}_t^\theta[\cdot] = \mathbb{E}_t[\cdot]$ in (18), (28), and (36)); diagnostic expectations affect the allocation only through county infrastructure investment $\mathbf{x}_t^\theta$.[10] Under Condition 3 of Section III.G the conditional dated equilibrium is unique (Proposition 6), and no selection rule is needed.

After early-stage plans are committed, the active producer's realized discovery $G_{it}(\ell) \geq 1$ enters the public knowledge pool, updating local pre-diffusion technology (Cai et al., 2022):

$$A_{it}^+(\ell) = A_{it}(\ell)\big[1 + e_{it}(\ell)(G_{it}(\ell) - 1)\big]. \tag{38}$$

The inactive sector ($e_{it}(\ell) = 0$) retains $A_{it}(\ell)$ at this step, and transitory aggregate gaps $\mathbf{z}_t$ do not enter permanent knowledge. Because spillovers decay with distance at rate $\delta > 0$ (Berkes et al., 2025; Keller, 2002), the technology frontier available at ℓ is the pointwise spatial supremum

$$F_{it}(\ell) = \sup_{r \in [0,1]} e^{-\delta|\ell - r|} A_{it}^+(r), \qquad \delta > 0. \tag{39}$$

Because the supremum is taken site by site, A_{it}^+ must be determined at every ℓ rather than almost everywhere, which is why Definition 1 requires sector assignment and research intensity to be

[10] Keeping within-period Bernoulli–Pareto discovery probabilities physical isolates the intertemporal network-investment channel, where construction must be committed a full period before operating returns realize.

optimal pointwise (Proposition 6); the supremum is finite almost surely because the Gaussian fields generating the discoveries in (24) have continuous sample paths (Online Appendix Lemma S.C.1).

Diffusing the frontier again creates no new knowledge: F_{it} is a fixed point of its own supremum because exponential decay compounds along the path. Relative to the initial reference network $T_{t_0}(\ell)$ (with baseline half-life $\log(2)/\lambda_0$ years), both sectors close a fraction of their gap to $F_{it}(\ell)$ via the Nelson and Phelps (1966) catch-up law at a speed set by installed route capacity (Berkes et al., 2026; Cinnirella et al., 2024):

$$A_{i,t+1}(\ell) = A_{it}^{+}(\ell) + (1 - e^{-\lambda_t(\ell)})[F_{it}(\ell) - A_{it}^{+}(\ell)], \tag{40}$$

$$\lambda_t(\ell) = \lambda_0 \left[\frac{T_t(\ell)}{T_{t_0}(\ell)}\right]^{1-\alpha_T}, \qquad \lambda_0 > 0. \tag{41}$$

Higher route capacity $T_t(\ell)$ accelerates local catch-up $\lambda_t(\ell)$ toward $F_{it}(\ell)$ without raising the frontier itself.

PROPOSITION 3: *For bounded positive post-discovery knowledge and finite nonnegative adoption rates, $A_{it}^{+}(\ell) \le A_{i,t+1}(\ell) \le F_{it}(\ell)$. Holding discoveries and reference levels fixed, more inherited capacity weakly raises next-period productivity. If no further discoveries occur and the spillover decay δ is fixed, the available frontier remains unchanged and*

$$F_{it}(\ell) - A_{i,t+h}(\ell) = [F_{it}(\ell) - A_{it}^{+}(\ell)] \exp\left[-\sum_{k=t}^{t+h-1} \lambda_k(\ell)\right], \qquad h \ge 1. \tag{42}$$

[Proof →]

Localized research in industrial hubs pushes out $F_{it}(\ell)$, while county transport investments raise $\lambda_t(\ell)$ and propagate discoveries across space. Because county infrastructure investors take the aggregate knowledge path as given (Definition 2(ii)) and private entrants hold no claim on future adoption, (40)–(41) generate a dynamic spatial adoption externality whose one-period marginal return is (66) and whose cumulative effect is the exponent in (42). Online Appendix S.C.2 proves that the frontier is finite with finite moments of every order below a_i and expresses the conditional mean of next-period knowledge in terms of the frontier mean $\mathbb{E}_t[F_{it}(\ell)]$.

E Transport and Distribution

Moving goods over land requires both durable route infrastructure (Donaldson, 2018; Redding and Turner, 2015) and operating carriage services priced in local transport markets (Brancaccio et al., 2020). Transport firms combine installed route capacity $T(\ell, t)$ with hired labor $L_T(\ell, t)$ committed at the local contract wage $w(\ell, t)$ before private discoveries, selling operating services ex post at the county toll τ_j:

$$Y_T(\ell, t) = Z_T(t) L_T(\ell, t)^{\alpha_T} T(\ell, t)^{1-\alpha_T}, \qquad Z_T(t) > 0, \quad 0 < \alpha_T < 1. \tag{43}$$

Each county $\mathcal{C}_j$ operates a competitive operating-service market at a common toll $\tau_{j(\ell)}$: services produced anywhere in $\mathcal{C}_j$ meet distribution or transit demand anywhere within $\mathcal{C}_j$ without additional cost, whereas services cannot be dispatched across county boundaries.

Shipping good i across space incurs both physical transit degradation at rate $\kappa(\ell, t)$ and r_i units of county operating services per unit of distance at toll $\tau_{j(\ell)}(t)$. A unit of good i carried a distance $d\ell$ loses $\kappa\, d\ell$ in transit and uses $r_i\, d\ell$ services at toll $\tau_{j(\ell)}$, so a trader earns zero profit only if the delivered price rises at the rate $\kappa p_i + r_i \tau_{j(\ell)}$ along routes with positive flow and no faster

where flows are zero. This yields the effective ad valorem transport-cost rate $\tilde{\kappa}_i(\ell,t)$ and the spatial no-arbitrage law $|\partial_\ell \log p_i| \le \tilde{\kappa}_i$ for signed trade flow $H_i(\ell,t)$ (positive rightward):

$$\tilde{\kappa}_i(\ell,t) = \kappa(\ell,t) + \frac{r_i \tau_{j(\ell)}(t)}{p_i(\ell,t)}, \tag{44}$$

$$|\partial_\ell p_i| \le \kappa p_i + r_i \tau_{j(\ell)}, \qquad \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)}) \operatorname{sign}(H_i) \quad \text{if } H_i \neq 0. \tag{45}$$

For the absolutely continuous price versions of Definition 1, (45) bounds the price gradient by $\kappa p_i + r_i \tau_{j(\ell)}$, so spot prices $p_i(\cdot,t)$ are Lipschitz across space: a step in $\tau_{j(\ell)}$ across a county border kinks $\partial_\ell p_i$ without creating a price jump.

Installed infrastructure reduces the physical loss rate and the bilateral iceberg delivery factor $d_t(\ell,r)$ (the units dispatched at r per unit delivered at ℓ):

$$\kappa(\ell,t) = \kappa_0 e^{-T(\ell,t)}, \qquad \kappa_0 > 0, \quad T(\ell,t) > 0, \tag{46}$$

$$d_t(\ell,r) = \exp\left\{ \kappa_0 \left| \int_r^\ell e^{-T_t(u)}\, du \right| \right\}. \tag{47}$$

Along a rightward flow path from r to ℓ, (45) integrates to the delivered price $p_i(\ell,t) = d_t(\ell,r)\, p_i(r,t) + r_i \int_r^\ell d_t(\ell,u)\, \tau_{j(u)}(t)\, du$: carriage adds a distance-weighted service charge grossed up by downstream losses rather than scaling d_t.

Each unit of marketed output $Y_i(\ell,t)$ requires r_i units of county service for local origin distribution and r_i units per unit of distance shipped ($r_M = 1$, $r_A > 0$), so county-wide service demand and operating capacity are

$$D_j(\omega) = \int_{\mathcal{C}_j} \sum_{i \in \{A,M\}} r_i \big(Y_i(\ell,\omega) + |H_i(\ell,\omega)|\big)\, d\ell, \qquad K_j = \int_{\mathcal{C}_j} Y_T(\ell,t)\, d\ell. \tag{48}$$

Carrier optimization and county service clearing.— Before private discoveries, a competitive carrier at $\ell \in \mathcal{C}_j$ chooses $L_T(\ell)$ taking $T(\ell)$, $w(\ell)$, and the distribution of τ_j as given:

$$\max_{L_T \ge 0} \mathbb{E}_t^\theta [\tau_j Y_T - w L_T], \qquad Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}, \tag{49}$$

$$w = \alpha_T Z_T L_T^{\alpha_T - 1} T^{1-\alpha_T}\, \mathbb{E}_t^\theta[\tau_j] \quad \Longleftrightarrow \quad L_T^*(w,\ell) = \left[\frac{\alpha_T Z_T\, \mathbb{E}_t^\theta[\tau_{j(\ell)}]}{w(\ell)} \right]^{\frac{1}{1-\alpha_T}} T(\ell), \tag{50}$$

where $\mathbb{E}_t^\theta[\tau_j] = \mathbb{E}_t[\tau_j]$ conditional on $(\Omega_t, \mathbf{x}_t)$ by Proposition 2. After private discoveries realize, predetermined county capacity K_j clears against county service demand D_j via complementary slackness, and carrier operating surplus accrues to the household portfolio:

$$D_j \le K_j, \qquad \tau_j \ge 0, \qquad \tau_j (K_j - D_j) = 0, \tag{51}$$

$$\Pi_T(\ell,t) = \tau_{j(\ell)}(t) Y_T(\ell,t) - w(\ell,t) L_T(\ell,t). \tag{52}$$

F County Infrastructure Companies

Local infrastructure companies built British turnpikes, canals, and railways under county parliamentary acts, rather than under a centralized national planner (Allen and Arkolakis, 2022; Bogart, 2014; Fajgelbaum and Schaal, 2020). Each county $j \in \{1, \dots, J\}$ has an infrastructure company holding a long-lived claim on land-lease rents inside $\mathcal{C}_j$. Roads, waterways, and railways accumulate into a composite route stock $T(\ell,t)$ under one-period time-to-build (Kydland and Prescott,

1982): before private discoveries at date t, company j commits completion intensity $x_{jt} \in [0, \bar{x}_{jt}]$ within the resource-feasible statutory box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}] \subseteq \prod_j [0, o_{jt}]$ ($o_{jt} \in \{0, 1\}$), and completed increments enter operating route capacity at date $t + 1$:

$$x_{jt} \in [0, \bar{x}_{jt}], \qquad \operatorname{supp} \iota_{jt} \subseteq \mathcal{C}_j, \tag{53}$$

$$T_{t+1}(\ell) = T_t(\ell) + \sum_{j=1}^{J} x_{jt} \iota_{jt}(\ell), \qquad T_0(\ell) \geq \underline{T} > 0. \tag{54}$$

Construction uses manufactured goods with quadratic adjustment costs on the county engineering profile (e_{jt}, k_{jt}) ($\operatorname{supp} k_{jt} \subseteq \mathcal{C}_j$, $e_{jt} > 0$):

$$I_t(\ell; \mathbf{x}_t) = \sum_{j=1}^{J} m_I e_{jt} k_{jt}(\ell) \left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2 \right), \qquad m_I > 0, \quad \chi_I > 0, \tag{55}$$

purchased at the realized spot price $p_M(\ell, t, \omega)$. Full-package physical increments grow at two percent per year, $\iota_{jt} = 1.02^{t - t_0} \iota_{j,t_0}$; after the last observed schedule year $t = 1881$, each county repeats its 1881 opportunity $o_{j,1881}$ and engineering profile $(e_j k_j)_{1881}$ at the same (m_I, χ_I) while physical increments continue to grow as $\iota_{j,1881+h} = 1.02^h \iota_{j,1881}$.

Investor optimization, marginal-q, and stage Nash equilibrium.— Taking rival counties' intensities $\mathbf{x}_{-j,t}$ as given, company j's current payoff is expected county land rent net of spot manufacturing construction expenditure:

$$R_{jt}(\mathbf{x}; \Omega_t) = \mathbb{E}_t^{\theta} \left[\int_{\mathcal{C}_j} \{ R_t(\ell; \mathbf{x}, \Omega_t) - p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t) I_t(\ell; \mathbf{x}) \} \, d\ell \right]. \tag{56}$$

By Proposition 2, current returns conditional on $(\Omega_t, \mathbf{x})$ depend only on physical private discoveries ($\mathbb{E}_t^{\theta} = \mathbb{E}_t$ in (56)), whereas future aggregate shocks enter the continuation value under $\mathbb{E}_t^{\theta}$ with perceived reference $\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^{\theta})$ from (13) (while the physical reference at $t + 1$ resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho} \mathbf{z}_t)$):

$$\Pi_{jt}^{\theta}(\mathbf{x}; \Omega_t) = R_{jt}(\mathbf{x}; \Omega_t) + \beta \mathbb{E}_t^{\theta} \left[V_{j,t+1}^{\theta}(\Omega_{t+1}) \right], \qquad V_{jt}^{\theta}(\Omega_t) = \Pi_{jt}^{\theta} \left(\mathbf{x}_t^{\theta}(\Omega_t); \Omega_t \right), \tag{57}$$

$$x_{jt}^{\theta}(\Omega_t) \in \arg \max_{0 \leq u \leq \bar{x}_{jt}} \Pi_{jt}^{\theta}(u, \mathbf{x}_{-j,t}^{\theta}(\Omega_t); \Omega_t), \qquad j = 1, \ldots, J. \tag{58}$$

If the pure-strategy Nash set $\mathcal{N}_t^{\theta}$ contains multiple profiles, parliamentary coordination selects the profile maximizing $\sum_{j=1}^{J} \Pi_{jt}^{\theta}$ (with deterministic tie-breaking). Because the aggregate knowledge trajectory $(A_{As}, A_{Ms})_{s \geq t+1}$ reflects atomistic discoveries and frontier adoption across the continuum, each county company takes it as external to its own choice x_{jt}: the expectation in (57) integrates $(A_{A,t+1}, A_{M,t+1})$ under a conjectured law conditional on Ω_t that does not vary with $\mathbf{x}$ (while $T_{t+1} = T_t + \sum_k x_{kt} \iota_{kt}$ does), and in dynamic equilibrium (Definition 2) that conjectured law coincides with the law generated by $\mathbf{x}_t^{\theta}$.

Where differentiable, county j equates its perceived marginal continuation value q_{jt}^{θ} (with respect to own route capacity T_{t+1} at the given knowledge path) to marginal construction cost c_{jt} (Hayashi, 1982):

$$q_{jt}^{\theta}(\mathbf{x}) = \beta \partial_{x_j} \mathbb{E}_t^{\theta}[V_{j,t+1}^{\theta}(\Omega_{t+1})], \qquad c_{jt}(\mathbf{x}) = -\partial_{x_j} R_{jt}(\mathbf{x}; \Omega_t), \tag{59}$$

$$q_{jt}^{\theta}(\mathbf{x}_t^{\theta}) = c_{jt}(\mathbf{x}_t^{\theta}) \tag{60}$$

at an interior best response $x^\theta_{jt} \in (0, \bar{x}_{jt})$, with complementary slackness at $\{0, \bar{x}_{jt}\}$. Because $\operatorname{supp} k_{kt} \subseteq \mathcal{C}_k$, county j pays only for its own construction, so $R_{jt}(\mathbf{x}; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}) - m_I(x_{jt} + \frac{\chi_I}{2} x_{jt}^2) D^I_{jt}(\mathbf{x})$ with gross county land rent and construction-price index

$$\widetilde{R}_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} R_t(\ell; \mathbf{x}, \Omega_t)\, d\ell, \qquad D^I_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} \mathbb{E}_t[p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t)] e_{jt} k_{jt}(\ell)\, d\ell, \tag{61}$$

where $D^I_{jt} > 0$ whenever $o_{jt} = 1$ (counties with $\bar{x}_{jt} = 0$ set $x_{jt} = 0$ and drop out of (62) and (63)–(64)). Differentiating yields c_{jt} and the gross marginal value $\widetilde{q}^\theta_{jt}$:

$$\begin{aligned} c_{jt} &= m_I(1 + \chi_I x_{jt}) D^I_{jt} - \partial_{x_j} \widetilde{R}_{jt} + m_I\Big(x_{jt} + \frac{\chi_I}{2} x_{jt}^2\Big) \partial_{x_j} D^I_{jt}, \\ \widetilde{q}^\theta_{jt} &= q^\theta_{jt} + \partial_{x_j} \widetilde{R}_{jt} - m_I\Big(x_{jt} + \frac{\chi_I}{2} x_{jt}^2\Big) \partial_{x_j} D^I_{jt}. \end{aligned}$$

Condition (60) reads $\widetilde{q}^\theta_{jt} = m_I(1+\chi_I x_{jt}) D^I_{jt}$, which with complementary slackness yields the clipped investment rule

$$x^\theta_{jt} = \min\left\{\bar{x}_{jt},\ \max\left\{0,\ \frac{1}{\chi_I}\left[\frac{\widetilde{q}^\theta_{jt}(\mathbf{x}^\theta_t)}{m_I D^I_{jt}(\mathbf{x}^\theta_t)} - 1\right]\right\}\right\}. \tag{62}$$

A county builds when the marginal value of capacity per unit of construction outlay exceeds one, and diagonal dominance of own adjustment-cost curvature over cross-county spillovers yields a unique stage equilibrium (Vives, 1999):

PROPOSITION 4: *Fix $(\Omega_t, \mathcal{E}_t)$ and $\theta \geq 0$ on a nonempty compact box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ where every profile has a unique admissible private allocation and continuous county payoffs Π^θ_{jt}. If each Π^θ_{jt} is concave in x_{jt}, $\mathcal{N}^\theta_t$ is nonempty and compact. If, in addition, Π^θ_{jt} is C^2 and there exist $m_j > 0$, $L_{jk} \geq 0$, and $v_j > 0$ on $\mathcal{X}_t$ such that*

$$-\partial_{jj}\Pi^\theta_j \geq m_j > 0, \qquad |\partial_{jk}\Pi^\theta_j| \leq L_{jk} \quad (k \neq j), \tag{63}$$

$$\gamma_I = \max_j \frac{\sum_{k \neq j} L_{jk} v_k}{m_j v_j} < 1, \tag{64}$$

then the county Nash equilibrium is unique and simultaneous best responses converge to it from every initial profile in $\mathcal{X}_t$. If instead Π^θ_{jt} is C^1 and the pseudogradient $\mathbf{g}^\theta_t = (\partial_{x_j}\Pi^\theta_{jt})_j$ is strongly monotone on $\mathcal{X}_t$, $\sum_j (x_j - y_j)[g^\theta_{jt}(\mathbf{y}) - g^\theta_{jt}(\mathbf{x})] \geq \mu \|\mathbf{x} - \mathbf{y}\|^2$ for some $\mu > 0$, the Nash equilibrium is likewise unique. In either case, if Π^θ_{jt} is jointly measurable in Ω_t, the selection $\Omega_t \mapsto \mathbf{x}^\theta_t(\Omega_t)$ is measurable. [Proof →]

By Proposition 2, θ enters the dated allocation only through $q^\theta_{jt}(\mathbf{x})$ in (60). Differentiating (11) in θ gives $\partial_\theta \mathbb{E}^\theta_t[X] = \operatorname{Cov}^\theta_t(X, \log \Lambda_t(\mathbf{z}_{t+1}))$ with $\log \Lambda_t(\mathbf{z}_{t+1}) = \mathbf{z}'_{t+1} \boldsymbol{\Sigma}^{-1} \boldsymbol{\rho} \boldsymbol{\varepsilon}_t + \text{const}$. Holding $V^\theta_{j,t+1}$ and its reference argument $M(\mathbf{m}^\theta_t)$ fixed, current news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$ raises perceived marginal-q whenever the marginal value of capacity has positive covariance with the shocks signaled by the news. Adding the continuation shift $dq^\theta_{jt}/d\theta = \partial_\theta q^\theta_{jt}\big|_{V_{t+1}} + \beta \mathbb{E}^\theta_t[\partial_{x_j} \partial_\theta V^\theta_{j,t+1}]$ (where $\partial_\theta V^\theta_{j,t+1}$ collects next-period policies, beliefs, and the reference shift $\partial_{\mathbf{m}} M(\mathbf{m}) = \operatorname{Var}_{\mathrm{TN}_{\mathcal{B}(\mathbf{m},\boldsymbol{\Sigma})}}(\mathbf{z})\, \boldsymbol{\Sigma}^{-1}$) and applying the implicit function theorem on the interior set $\mathcal{I}_t = \{j : x^\theta_{jt} \in (0, \bar{x}_{jt})\}$ gives

$$\begin{aligned} \partial_\theta q^\theta_{jt}(\mathbf{x})\Big|_{V_{t+1}} &= \beta \operatorname{Cov}^\theta_t\Big(\partial_{x_j} V^\theta_{j,t+1}(\Omega_{t+1}),\ \mathbf{z}'_{t+1} \boldsymbol{\Sigma}^{-1} \boldsymbol{\rho} \boldsymbol{\varepsilon}_t\Big), \\ \frac{d\mathbf{x}^\theta_{t,\mathcal{I}_t}}{d\theta} &= -\left[D_{\mathbf{x}_{\mathcal{I}_t}}(\mathbf{q}^\theta_{t,\mathcal{I}_t} - \mathbf{c}_{t,\mathcal{I}_t})\right]^{-1} \frac{d\mathbf{q}^\theta_{t,\mathcal{I}_t}(\mathbf{x}^\theta_t)}{d\theta}, \end{aligned} \tag{65}$$

under strict complementary slackness outside $\mathcal{I}_t$ (since c_{jt} is θ-free by Proposition 2, and the interior block $[\partial_{jk}\Pi^{\theta}_{jt}]_{j,k\in\mathcal{I}_t}$ is invertible under either certificate of Proposition 4).

Each county company captures only local land rents inside $\mathcal{C}_j$ and omits three social returns to route capacity: (i) rival-county rent gains $\sum_{k\neq j}\partial_{x_j}\Pi^{0}_{kt}$ from cheaper transit through $\mathcal{C}_j$ (Allen and Arkolakis, 2022; Donaldson and Hornbeck, 2016; Fajgelbaum and Schaal, 2020; Fogel, 1964; Krugman, 1991); (ii) the allocative gain $\int_0^1 \psi_M(\ell)w(\ell)\partial_{x_j}L_{Mt}(\ell)\,d\ell$ from labor pulled into wedge-distorted manufacturing (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008); and (iii) faster spatial knowledge diffusion (Desmet and Rossi-Hansberg, 2014; Lucas, 1988; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990), since higher $T_s(\ell)$ at $s \geq t+1$ raises next-period knowledge at given discoveries by

$$\frac{\partial A_{i,s+1}(\ell)}{\partial T_s(\ell)} = (1-\alpha_T)\frac{\lambda_s(\ell)}{T_s(\ell)}e^{-\lambda_s(\ell)}\left[F_{is}(\ell) - A^{+}_{is}(\ell)\right] \geq 0, \tag{66}$$

which is positive wherever the available frontier exceeds local technology. While (i) and (ii) can take either sign in general and are positive when cheaper transit raises rival rents and manufacturing employment, (iii) is nonnegative by (66); rational investors ($\theta = 0$) omit all three and underinvest whenever their sum is positive.

Diagnostic expectations ($\theta > 0$) expand buildout after favorable news when county investments are complements or cross-county spillovers are small. By Stein's identity for the truncated Gaussian kernel, the kernel term in (65) equals

$$\partial_\theta q^{\theta}_{jt}\Big|_{V_{t+1}} = \beta \sum_{s\in\{A,M,T\}} \rho_s \varepsilon_{st}\, \mathbb{E}^{\theta}_t\Big[\partial_{z_{s,t+1}}\partial_{x_j} V^{\theta}_{j,t+1}(\Omega_{t+1})\Big]$$

up to boundary terms on $\mathcal{B}$ that vanish as $c \to \infty$, so favorable manufacturing and transport news ($\varepsilon_{Mt}, \varepsilon_{Tt} > 0$) raises perceived marginal-q when $\rho_M, \rho_T > 0$, the marginal continuation value of capacity is nondecreasing in $z_{M,t+1}$ and $z_{T,t+1}$, and the agricultural term does not offset them. A higher q^{θ}_{jt} then expands every interior county's buildout in (65) when county investments are strategic complements ($\partial_{jk}\Pi^{\theta}_{jt} \geq 0$ for $k \neq j$, making the negated interior block an M-matrix with nonnegative inverse) or when cross-county spillovers are small relative to own curvature; otherwise the sign of $dx^{\theta}_{jt}/d\theta$ is quantified in Section IV.

G Equilibrium and State Transitions

Fix $(\Omega_t, \mathcal{E}_t)$ and a feasible county investment profile $\mathbf{x}_t \in \mathcal{X}_t$. Before private discoveries, pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w)$ clear local and national labor markets, while ex-post goods flows $H_i(\ell, t)$ satisfy spatial conservation with closed borders:

$$n = \sum_{i\in\{A,M\}} e_i L_i + L_T, \tag{67}$$

$$\int_0^1 n\,d\ell = \bar{L}_t, \tag{68}$$

$$\partial_\ell H_i = Y_i - nc_i - e_i\frac{\Psi_i}{p_i} - I_i - \kappa|H_i|, \qquad H_i(0,t) = H_i(1,t) = 0, \quad i \in \{A, M\}. \tag{69}$$

Here $I_A = 0$ and $I_M = I_t(\ell; \mathbf{x}_t)$ is manufactured construction demand (55). Realized producer net profit Π_F, contractual land rent R, origin carrier surplus Π_T, manufacturing wedge rebates Π_ψ, net

land rent R^{net}, and the common portfolio dividend b_t are

$$\Pi_F = \sum_i e_i \Big\{ (p_i - r_i \tau_{j(\ell)}) Y_i - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell) \Big\} - R, \tag{70}$$

$$R = \sum_i e_i R_i, \qquad \Pi_T = \tau_{j(\ell)} Y_T - w L_T, \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i, \tag{71}$$

$$R^{\text{net}} = R - p_M I, \tag{72}$$

$$b_t = \frac{1}{\bar{L}_t} \int_0^1 \{ R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi \} \, d\ell. \tag{73}$$

Free entry implies $\mathbb{E}_t[\Pi_F] = 0$, while ex-post windfalls $\Pi_F(\ell, \omega)$ accrue to households. Carrier tolls and wedge rebates are pecuniary transfers, whereas research goods Ψ_i/p_i, construction I, and iceberg degradation $\kappa|H_i|$ absorb physical output.

DEFINITION 1 (Conditional dated equilibrium): Given $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$, a conditional dated equilibrium consists of pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w, R, I)$ and a measurable post-discovery family $(Y_i, c_i, p_i, \tau_j, H_i, b_t)$ such that: (i) entrants solve (29), land is assigned by (34), and utilization solves (27); (ii) households choose residence by (17) and consumption by (16); (iii) carriers solve (49) and county services clear by (51); and (iv) for almost every realization, spatial no-arbitrage, goods and labor clearing, and (73) hold. At positive, absolutely continuous versions of $p_i(\cdot, \omega)$ (hence Lipschitz, by (45)), absolutely continuous $H_i(\cdot, \omega)$, and continuous $w(\cdot)$ on $[0,1]$ (with $p_A(1,t) = 1$), entrant optimality (29) and pure-site sorting (34) hold at every $\ell \in [0,1]$ (breaking isolated ties to the left sector), differential flow and no-arbitrage relations hold almost everywhere, subsistence is feasible statewise, and expectations in agent objectives are finite.

The pointwise requirement matters because the knowledge transition (38) and the frontier (39) are evaluated site by site, so an allocation fixed only almost everywhere would leave next-period knowledge undetermined on sets the supremum can see. In a competitive continuum economy (Aumann, 1966), individual entrants and carriers evaluate alternative plans at the quoted price and contract-wage functions (w, p, τ) without recomputing local market clearing, while equilibrium announcements jointly clear labor, goods, and county service markets.

PROPOSITION 5: *For every realization with positive, absolutely continuous prices and absolutely continuous flows, in which each term of the ledgers (70)–(72) and of (69) is integrable on $[0,1]$, satisfying local labor clearing (67), spatial goods balance (69), spatial no-arbitrage (45), county service clearing (51), and closed national boundaries, the ledgers (70)–(73) imply the aggregate household budget identity $\int_0^1 \sum_i p_i n c_i \, d\ell = \int_0^1 wn \, d\ell + b_t \bar{L}_t$. Since each household spends $\sum_i p_i c_i = w + b_t$ by (14), this identity is equivalent to $b_t(\bar{L}_t - \int_0^1 n \, d\ell) = 0$: the site-by-site household budgets are consistent with spatial goods balance, and whenever $b_t \neq 0$ national labor clearing (68) is implied by the remaining conditions (Walras' law), so no additional national resource constraint need be imposed.* [Proof →]

Spatial operator and existence of the dated equilibrium.— Building on classical general equilibrium analysis (Arrow and Hahn, 1971) and the nonlinear spatial operator representation of Allen and Arkolakis (2014) and Allen et al. (2024), substituting the closed-form household demand and mobility rules (16)–(18), producer hiring and research rules (30)–(32), land sorting (34), and carrier hiring (50) into market clearing (67)–(73) reduces the conditional dated equilibrium to a fixed point $(w, p, \tau) = \mathcal{T}_t(w, p, \tau)$ on the quote domain $\mathcal{U}_t \equiv \mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ of positive continuous wages $w \in C([0,1]; [\underline{w}_t, \overline{W}_t])$, statewise continuous spot prices $p = (p_A, p_M)$ with $p_A(1, \omega) = 1$ and

$\underline{p}_i \le p_i(\ell,\omega) \le \overline{p}_i$ for fixed $0 < \underline{p}_i < \overline{p}_i < \infty$, and county tolls $\tau = (\tau_1, \dots, \tau_J)$ with $0 \le \tau_j(\omega) \le \overline{\tau}_{jt}$ and positive expected tolls $\mathbb{E}_t[\tau_j] \ge \underline{\tau}_{jt} > 0$:

$$\begin{aligned} \widetilde{w}(\ell) = (\mathcal{T}_t^w(w,p,\tau))(\ell) &= \frac{\bar{u}_t(w,p,\tau) - \mathbb{E}_t\big[(b_t(\omega;w,p,\tau) - \gamma p_A(\ell,\omega))/\mathcal{P}(\ell,\omega)\big]}{\mathbb{E}_t\big[1/\mathcal{P}(\ell,\omega)\big]}, \\ (\widetilde{p}(\cdot,\omega), \widetilde{\tau}(\omega)) &= \mathcal{T}_t^p(w,p,\tau;\omega), \\ \bar{L}_t &= \int_0^1 \Big[\sum_{i\in\{A,M\}} e_i(\ell) L_i^*(\phi_i^*(\widetilde{w}(\ell),\ell);\widetilde{w}(\ell),\ell) + L_T^*(\widetilde{w}(\ell),\ell) \Big] d\ell, \end{aligned} \tag{74}$$

National labor clearing at $\widetilde{w} = \mathcal{T}_t^w(w,p,\tau)$ pins down $\bar{u}_t(w,p,\tau)$, while $\mathcal{T}_t^p(w,p,\tau;\omega)$ maps predetermined capacities and construction demand $I_t(\cdot;\mathbf{x}_t)$ into the spot prices and county tolls supporting the concave Negishi-weighted log-surplus spatial dispatch problem under (45), (51), and (69)–(73).

The dispatch program evaluates its Negishi weights $w + b_t - \gamma p_A$ and research demand $e_i \Psi_i / p_i$ at the input quotes, keeping the inner program concave while enforcing weight-multiplier consistency in the outer fixed point; its image consists of Lipschitz prices, as Definition 1 requires. The domain $\mathcal{U}_t$ is convex, with generic element $u = (w,p,\tau)$, and we equip it with the norm $\|(w,p,\tau)\|_{\mathcal{U}_t} = \max\{\|w\|_\infty,\ \operatorname{ess\,sup}_\omega \|p(\cdot,\omega)\|_\infty / v_p,\ \max_j \operatorname{ess\,sup}_\omega |\tau_j(\omega)|/v_j\}$ for fixed weights $v_p, v_j > 0$. By Proposition 5, the aggregate household budget identity holds at any fixed point, so imposing the dispatch conditions together with the dividend ledger (73) does not overdetermine it.

Regularity conditions on the quote domain.— Existence and uniqueness of the conditional dated equilibrium rest on three conditions on the quote domain $\mathcal{U}_t$ of (74); Part 3 of Proposition 6 gives primitive sufficient conditions for the first two under small frictions.

1. *(Resource feasibility, subsistence slack, and invariance.)* Let $\underline{w}_t$ be a uniform wage at which zero-research labor demand $\int_0^1 [\sum_i e_i L_i^*(0;w,\ell) + L_T^*(w,\ell)]\, d\ell$ exceeds $\bar{L}_t$ and $\overline{W}_t$ a uniform wage at which the research-inclusive envelope $\int_0^1 [\max_i (\alpha_i Z_i \bar{m}_i / ((1+\psi_i)w))^{1/(1-\alpha_i)} + L_T^*(w,\ell)]\, d\ell$ ($\bar{m}_i$ from (A.5)) falls below $\bar{L}_t$, both at every $(p,\tau) \in \mathcal{P}_t \times \mathcal{M}_t$; these define the contract-wage collar $[\underline{w}_t, \overline{W}_t]$. On $\mathcal{U}_t$, for every $\mathbf{x}_t \in \mathcal{X}_t$ and almost every realization: deliverable agricultural capacity exceeds subsistence ($\underline{Y}_{At} > \gamma \bar{L}_t$) and deliverable manufacturing capacity exceeds construction demand ($\underline{Y}_{Mt} > \int_0^1 I_t(\ell;\mathbf{x}_t)\, d\ell$), where $\underline{Y}_{it}$ is the infimum over $\mathcal{U}_t$ and over realizations of sector-i output net of research goods and of the worst-case route loss $e^{-\int_0^1 \kappa\, d\ell}$; nominal subsistence slack $w(\ell) + b_t(\omega;w,p,\tau) - \gamma p_A(\ell,\omega) > 0$ holds at every site; and the quote domain is invariant: the dispatch program at the predetermined capacities $K_j(w, \mathbb{E}_t[\tau_j])$ admits an allocation with $c_A > \gamma$, $c_M > 0$, and $D_j < K_j$ (so multipliers exist), its multipliers are unique (the prices at every occupied site, and the toll of a binding county whenever a site in it trades or partially utilizes capacity) and satisfy $\widetilde{p}_i \in [\underline{p}_i, \overline{p}_i]$, $\widetilde{\tau}_j \in [0, \overline{\tau}_{jt}]$, and $\mathbb{E}_t[\widetilde{\tau}_j] \ge \underline{\tau}_{jt}$, and the mobility-implied wage satisfies $\widetilde{w}(\ell) \in [\underline{w}_t, \overline{W}_t]$ for all ℓ. Finally, $a_i > 1/(1-\alpha_i)$ for $i \in \{A, M\}$.

2. *(Innovation-cost convexity and Ricardian single crossing.)* $\chi_i > \bar{\chi}_{it}$ (Appendix A.6.1, under the maintained independence of the own-site success index from $(V_i, p_i, \tau_{j(\ell)})$ stated there), so that active research (31) has at most one interior maximizer; for every $u \in \mathcal{U}_t$ the activation margin $\pi_i^0(w(\ell),\ell) - \underline{\pi}_i(\ell)$ vanishes on a Lebesgue-null set of sites (for instance, it changes sign at most finitely many times); and the comparative-advantage profile $\frac{1}{1-\alpha_M} \log Z_{Mt}(\ell) - \frac{1}{1-\alpha_A} \log Z_{At}(\ell) - \frac{\alpha_M}{1-\alpha_M} \log(1+\psi_M(\ell))$ rises enough relative to the spatial variation of factor prices, net prices, and research gains admitted by $\mathcal{U}_t$ that, for every $u \in \mathcal{U}_t$, the bid-rent difference satisfies $R_A(\ell) - R_M(\ell) > 0$ on $[0, b_{ct})$ and $R_A(\ell) - R_M(\ell) < 0$ on $(b_{ct}, 1]$ for a unique cutoff $b_{ct} \in (0,1)$.

3. *(Norm contraction.)* $\mathcal{T}_t$ maps $\mathcal{U}_t$ into itself and $\|\mathcal{T}_t(u_1) - \mathcal{T}_t(u_0)\|_{\mathcal{U}_t} \le \bar{\rho}_t \|u_1 - u_0\|_{\mathcal{U}_t}$ for all $u_0, u_1 \in \mathcal{U}_t$, with $\bar{\rho}_t < 1$: direct local labor-demand elasticities $1/(1-\alpha_i)$ dominate indirect spatial trade, toll, and dividend feedbacks. This is a restriction on the equilibrium operator at the actual frictions; it is not implied by small frictions (Appendix A.6.3).

PROPOSITION 6: *Consider the economy of Definition 1 with* $0 < \alpha_A < \alpha_M < 1$, $\alpha_T \in (0,1)$, *bounded positive measurable* (A_{At}, A_{Mt}), $\eta \in (0,1)$, *and* $\gamma > 0$.

1. (Operator continuity, active-research threshold, and pure-site sorting.) *Under Conditions 1 and 2,* $\mathcal{T}_t$ *is a continuous self-map of the convex set* $\mathcal{U}_t$ *for every* $\mathbf{x}_t \in \mathcal{X}_t$, *and every conditional dated equilibrium with quotes in* $\mathcal{U}_t$ *is a fixed point of* $\mathcal{T}_t$. *At the absolutely continuous price versions and continuous wage version of Definition 1, optimal research* $\phi^*_{it}(\ell) \in [0,1)$ *is uniquely determined at every* $\ell \in [0,1]$ *by the threshold rule (32)–(33), and land partitions at the unique cutoff* $b_{ct} \in (0,1)$ *of Condition 2, with agriculture on* $[0, b_{ct}]$ *and manufacturing on* $(b_{ct}, 1]$, *uniquely determining* $A^+_{it}(\ell, \omega)$ *and the almost surely finite frontier* $F_{it}(\ell, \omega)$ *(Online Appendix Lemma S.C.1) everywhere on* $[0,1]$.

2. (Existence and uniqueness on the quote domain.) *Under Conditions 1–3,* $\mathcal{T}_t$ *has a unique fixed point in* $\mathcal{U}_t$, *so for each* $\mathbf{x}_t \in \mathcal{X}_t$ *a conditional dated equilibrium with quotes in* $\mathcal{U}_t$ *exists and is unique; uniqueness among equilibria whose quotes leave* $\mathcal{U}_t$ *is not claimed.*

3. (Small-friction sufficient condition for Conditions 1 and 2.) *Suppose the comparative-advantage profile of Condition 2 is strictly increasing in* ℓ, *research is dormant in the frictionless benchmark economy* ($\chi_i > \bar{f}_i$ *for* $i \in \{A, M\}$, *with* $\bar{f}_i$ *defined in Appendix A.6.3), and that in this benchmark—goods move without loss, require no county services, and construction demand is absent—the bid rents satisfy* $R_A(0) > R_M(0)$ *and* $R_A(1) < R_M(1)$ *and the resource inequalities of Condition 1 hold strictly. Then there is* $\bar{\varepsilon}_t > 0$ *such that, whenever* $\kappa_0 e^{-\underline{T}}$, *the toll ceilings* $\overline{\tau}_{jt}$, *and* m_I *lie below* $\bar{\varepsilon}_t$, *a quote domain* $\mathcal{U}_t$ *with a wage collar* $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ *independent of the accumulated stock can be chosen on which research is dormant and the resource, subsistence, wage- and price-invariance, and single-crossing clauses of Conditions 1 and 2 hold for every admissible county infrastructure path* $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$ *(Appendix A.6.3). The toll-invariance clause of Condition 1 and Condition 3 are not implied by small frictions, and the argument establishes the existence of* $\bar{\varepsilon}_t$ *without bounding it.* [Proof →]

DEFINITION 2 (Dynamic equilibrium): Given Ω_0, diagnosticity $\theta \ge 0$, $\{\mathcal{E}_t\}_{t \ge 0}$, and the physical shock laws, a dynamic equilibrium is an adapted process of states, investments, conditional dated equilibria, and county values such that: (i) each dated allocation satisfies Definition 1; (ii) county policies satisfy (58) under values (57) and perceived forecast-reference update (13), taking the aggregate knowledge trajectory (A_{At}, A_{Mt}) as given; and (iii) realized states follow (10), (38)–(40), and (54) with observed reference $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$. Attention is restricted to Markov strategies and county values measurable in Ω_t.

Substituting the unique conditional dated equilibrium rent and manufacturing price schedules from Proposition 6 into (56) yields county j's reduced stage payoff

$$R_{jt}(\mathbf{x}_t; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}_t) - m_I \left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2 \right) D^I_{jt}(\mathbf{x}_t),$$

with gross county land rent $\widetilde{R}_{jt}$ and construction-price index D^I_{jt} from (61), county mean route stock $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt} x_{jt}$ ($\overline{T}_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} T_t(\ell)\, d\ell$ and $d_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} \iota_{jt}(\ell)\, d\ell$ by (54)) and

knowledge transition (38)–(40). After 1881, household mass remains at $\bar{L}_{1881+h} = \bar{L}_{1881}$ while the 1881 project schedule repeats with physical package growth $g = 1.02$. Write $\bar{Z}_T \equiv A_T/\mathbb{E}_\pi[e^{z_T}]$ and $\varepsilon = e^{-z_{Tt}/\alpha_T}/(\bar{Z}_T)^{1/\alpha_T} = Z_T(t)^{-1/\alpha_T}$ for the carrier-friction scale: by (50) and (51), the expected toll at which carrier capacity clears county service demand D_j at a realization where capacity binds is, at a uniform wage w, $\mathbb{E}_t[\tau_j] = (w/\alpha_T)\,\varepsilon\,[D_j/(|\mathcal{C}_j|\overline{T}_{jt})]^{(1-\alpha_T)/\alpha_T}$, so ε scales the toll channel through which route stocks enter county rents, with elasticity $\partial_{\overline{T}_j} \log \mathbb{E}_t[\tau_j] = -(1-\alpha_T)/(\alpha_T \overline{T}_{jt})$ at given service demand. Write $\bar{\varepsilon} = \sup_{\mathbf{z}\in\mathcal{B}} \varepsilon$ and $D_{\min} = \inf_t \min_{j:\bar{x}_{jt}>0} D^I_{jt} > 0$ for the lower bound on the construction-price index over county-years with an open project.

Theorem A.1 (Appendix A.7.1) is a contraction argument in a norm $\|\cdot\|_{W,1}$ that controls county values together with their derivatives in county route stocks, the derivative being what county policies respond to through marginal-q (62). Its constants are the one-step drift κ_W of a state weight W under the forecasting kernel, the Lipschitz moduli L_a and L'_a of the stage Nash profile and of its stock derivatives in the continuation value, the bound A_1 on those stock derivatives, and the moduli B and B_2 of the stage payoff and of its rival-action gradient; it establishes uniqueness within a closed class $\mathcal{C}$ of Markov county values.

PROPOSITION 7: *Suppose the conditions of Proposition 6 hold along feasible state histories.*

1. (Conditional Markov perfect equilibrium.) *Suppose that at every date and state the stage pseudogradient* $\mathbf{g}^{(n),\theta}_t = (\partial_{x_j}\Pi^{(n),\theta}_{jt})_j$ *is strongly monotone on* $\mathcal{X}_t$ *with a modulus* $\mu > 0$ *common to all stages* n*, and that the hypotheses of Theorem A.1 hold on a class* $\mathcal{C} \ni \mathbf{0}$ *with contraction constant* $\rho = \beta\kappa_W + B(L_a + L'_a) + A_1B_2 < 1$*. Then for every finite horizon* $H \geq 1$ *backward induction yields a unique pure-strategy Markov perfect equilibrium* $(\mathbf{x}^{(H),\theta}_t, \mathbf{V}^{(H),\theta}_t)_{t=0}^{H}$*, which converges geometrically in* $\|\cdot\|_{W,1}$ *as* $H \to \infty$ *to the unique infinite-horizon Markov perfect equilibrium* $(\mathbf{x}^\theta_t, \mathbf{V}^\theta_t)_{t\geq 0}$ *within* $\mathcal{C}$*.*

2. (Small-friction corridor.) *Under the terminal schedule, restrict attention to counties with an open 1881 project (*$o_{j,1881} = 1$*; the others set* $x_{jt} = 0$*), and consider the small-friction regime of Proposition 6(3) in the no-research neighborhood of the benchmark (*$\phi^*_{it} \equiv 0$*, knowledge evolving by diffusion alone), with* $m_I = c\,\bar{\varepsilon}$ *and* $\kappa_0 e^{-\underline{T}} \leq c'\bar{\varepsilon}$ *for fixed* $c, c' > 0$*, the dated-equilibrium map* $\mathbf{x} \mapsto (\widetilde{R}_{jt}, D^I_{jt})$ *twice continuously differentiable, and* χ_I *in the two-sided range* $\underline{\chi} \leq \chi_I \leq \bar{m}/m_I$ *of Appendix A.7, Steps 1 and 3. Then there are* $\bar{\varepsilon}_0 > 0$ *and constants* $c_0, c_1, C_1, C_2, c_R, C_u > 0$ *depending only on primitives such that for* $\bar{\varepsilon} < \bar{\varepsilon}_0$*, at every date and at every finite-horizon stage whose continuation value is concave in own county stock with stock curvature bounded as in Step 1: (a) the stage pseudogradient is strongly monotone on* $\mathcal{X}_t$ *with modulus* $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$*, so the stage Nash profile is unique (Proposition 4); and (b) on each active interior set the profile responds to inherited county stocks only through the toll channel,*

$$\left\| \frac{\partial \mathbf{x}^{(H),\theta}_t}{\partial \overline{T}_{it}} \right\| \leq \frac{C_u \varepsilon}{\mu\, \overline{T}_{it}^{1/\alpha_T}}, \tag{75}$$

with C_u *collecting the toll sensitivity of the construction-price index and of marginal rents together with the continuation curvature constants (Appendix A.7, Step 1). Appendix A.7, Steps 2–3, argues by backward induction that the continuation bounds propagate across stages and that, on the lower stock set* $\prod_j [L_j g^{\alpha_T(t-1881)}, \infty)$ *left invariant by the clipped rule under a primitive inequality at its lower edge, the stage constants are stationary in the rescaled stock, so that item 1 applies and the finite-horizon equilibria converge to the unique infinite-horizon Markov perfect equilibrium in* $\mathcal{C}$*, corner investment included; the induction and the uniformity in the date of the constants of Theorem A.1 are argued rather than computed there.* [Proof →]

IV Quantitative Analysis

This section estimates the model on historical evidence from England and Wales over 1710–1881 and uses the estimated model to quantify the contribution of diagnostic expectations to transport accumulation, structural transformation, real wages, prices, and frontier knowledge during the industrial takeoff (Broadberry et al., 2014; Crafts and Harley, 1992). Spatial allocations are computed on $N = 500$ locations partitioned into $J = 10$ county jurisdictions, the corridor on which Proposition 7(2) is verified, solving at each annual date for the unique conditional dated equilibrium (Proposition 6) and the unique Markov perfect county investment profile (Proposition 7), so that each primitive shock history generates a unique path (Corollary A.1).

A From Historical Districts to Model Space

Because England and Wales is oriented north to south, latitude provides a natural one-dimensional ordering of regional economic activity on $\ell \in [0, 1]$ (Bogart et al., 2022; Trew, 2020). For each of twenty-five equal-latitude bands, historical employment and route quantities are summed across the band's east–west extent and divided by land width, with $\ell = 0$ at the southern coast and $\ell = 1$ at the northern border.

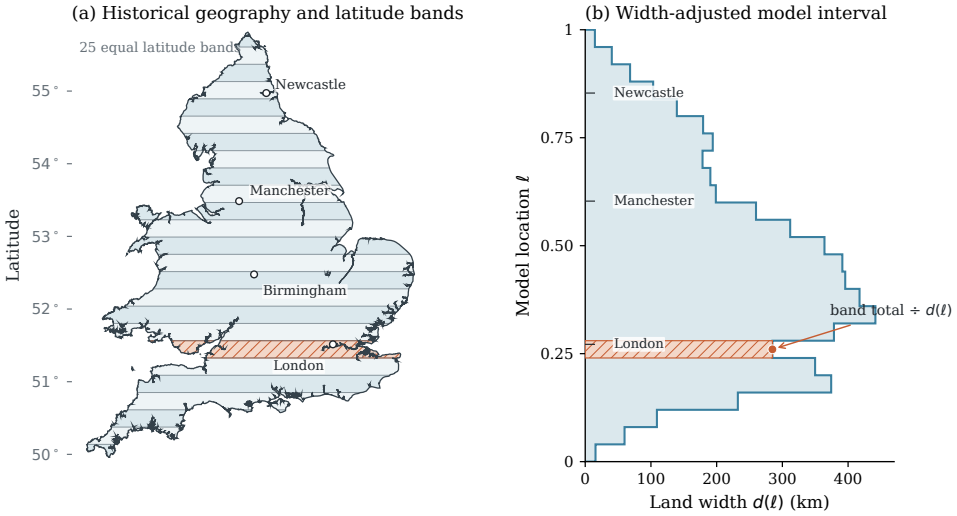


Figure 6: From Historical Geography to the Model Interval

Notes: Panel (a) partitions England and Wales into 25 equal-latitude bands. Panel (b) divides band totals by east–west land width $d(\ell)$ to form densities on $\ell \in [0, 1]$.

This projection captures the relative positions of metropolitan London in the South, the West Midlands, and the northern coalfield belt, together with the reallocation of labor across them, while abstracting from east–west variation within a latitude band (Online Appendix S.B).

B Data and External Calibration

Aggregate household mass $\bar{L}_t$ follows the annual England and Wales population series of Thomas and Dimsdale (2017) over 1710–1881, normalized to $\bar{L}_{1710} = 100$ and held constant after 1881 ($\bar{L}_{1881+h} = \bar{L}_{1881}$). Installed transport capacity is reconstructed each year across the twenty-five latitude bands from turnpike, waterway, and railway GIS records. Letting M^r_{hy}, M^w_{hy}, and M^g_{hy}

denote permanent route miles of turnpike road (Rosevear et al., 2026), navigable waterway (Cambridge Group for the History of Population and Social Structure, 2016a), and railway (Cambridge Group for the History of Population and Social Structure, 2016b; Coll Ardanuy et al., 2021; Mitchell, 1988) in band h and year y, and d_h the band's mean east–west land width, the measured transport stock is

$$\mathcal{I}^T_{hy} = \frac{(M^r_{hy} + 10M^w_{hy} + 50M^g_{hy})/d_h}{25^{-1}\sum_{k=1}^{25}(M^r_{k,1710} + 10M^w_{k,1710} + 50M^g_{k,1710})/d_k}, \tag{76}$$

where the service-capacity weights $(1, 10, 50)$ follow Trew (2020) and the 1710 cross-band mean is normalized to unity. The 1710 cross-section initializes $T_0(\ell)$, while subsequent stocks emerge from county investment (53)–(54).

Following Hornbeck and Rotemberg (2024), the manufacturing operating wedge $\psi_M(\ell) = \bar{\psi}_M \omega_M(\ell)$ combines a scale parameter $\bar{\psi}_M$ with a predetermined spatial exposure profile $\omega_M(\ell)$ from geocoded fire-insurance registers (Study 1838), normalized to unit 1710 manufacturing-labor-weighted mean ($\frac{\int_0^1 \omega_M(\ell)\omega_{M0}(\ell)\,d\ell}{\int_0^1 \omega_{M0}(\ell)\,d\ell} = 1$ with $\omega_{M0}(\ell) \equiv e_{M,1710}(\ell)L_{M,1710}(\ell)$). By the conditional hiring rule (30), an active manufacturing site with payroll $w(\ell)L_M(\ell)$ and expected gross operating surplus $\mathbb{E}_t[F_{M\ell}] = \mathbb{E}_t\big[[p_M(\ell) - r_M\tau_{j(\ell)}]_+ Y_M(\ell)\big] - w(\ell)L_M(\ell)$ satisfies

$$1 + \psi_M(\ell) = \alpha_M\left(1 + \frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)}\right), \qquad \text{or} \qquad \frac{\alpha_M}{1-\alpha_M}\frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)} = 1 + \frac{\psi_M(\ell)}{1-\alpha_M}, \tag{77}$$

which calibrates $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from the Clark (2002) nonagricultural labor share, with digitized firm operating accounts (Clough records) providing an external check on $\mathbb{E}_t[F_{M\ell}]/[w(\ell)L_M(\ell)]$.

C Estimation and Identification

We estimate the model by the Simulated Method of Moments (Bordalo et al., 2026; Hansen, 1982) across $p = 26$ free structural parameters under diagnostic expectations ($\theta \geq 0$, the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters, and the 9 sectoral shock parameters in Table 5). We calibrate the manufacturing wedge scale $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from (77), concentrate out the eight shape parameters of the five-knot 1710 productivity profiles $A_A(\ell, 1710)$ and $A_M(\ell, 1710)$ ($\int_0^1 A_i(\ell, 1710)\,d\ell = 1$) on the 1710 occupational cross-section, and re-estimate the model under the rational expectations restriction ($\theta = 0$, $p = 25$ free parameters). Sharing the fixed parameters in Table 3, both specifications stack the $M = 30$ historical moments of Panels A–E of Table 6 into $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ and weight each residual by its inverse empirical variance $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \ldots, 1/\widehat{\text{SE}}_{30}^2)$ across B common primitive shock histories $\{u_b\}_{b=1}^B$:

$$Q(\Theta) = \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big]' W \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big], \qquad \overline{\mathbf{m}}(\Theta) = \frac{1}{B}\sum_{b=1}^{B}\mathbf{m}(\Theta; u_b). \tag{78}$$

The criterion $Q(\Theta) = \sum_{k=1}^{30}\big[(\overline{m}_k(\Theta) - m_k^{\text{data}})/\widehat{\text{SE}}_k\big]^2$ thus measures fit in studentized units (Online Appendix S.D and Table S.D.2 report each moment formula, standard error $\widehat{\text{SE}}_k$, and weight W_{kk}).

Table 4 summarizes how the moments discipline each parameter block. By Proposition 2 and (74), the dated allocation given $(\Omega_t, \mathbf{x}_t)$ is the fixed point of $\mathcal{T}_t$, which depends on (m_I, χ_I) only through construction demand $I_t(\cdot; \mathbf{x}_t)$ and on θ not at all; conditional on the network, within-period margins discipline the static parameters, while network dynamics discipline beliefs and construction costs (the unconditional moment map is not block triangular, since θ moves every

Table 3: Externally Fixed Parameters and Normalizations

Parameter	Value	Description
β	1/1.05	Annual discount factor (real interest rate of 5%)
$\alpha_A,\ \alpha_M$	0.390, 0.710	Production labor elasticities (agriculture, manufacturing)
r_M	1.000	Manufacturing distribution requirement (normalization)
c	3	Truncation bound for aggregate productivity innovations (in s.d.)
d_0	0.100	Spatial Ornstein–Uhlenbeck correlation length for discovery draws
g_m	log(1.02)	Post-1881 growth rate of physical project increments
$\bar{L}_{1710}$	100	Initial aggregate labor supply (normalization)
$N,\ J$	500, 10	Number of spatial grid cells and county jurisdictions

Notes: Numeraire $p_A(1,t) = 1$; initial productivity profiles satisfy $\int_0^1 A_{i,1710}(\ell)\,d\ell = 1$ $(i \in \{A, M\})$; and the 1710 transport stock has cross-band mean one. Production labor elasticities (α_A, α_M) follow sectoral value-added shares (Herrendorf et al., 2015) (Proposition S.E.2), and demographic mass $\bar{L}_t$ follows (9).

moment through the investment path; Online Appendix S.E.4). Sectoral labor shares fix (α_A, α_M), and the nonagricultural labor share, given fire-insurance exposure, calibrates $\bar{\psi}_M$ (77). The 1710 occupational shares, relative prices, and dated spatial employment covariances discipline (η, γ), $(\alpha_T, \kappa_0, r_A)$, and $A_i(\ell, 1710)$; because manufacturing is confined to $(b_{ct}, 1]$ by Proposition 6, the Panel C covariances trace b_{ct} and the comparative-advantage slope in (34). Aggregate growth and structural change identify $(a_i, f_i, \chi_i, \omega_E, \delta, \lambda_0)$, while HP-filtered sectoral output variances, autocovariances, and cross-covariances pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$.

Transport-stock growth rates (Panel B of Table 6) and investment-cycle regressions (Panel E) separate θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Under $\theta = 0$, lowering χ_I or raising $\boldsymbol{\rho}$ raises both investment–output comovement and the first-order autocorrelation ρ_1 of cyclical investment, whereas $\theta > 0$ amplifies boom-time marginal q, raises 1830–1850 transport growth, and lowers ρ_1 as transitory news subsides. The two θ signs follow, to first order in θ and holding the continuation value fixed, from the clipped investment rule (Online Appendix S.E.4); all four sign restrictions are heuristic and are to be checked numerically once the model is estimated.[11]

Following Bordalo et al. (2026), Table 5 reports the parameter estimates and standard errors under diagnostic (DE) and rational (RE, $\theta = 0$) expectations, and Table 6 compares the empirical and simulated moments across all five blocks. Diagnosticity is estimated at $\widehat{\theta} = 0.739$ (SE = 0.090), rejecting $\theta = 0$ ($t = 8.21$, $p < 0.0001$). This estimate lies within the benchmark range across financial and macroeconomic settings (Bordalo et al., 2026): just below the $\theta \approx 0.8$–0.9 estimated from earnings expectations and credit spreads (Bordalo et al., 2018, 2019), above the $\theta \approx 0.5$–0.6 from macroeconomic surveys (Bordalo et al., 2020), and aligned with the $\theta \approx 0.74$–1.08 in structural models of investment and credit cycles (Bianchi et al., 2024; Bordalo et al., 2026; Maxted, 2023). When the remaining $p = 25$ parameters are re-estimated under $\theta = 0$, the restricted RE estimator lowers adjustment convexity ($\widehat{\chi}_I = 1.579$ versus 1.920) and construction scale ($\widehat{m}_I = 0.020$ versus 0.023), raises manufacturing persistence ($\widehat{\rho}_M = 0.561$ versus 0.533) and volatility ($\widehat{\sigma}_M = 0.042$ versus 0.039), and adjusts spatial catch-up and transport costs ($\widehat{\lambda}_0 = 0.175$ versus 0.160, $\widehat{\kappa}_0 = 0.073$ versus 0.080, $\widehat{\delta} = 1.245$ versus 1.350) to compensate for the absence of extrapolation.

Because those parameter shifts distort other margins, the fit improvement under DE spans Table 6. Both models match long-run output and employment trends (Panel A) and dated spatial sorting covariances (Panel C). The restricted RE model, however, underpredicts the linear and

[11] Online Appendix S.E states the local rank conditions (Andrews et al., 2017; Rothenberg, 1971) and sign restrictions; Replication Appendix R.A details the sequence-space Jacobian evaluation of (78) (Auclert et al., 2021).

Table 4: Parameter Blocks and Identifying Moments

Parameter block	Parameters	Primary identifying moments
Preferences	$\eta,\ \gamma$	Linear trends in sectoral employment shares and log relative price $\ln(p_M/p_A)$
Operating wedge	$\bar{\psi}_M$	Nonagricultural labor share and fire-insurance exposure (77)
Innovation	$a_i,\ f_i,\ \chi_i,\ \omega_E$	Linear trends in log real GDP and $\ln(Y_M/Y_A)$; manufacturing share–energy covariance $C_{e,y}$
Spatial diffusion	$\delta,\ \lambda_0$	Manufacturing share–latitude covariance $C_{l,y}$ and spatial autocovariance $C_{N,y}$
Transport	$\alpha_T,\ \kappa_0,\ r_A$	Tertiary employment share trend, cross-band variance $V^{\text{occ}}_{T,y}$, and $\ln(p_M/p_A)$ trend
Construction costs	$m_I,\ \chi_I$	Linear and quadratic trends of log transport stock $\ln \overline{T}_t$; cross-band variance $V_{T,t}$
Diagnostic beliefs	θ	Subperiod transport growth (1830–1850), post-1850 slope change, investment–output slopes (b_0, b_1), autocorrelation (ρ_1), and county lag slopes (g_1, g_2)
Initial productivity	$A_i(\ell, 1710)$	1710 occupational shares and cross-sectional employment covariances
Productivity process	$\boldsymbol{\rho},\ \boldsymbol{\Sigma}$	Variances, first-order autocovariances, and cross-covariances of HP-filtered log sectoral output

Notes: Primary identifying moments for each parameter block (Table 6 and Online Appendix S.D); all blocks are estimated jointly via (78) under DE ($\theta \geq 0$) and RE ($\theta = 0$). Online Appendix S.E states the local rank and sign restrictions.

quadratic trends of the log transport stock (1.758 versus 1.948 percent per year and 2.144 versus 2.923 in Panel B), the 1830–1850 railway boom (4.801 versus 7.410 percent per year under DE and 7.470 in the data), and the post-1850 slowdown (−2.461 versus −5.060 and −5.101).

Raising $\widehat{\rho}_M$ and $\widehat{\sigma}_M$ under RE also distorts the cyclical moments in Panels D and E. It overstates manufacturing and transport cycle variances ($v_M = 12.12$ versus 9.96 under DE and 9.64 in the data; $v_T = 7.37$ versus 5.86 and 5.66 in Panel D), shifts investment comovement from current to lagged output ($b_0 = 2.063$ and $b_1 = 3.142$ under RE, versus 3.091 and 1.662 under DE and 2.954 and 1.585 in the data), and mutes both negative investment autocorrelation ($\rho_1 = -0.183$ versus −0.344 under DE and −0.325 in the data) and county reversals after population growth ($g_1 = -0.082$ and $g_2 = -0.144$ under RE, versus −0.190 and −0.484 under DE and −0.162 and −0.496 in the data; Panel E).

Online Appendix Table S.E.1 confirms that this fit gain does not stem from an extra parameter or moment weighting. The Rivers and Vuong (2002) equal-parameter test ($p = 25$ free parameters in every column) favors $\theta = 0.9$ ($Q = 5.54$) over $\theta = 0$ ($Q = 117.86$) with $z = 4.35$ ($p < 0.0001$). The Andrews et al. (2017) sensitivity measures show that Panels E and B supply 85.7 percent (49.8 and 35.9 percent) of the identifying variation for $\widehat{\theta}$, while leave-one-block-out estimations yield $\widehat{\theta} \in [0.666, 0.747]$ and equal-block studentized weights yield $\widehat{\theta}_{\text{eq}} = 0.748$ (SE = 0.104).

D Aggregate Takeoff and Spatial Reallocation

Figures 7 and 8 plot the historical transition alongside the mean simulated trajectories and 5th–95th percentile bands across 1,000 Monte Carlo draws under DE and RE. Total real GDP rose 14.33-fold between 1710 and 1881, averaging 1.56 percent per year (Figure 7, Panel a), as annual growth accelerated from 1.02 percent in 1710–1760 and 1.16 percent in 1760–1800 to 1.60 percent in 1800–1830 and 2.37 percent in 1830–1881. The industrial-to-agricultural output ratio Y_M/Y_A (Panel c)

Table 5: SMM Parameter Estimates: Diagnostic versus Rational Expectations

Parameter	Description	DE	RE ($\theta = 0$)
Panel A. Preferences, production, and trade costs			
η	Stone–Geary food expenditure share	0.520 (0.028)	0.540 (0.031)
γ	Subsistence food requirement	0.042 (0.003)	0.039 (0.003)
α_A	Agricultural labor elasticity	0.390 [fixed]	0.390 [fixed]
α_M	Manufacturing labor elasticity	0.710 [fixed]	0.710 [fixed]
$\bar{\psi}_M$	Manufacturing labor wedge scale	0.251 (0.032) [calibrated]	0.251 (0.032) [calibrated]
α_T	Transport labor elasticity	0.450 (0.029)	0.473 (0.033)
κ_0	Iceberg transport cost scale	0.080 (0.007)	0.073 (0.007)
r_A	Agricultural distribution requirement ($r_M = 1$)	0.200 (0.015)	0.216 (0.018)
Panel B. Innovation and spatial diffusion			
a_A	Agricultural innovation Pareto shape	2.200 (0.150)	2.122 (0.156)
a_M	Manufacturing innovation Pareto shape	1.800 (0.112)	1.887 (0.126)
f_A	Agricultural fixed innovation cost	0.040 (0.004)	0.043 (0.005)
f_M	Manufacturing fixed innovation cost	0.030 (0.003)	0.028 (0.003)
χ_A	Agricultural variable innovation cost	0.080 (0.008)	0.085 (0.009)
χ_M	Manufacturing variable innovation cost	0.060 (0.006)	0.055 (0.006)
ω_E	Coalfield innovation cost loading	0.0040 (0.0006)	0.0045 (0.0007)
λ_0	Baseline technological catch-up rate	0.160 (0.014)	0.175 (0.016)
δ	Spatial knowledge decay rate	1.350 (0.097)	1.245 (0.097)
Panel C. Infrastructure investment and expectations			
θ	Diagnostic expectations parameter	0.739 (0.090)	0 [fixed]
m_I	Infrastructure construction cost scale	0.023 (0.006)	0.020 (0.005)
χ_I	Infrastructure adjustment cost convexity	1.920 (0.233)	1.579 (0.213)
Panel D. Aggregate productivity process			
ρ_A	AR(1) persistence: Agriculture	0.168 (0.042)	0.174 (0.043)
ρ_M	AR(1) persistence: Manufacturing	0.533 (0.124)	0.561 (0.108)
ρ_T	AR(1) persistence: Transport	0.303 (0.076)	0.318 (0.079)
σ_A	Innovation s.d.: Agriculture	0.067 (0.004)	0.070 (0.004)
σ_M	Innovation s.d.: Manufacturing	0.039 (0.002)	0.042 (0.002)
σ_T	Innovation s.d.: Transport	0.023 (0.002)	0.025 (0.002)
ϱ_{AM}	Innovation correlation: (A, M)	−0.058 (0.015)	−0.082 (0.020)
ϱ_{AT}	Innovation correlation: (A, T)	0.094 (0.024)	0.077 (0.019)
$\varrho_{MT\|A}$	Partial innovation correlation: $(M, T \mid A)$	0.337 (0.084)	0.373 (0.093)

Notes: SMM estimates minimizing (78) across $M = 30$ moments (Table 6) under diagnostic (DE, $\theta \geq 0$, $p = 26$) and rational (RE, $\theta = 0$, $p = 25$) expectations across $B = 1{,}000$ common shock histories, with misspecification-robust sandwich standard errors (S.D.11) in parentheses. The wedge scale $\bar{\psi}_M = 0.251$ (SE = 0.032) is calibrated via (77), the eight 1710 productivity spline coefficients are concentrated out on the 1710 occupational cross-section, and fixed parameters are in Table 3.

follows the same two-phase timing, drifting to 0.60 times its 1710 level in 1781 as population growth raised food demand before accelerating at 2.61 percent per year after 1813 to 5.77 times its 1710 level by 1881.[12]

Because no moment of the real wage series enters the SMM objective in Table 6, Panel (b) of Figure 7 provides an untargeted check on the model's price and wage predictions. In the data, real wages $w/\mathcal{P}$ rose little in the early eighteenth century, plateaued at 1.35–1.55 times their 1710 level during the late eighteenth and early nineteenth centuries—the "Engels' pause" of Allen (2009)—and then accelerated after 1820 to 3.28 times their 1710 level by 1881 as transport integration and industrial productivity gains lowered delivered prices. The estimated DE model reproduces this two-phase acceleration in aggregate output, Y_M/Y_A, and the untargeted real wage path $w/\mathcal{P}$

[12]The relative price p_M/p_A (Online Appendix Figure S.D.1, Panel c) moves in the opposite direction, remaining above its 1710 level through much of the eighteenth century (peaking at 1.47 in 1733) before falling below unity after 1799 to 0.71 in 1800, 0.59 in 1830, 0.39 in 1855, and 0.49 in 1870.

Table 6: Empirical Moments and Model Fit

Empirical moment	Window	Data	DE	RE
Panel A. Aggregate growth and structural transformation				
Log real GDP trend (%/yr)	1710–1881	1.473 (0.210)	1.425	1.364
Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	1710–1881	1.085 (0.572)	0.970	0.799
Log relative price $\ln(p_M/p_A)$ trend (%/yr)	1710–1870	−0.757 (0.616)	−0.632	−0.496
Agricultural employment share trend (pp/yr)	1710–1881	−0.146 (0.023)	−0.139	−0.116
Tertiary employment share trend (pp/yr)	1710–1881	0.060 (0.007)	0.058	0.055
Panel B. Transport network growth and spatial dispersion				
Log transport stock: linear trend (%/yr)	1710–1881	1.948 (0.103)	1.943	1.758
Log transport stock: quadratic trend (×100)	1710–1881	2.923 (0.217)	2.952	2.144
Log transport stock: 1830–1850 growth (%/yr)	1830–1850	7.470 (0.759)	7.410	4.801
Log transport stock: post-1850 slope change (%/yr)	1830–1881	−5.101 (0.881)	−5.060	−2.461
Cross-band variance of log transport stock	1710–1881	0.469 (0.148)	0.437	0.410
Panel C. Cross-sectional employment covariances				
Manufacturing share–latitude covariance	1710–1881	0.077 (0.026)	0.070	0.065
Manufacturing share–energy covariance	1710–1881	−0.070 (0.024)	−0.064	−0.059
Spatial autocovariance of manufacturing share	1710–1881	0.030 (0.006)	0.028	0.026
Cross-band variance of tertiary share	1710–1881	0.0029 (0.0011)	0.0027	0.0025
Panel D. HP-filtered log sectoral output moments ($\times 10^4$)				
Variance: Agriculture (A)	1710–1881	34.79 (4.60)	35.81	39.80
Variance: Manufacturing (M)	1710–1881	9.64 (1.40)	9.96	12.12
Variance: Transport (T)	1710–1881	5.66 (0.93)	5.86	7.37
First-order autocovariance: Agriculture (A)	1710–1881	−7.27 (2.41)	−6.86	−6.75
First-order autocovariance: Manufacturing (M)	1710–1881	−0.49 (0.68)	−0.35	−0.08
First-order autocovariance: Transport (T)	1710–1881	−0.63 (0.39)	−0.55	−0.39
Cross-sector covariance: (A, M)	1710–1881	−2.59 (1.43)	−2.89	−3.89
Cross-sector covariance: (A, T)	1710–1881	−0.55 (0.83)	−0.71	−1.20
Cross-sector covariance: (M, T)	1710–1881	4.74 (0.78)	4.95	6.34
Panel E. Infrastructure investment dynamics				
Standard deviation of cyclical investment, $\sigma(c_t^I)$	1710–1880	0.431 (0.043)	0.427	0.345
Investment on current output cycle (b_0)	1711–1880	2.954 (1.431)	3.091	2.063
Investment on lagged output cycle (b_1)	1711–1880	1.585 (1.257)	1.662	3.142
First-order autocorrelation of investment (ρ_1)	1711–1880	−0.325 (0.074)	−0.344	−0.183
Lead output cycle on investment (b_{lead})	1711–1880	−0.0051 (0.0039)	−0.0049	−0.0035
County investment on population growth, lag 1 (g_1)	1791–1851	−0.162 (0.050)	−0.190	−0.082
County investment on population growth, lag 2 (g_2)	1791–1851	−0.496 (0.050)	−0.484	−0.144

Notes: Empirical moments (with source-bootstrap or Newey–West HAC standard errors in parentheses; Online Appendix S.D.8 and Table S.D.2) and simulated means across $B = 1{,}000$ common shock histories at the DE and RE estimates in Table 5. Sources: Broadberry et al. (2022) and Feinstein (1972) (Panels A, D), Shaw-Taylor et al. (2019) and Keibek (2017) (Panels A, C), and the measured transport stock (76) (Panels B, E).

(1.41 in 1760, 1.75 in 1820, and 3.11 in 1881), whereas the restricted RE trajectory matches the eighteenth-century plateau (1.35 in 1760 and 1.47 in 1820) but stalls at 1.72 by 1881 alongside a muted rise in Y_M/Y_A.

Transport capacity expanded in waves over the sample, rising 27.70-fold between 1710 and 1881 (76) (Figure 8, Panel c). Before 1830 it grew at 0.99 percent per year to 3.28 times its 1710 level, with turnpike-era bursts of 1.79 percent per year in 1750–1770 and 1.68 percent in the 1790s. Railway construction raised annual log growth to 5.90 percent in the 1830s and 8.43 percent in the 1840s (7.47 percent per year over 1830–1850), lifting the stock to 5.91 by 1840 and 13.73 by 1850, before annual growth slowed by 5.10 percentage points to 2.37 percent over 1850–1881 (24.36 in 1870). Disciplined by Panel B of Table 6, the DE path tracks eighteenth-century accumulation, the 1830–1850 surge, and post-1850 moderation, whereas the RE path is too smooth and misses the mid-century acceleration.

The 1881 spatial transport profile (Panel d) is multi-peaked. The West Riding and Lancashire

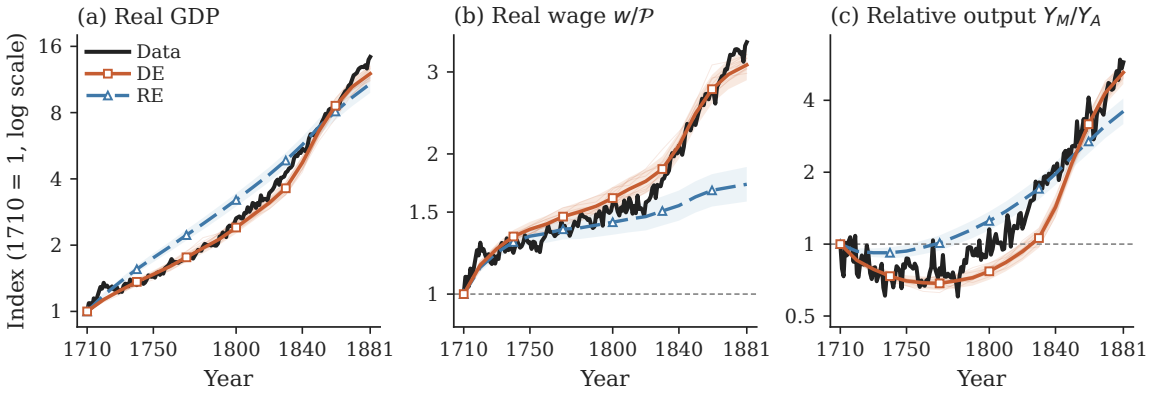


Figure 7: Aggregate Transition, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and restricted rational (RE, $\theta = 0$) expectations (1710 = 1, log scale). Panels (a) and (c) plot real GDP and relative industrial output Y_M/Y_A (Broadberry et al., 2022; Feinstein, 1972); Panel (b) plots the untargeted aggregate real wage $w/\mathcal{P}$ (Broadberry et al., 2022; Clark, 2002).

band carried 53.8 times the 1710 national mean, followed by London (46.0) and the Midlands–Northern industrial bands (35.1–40.1), while Cornwall (3.5) and Northumberland (8.2) remained least served. Cross-band log-stock variance $V_{T,y}$ fell from 0.713 in 1710 to 0.295 in 1881 in the data (0.658 to 0.240 under DE; Figure S.D.1, Panel f), with a sample mean of $\overline{V}_T = 0.469$ (0.437 under DE and 0.410 under RE; Panel B). Under DE, counties build more during the canal and railway booms across the Midlands and Northern corridor (bands 12–22) than under RE, matching the 1881 cross-sectional peaks.

Panels (a)–(b) of Figure 8 compare 1817–1881 changes in manufacturing and tertiary employment shares across the 25 latitude bands with DE and RE predictions (Online Appendix S.B). As the national manufacturing share rose from 49.9 to 62.1 percent, industrial employment diffused across space: the manufacturing share rose in 20 of 25 bands (correlation −0.81 with the 1817 share), gaining 20–26 percentage points in peripheral bands (bands 6, 8, and 20–22) while falling 5–7 points in the Lancashire–Yorkshire core (bands 16–18, which stayed above 79 percent manufacturing in 1881). Tertiary employment rose in 23 of 25 bands as the national transport and distribution share nearly doubled from 9.5 to 18.5 percent, led by the Southern, London, and Lancashire–West Riding corridors. Targeting only aggregate occupational series and dated spatial covariances in Panels A and C, the DE model matches manufacturing diffusion and tertiary expansion better than RE.[13]

E Counterfactual Analysis: Diagnostic versus Rational Expectations

Re-estimating the model under $\theta = 0$ alters technology and construction-cost parameters alongside beliefs. To isolate diagnostic expectations, the belief-neutralization counterfactual (R | DE) holds all other parameters at their diagnostic estimates, together with the 1710 state, statutory project opportunities, and primitive shock draws $\{u_b\}_{b=1}^{B}$, sets $\theta = 0$, and recomputes county investment policies $\mathbf{x}_t^0$ and the general equilibrium path from 1710 to 1881.

By Proposition 2, household and firm decisions and prices at each date are invariant to θ given county investment, so every difference between the DE and R | DE paths originates in county construction and propagates through the installed network. Construction absorbs manufactures

[13] Online Appendix S.D (Figure S.D.1) plots the complete historical and simulated series for every moment block.

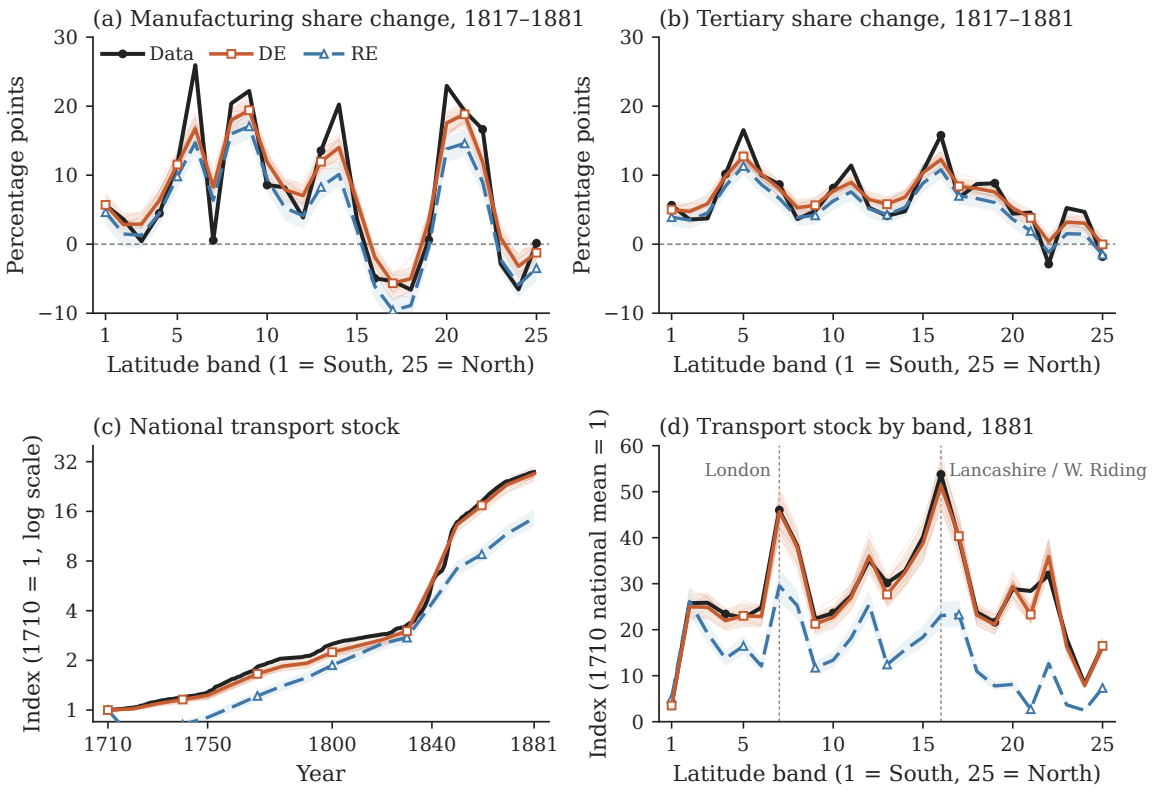


Figure 8: Spatial Reallocation and Infrastructure, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$). Panels (a)–(b) plot 1817–1881 percentage-point changes in manufacturing and tertiary employment shares across the 25 latitude bands (1 = South, 25 = North; Online Appendix S.B); Panels (c)–(d) plot the national mean transport stock (76) (1710 = 1, log scale) and the 1881 band profile (1710 national mean = 1).

when built; completed capacity lowers route losses $\kappa(T)$ and expands carrier service capacity at later dates; and, by Proposition 3, higher capacity accelerates spatial knowledge catch-up.[14]

Figure 9 and Table 7 report the counterfactual. By 1881, diagnostic expectations expand the aggregate transport stock $\overline{T}$ by +43.79 percent (3.001 versus 2.087; Panel e), with regional gains of +29.53 percent in the Southern belt (Counties 1–5), +54.47 percent in the Midlands–North corridor (Counties 6–9), and +61.33 percent in the Northern border (County 10). This network expansion raises 1881 real output Y (and output per worker y) by +20.96 percent (4.774 versus 3.947; Panel a) and lifts the population-weighted real wage $w/\mathcal{P}$ by +12.97 percent (1.158 versus 1.025; +12.78 to +13.59 percent across regions; Panel b).

Larger route capacity also accelerates structural transformation and spatial diffusion (Trew, 2020). It lowers the 1881 agricultural employment share by −12.9 percentage points (13.0 versus 25.9 percent) and shifts +11.4 points into manufacturing (64.0 versus 52.6 percent) and +1.5 points into tertiary services (23.0 versus 21.5 percent; Panel c). Faster spatial diffusion lifts manufacturing frontier productivity F_M by +10.90 percent (2.657 versus 2.396; Panel d), while cheaper distribution and higher manufacturing productivity lower p_M/p_A by −13.79 percent (0.492 versus 0.571) and the Stone–Geary price index $\mathcal{P}$ by −5.97 percent (0.720 versus 0.765; Panel f).

[14]Level indices in Figure 9 and Table 7 are in balanced-growth-detrended units (1710 = 1), and employment shares are model-domain shares $\mathcal{L}_o(t)/\bar{L}_t$ (S.D.2); percentage differences $100\,(\mathrm{DE/R} \mid \mathrm{DE} - 1)$ are identical in un-detrended

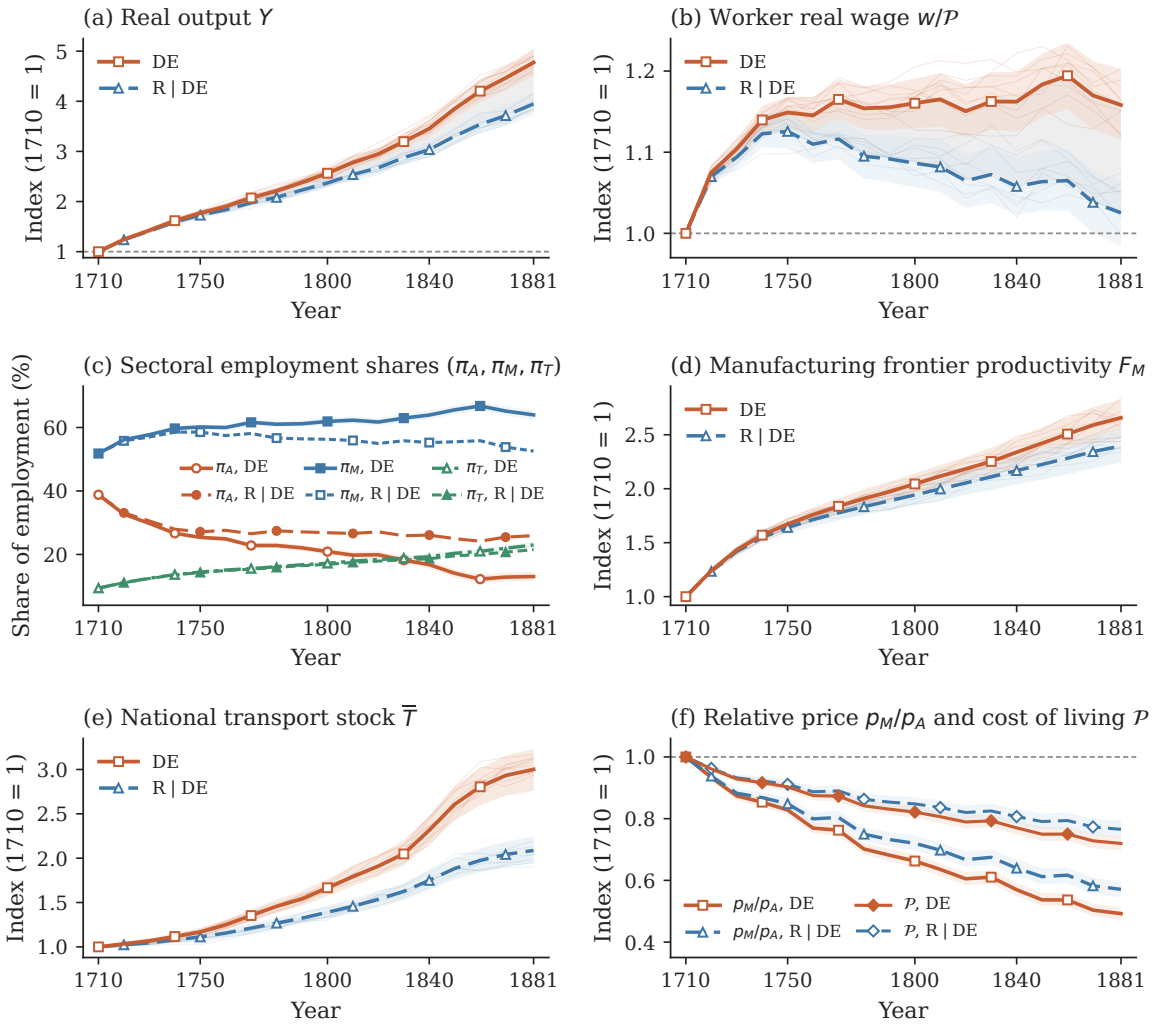


Figure 9: Counterfactual Effects of Diagnostic Expectations, 1710–1881

Notes: Simulated means and 5th–95th percentile bands (1,000 draws) under the baseline diagnostic economy (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE: $\theta = 0$ holding all other parameters and shock histories fixed). Panels (a), (b), (d), (e), and (f) report balanced-growth-detrended indices (1710 = 1) for real output Y, real wage $w/\mathcal{P}$, manufacturing frontier F_M, transport stock $\overline{T}$, and prices ($p_M/p_A, \mathcal{P}$); Panel (c) reports sectoral employment shares π_A, π_M, π_T (percent).

Regional productivity patterns in Panel B of Table 7 separate frontier shifts from local adoption. The counterfactual percentage gain in the available frontier $F_{Mt}(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell-r|} A^{+}_{Mt}(r)$ is uniform at +10.90 percent across all three regions because, along the simulated path, a single northern site r^* attains the supremum in (39) for every ℓ, so that $F_{Mt}(\ell) = e^{-\delta|\ell-r^*|} A^{+}_{Mt}(r^*)$ and any proportional change in $A^{+}_{Mt}(r^*)$ shifts $\log F_{Mt}$ uniformly. Local applied manufacturing productivity $A_{Mt}(\ell)$, governed by the transport-dependent catch-up rate $\lambda_t(\ell) = \lambda_0[T_t(\ell)/T_{t_0}(\ell)]^{1-\alpha_T}$ in (40)–(41), expands most where route buildout is largest (+7.84 percent in the Southern belt, +13.42 percent in the Midlands–North corridor, and +14.18 percent in the Northern border, or +11.65 percent in the aggregate).

Decomposing the +20.96 percent 1881 output gain across the three channels omitted from

units.

Table 7: Counterfactual Effects of Diagnostic Expectations in 1881

1881 outcome (1710 = 1, except shares)	DE ($\widehat{\theta} = 0.739$)	R | DE ($\theta = 0$)	Difference
Panel A. Aggregate outcomes			
Real output (Y)	4.774	3.947	+20.96%
Population-weighted real wage ($w/\mathcal{P}$)	1.158	1.025	+12.97%
Agricultural employment share (π_A, %)	13.0	25.9	−12.9 pp
Manufacturing / tertiary shares (π_M/π_T, %)	64.0 / 23.0	52.6 / 21.5	+11.4 / +1.5 pp
Manufacturing frontier productivity (F_M)	2.657	2.396	+10.90%
Aggregate transport stock ($\overline{T}$)	3.001	2.087	+43.79%
Relative price of manufactures (p_M/p_A)	0.492	0.571	−13.79%
Stone–Geary price index ($\mathcal{P}$)	0.720	0.765	−5.97%
Panel B. Regional outcomes ($w/\mathcal{P}$ / F_M / $\overline{T}$)			
Southern belt (Counties 1–5)	1.188 / 1.888 / 2.470	1.053 / 1.703 / 1.907	+12.78% / +10.90% / +29.53%
Midlands–North corridor (Counties 6–9)	1.156 / 3.299 / 3.549	1.023 / 2.975 / 2.298	+13.00% / +10.90% / +54.47%
Northern border (County 10)	1.116 / 3.935 / 3.463	0.982 / 3.548 / 2.146	+13.59% / +10.90% / +61.33%

Notes: 1881 outcomes averaged across $B = 1{,}000$ common shock histories under baseline diagnostic expectations (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE, $\theta = 0$ holding all other parameters and shock histories fixed). Level indices are in balanced-growth-detrended units (1710 = 1; percentage differences $100\,(\text{DE}/\text{R} \mid \text{DE} - 1)$ are identical in historical units), and employment shares are in percent (differences in percentage points).

private marginal-q in Section III.F shows why static "social savings" calculations (Fogel, 1964) capture only a fraction of the long-run return to connective infrastructure. Holding spatial sorting and technology at their R | DE paths, direct carrier and iceberg transit savings (Donaldson and Hornbeck, 2016; Fogel, 1964) account for +5.42 percentage points (25.9 percent of the gain). Spatial labor reallocation into wedge-distorted manufacturing ($\psi_M(\ell) > 0$; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024; Hsieh and Klenow 2009) adds +6.18 percentage points (29.5 percent), and transport-mediated spatial knowledge diffusion and induced innovation (66) (Desmet and Rossi-Hansberg, 2014; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990) add +9.36 percentage points (44.7 percent). Nearly three-quarters of the long-run output gain from corrective overoptimism thus arises from allocative reallocation (29.5 percent) and spatial idea diffusion (44.7 percent) rather than static transit savings (25.9 percent).

V Concluding Remarks

A great arterial system outlives the illusions of those who built it. Across the eighteenth and nineteenth centuries, waves of speculative enthusiasm drew private capital into the turnpikes, canals, and railways that stitched the parishes of England and Wales into a single national market. When expectations outran reality and reverted, investors bore heavy losses; yet the stone roadbeds, cut waterways, and iron rails remained in the landscape. Because physical infrastructure is sunk while market power and knowledge spillovers cause rational investors to build too little, those bursts of diagnostic overoptimism left the economy with a larger network, higher industrial output, and higher real wages than sober calculation would have delivered. Most of that lasting dividend came not from cheaper freight along existing routes, but from the movement of labor into industry and the quiet, cumulative diffusion of ideas across connected towns and shires.[15]

[15] In the literature on rational bubbles, speculative asset valuations promote growth by relaxing collateral and borrowing constraints (Guerron-Quintana et al., 2023; Hirano and Yanagawa, 2016; Martin and Ventura, 2012; Miao and Wang, 2014). Here the mechanism operates even without financial frictions, as a second-best offset to market power and spatial and knowledge externalities under irreversible investment (Desmet and Rossi-Hansberg, 2014;

Infrastructure booms create a commons where communities share ideas and catalyze mutual growth. This force is most potent where markets are incomplete, state capacity is modest, and capital is irreversible—conditions that shaped the Great Divergence. Britain entered the Industrial Revolution with secure property rights and parliamentary protection for private tolls, yet with a fiscal state that committed almost no public treasure to domestic roads, canals, or railways. Corrective overoptimism bridges these two institutional facts without invoking central planning: where the law lets private investors claim local land and toll rents, recurrent waves of optimism can summon the sunk network buildouts that a weak exchequer cannot finance, turning private speculation into a decentralized big push.[16]

Not every speculative mania leaves such an inheritance behind. For overoptimism to serve the common good, the capital it creates must endure and the social returns it unlocks must exceed what builders can capture in tolls and prices. Many compare contemporary booms in computation and artificial intelligence to the railway and telegraph manias of the nineteenth century, hoping that today's exuberance will leave a similar bounty.[17] Yet that parallel rests on conditions that do not always hold. Where capital depreciates in a few years rather than over generations—as with short-lived computing equipment compared to stone viaducts, canals, or copper and glass cables—the installed stock decays before slow-moving spillovers across space and industry can compound. Moreover, where modern contracts and fine price discrimination allow firms to segment buyers and extract a far larger share of surplus, the initial underinvestment gap is narrow, so that diagnostic exuberance can overshoot the social optimum and—in an inversion of the historical mechanism—generate wasteful overbuilding.

Wider horizons open once we view speculative booms through the lens of durable commons. In the study of development, each stage of industrial transformation requires matching connective infrastructure, and institutions that channel private waves of optimism into enduring networks may help societies cross thresholds that state budgets alone cannot reach. Within macroeconomics, a unified account of diagnostic expectations must weigh productive booms in shared arteries against booms in housing or financial claims that absorb savings without expanding the economy's productive base, while tracing how sunk buildouts interact with social contagion and shifting degrees of overreaction across the business cycle.[18]

The mechanism identified here is not specific to turnpikes and railways. The electric grid, the fiber-optic network, and future infrastructure not yet built share the same structural features: large sunk costs, spatial and knowledge externalities, and investors who must form expectations about a transformed economy they cannot yet observe. Long after the speculative fever subsides and time erases the balance sheets of the builders, the arteries remain in the earth, linking quiet homes and distant parishes into the life of the nation. From corrective overoptimism can arise a force for each to stand on their neighbor's shoulders.

Lucas, 1988; Hornbeck and Rotemberg, 2024; Lipsey and Lancaster, 1956; Romer, 1990).

[16] On property rights and Parliamentary enclosure and transport acts, see North (1990), Acemoglu et al. (2005), and Bogart (2014); on limited domestic fiscal capacity for public works, see Milward and Saul (1973), Besley and Persson (2009), and Dincecco (2011); and on coordinated big-push industrialization, see Rosenstein-Rodan (1943), Murphy et al. (1989), Kline and Moretti (2014), Moneke (2024), and Garg (2025).

[17] See, for example, Hobart and Huber (2024) on productive technology bubbles, as well as contemporary commentary comparing artificial-intelligence capital expenditure to nineteenth-century railway buildouts.

[18] These questions connect to new structural economics and big-push development policy (Garg, 2025; Lin, 2011; Moneke, 2024), to diagnostic analyses of housing and credit booms (Bordalo et al., 2019; Chodorow-Reich et al., 2024), and to models of belief contagion and endogenous overreaction over the cycle (Bianchi et al., 2024; Burnside et al., 2016; Flynn and Sastry, 2025).

subsection

A Proofs of Main Results

A.1 Proof of Proposition 1

Proof. Because $\mathbb{E}_t^\theta$ is the expectation under the tilted probability in (11), it is linear, so the objective in (14) equals $\mathbb{E}_t^\theta[U(c_A(t), c_M(t))] + \mathbb{E}_t^\theta[\sum_{s\geq t+1} \beta^{s-t} U(c_A(s), c_M(s))]$. Goods are nonstorable, households neither save nor borrow, relocation is costless, portfolio income b_s is residence independent, and households are atomistic, so the set of feasible continuation plans $\{\ell_s, c_A(s), c_M(s)\}_{s\geq t+1}$ does not depend on $(\ell_t, c_A(t), c_M(t))$ and the supremum C_t of the second term is the same for every current residence. Ex-post consumption optimization gives $U(c_A(t), c_M(t)) = u_t(\ell_t, \omega)$ statewise by (16), so $W_t = \sup_{\ell_t \in [0,1]} \mathbb{E}_t^\theta[u_t(\ell_t)] + C_t$, where every term is nonnegative because $U \geq 0$, so finiteness of W_t makes C_t a finite constant and every $\mathbb{E}_t^\theta[u_t(\ell)]$ finite. Hence occupied locations maximize $\mathbb{E}_t^\theta[u_t(\ell)]$ at the common value $\bar{u}_t$ and unoccupied locations satisfy $\mathbb{E}_t^\theta[u_t(\ell)] \leq \bar{u}_t$. No Bellman recursion is used: diagnostic expectations are not dynamically consistent, so $\mathbb{E}_t^\theta$ applied to the date-$(t+1)$ value formed under $\mathbb{E}_{t+1}^\theta$ need not equal the date-t perceived continuation.[← Back to Proposition 1] □

A.2 Proof of Proposition 2

Proof. Let $\mathcal{F}_t^+$ be the σ-field generated by $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$ and the date-t discovery fields $(X_{it}^s, X_{it}^g)_i$. By the selection rule in the statement, the conditional dated equilibrium of Definition 1 is $\mathcal{F}_t^+$-measurable, and $\mathbf{z}_{t+1}$ is independent of $\mathcal{F}_t^+$ conditional on Ω_t because the discovery fields are independent across dates and of the aggregate gaps. In (11) the likelihood ratio $\Lambda_t(\mathbf{z}_{t+1})^\theta$ is a function of $\mathbf{z}_{t+1}$ alone and is bounded above and below by positive constants, because the truncated Gaussian density is continuous and strictly positive on the compact set $\mathcal{B} \times \mathcal{B}$ (Online Appendix S.C.1); hence $0 < \mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta] < \infty$ for every $\theta \geq 0$, $X_t \Lambda_t(\mathbf{z}_{t+1})^\theta$ is integrable whenever X_t is, and for every integrable $\mathcal{F}_t^+$-measurable X_t, $\mathbb{E}_t[X_t \Lambda_t(\mathbf{z}_{t+1})^\theta] = \mathbb{E}_t[X_t]\,\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]$ and $\mathbb{E}_t^\theta[X_t] = \mathbb{E}_t[X_t]$. Every within-period objective in Section III—the worker's expected indirect utility (36), the producer's expected net revenue (28), the carrier's expected toll (50), and the county current dividend (56)—is of this form, so conditions (i)–(iv) of Definition 1 read identically for every $\theta \geq 0$; the set of conditional dated equilibria at $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$ therefore does not depend on θ, the common selection rule, which does not depend on θ, returns the same element, and the selected conditional dated equilibrium is invariant to θ once $\mathbf{x}_t$ is fixed. The next aggregate state Ω_{t+1} follows from the same discovery fields and the same $\mathbf{z}_{t+1}$ through (38)–(40) and (54), with the observed reference resetting to $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$ (Definition 2(iii)); the θ-dependent forecast reference $M(\mathbf{m}_t^\theta)$ in (13) enters only county continuation values. [← Back to Proposition 2] □

A.3 Proof of Proposition 3

Proof. Let $(\mathcal{D}A^+)(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|} A^+(r)$ and $F = \mathcal{D}A^+$. For any nonnegative g, taking $r = \ell$ gives $g \leq \mathcal{D}g$, so $A^+ \leq F \leq \mathcal{D}F$; the triangle inequality $|\ell - s| \leq |\ell - r| + |r - s|$ and $A^+ \geq 0$ give $\mathcal{D}F \leq F$, hence $\mathcal{D}F = F$. Because $A' = e^{-\lambda}A^+ + (1 - e^{-\lambda})F$ is a convex combination of A^+ and F for $\lambda \geq 0$, we have $A^+ \leq A' \leq F$ and hence $F = \mathcal{D}A^+ \leq \mathcal{D}A' \leq \mathcal{D}F = F$. If $G_{i,t+1} \equiv 1$, then $A_{i,t+1}^+ = A_{i,t+1} = A'$ whatever the sector assignment, so $F_{i,t+1} = \mathcal{D}A' = F$; since each later $A_{i,t+h}$ again lies in $[A_{i,t+h-1}, F]$, induction gives $F_{i,t+h} = F$ and $F - A_{i,t+h} = e^{-\lambda_{t+h-1}}(F - A_{i,t+h-1})$, which iterates to (42); since $\lambda_t(\ell)$ increases in $T_t(\ell)$ and $F - A^+ \geq 0$, higher $T_t(\ell)$ weakly raises $A_{i,t+1}(\ell)$ at fixed A^+ and F (plans and prices may also respond to T_t). [← Back to Proposition 3] □

A.4 Proof of Proposition 4

Proof. Let $B(\mathbf{x}) = \prod_j \arg\max_{u\in[0,\bar{x}_{jt}]} \Pi^\theta_{jt}(u, \mathbf{x}_{-j})$. Since $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ is nonempty, compact, and convex, B is nonempty-, compact-, and convex-valued (Weierstrass; concavity of Π^θ_{jt} in x_{jt}) with closed graph (Berge's maximum theorem, Π^θ_{jt} being jointly continuous), so Kakutani's fixed-point theorem yields $\mathbf{x} \in B(\mathbf{x})$; the Nash set $\mathcal{N}^\theta_t = \{\mathbf{x} : \mathbf{x} \in B(\mathbf{x})\}$ is closed in the compact $\mathcal{X}_t$, hence nonempty and compact. When Π^θ_{jt} is differentiable, concavity makes the variational inequality

$$g^\theta_{jt}(\mathbf{x}^*)(u - x^*_j) \le 0 \quad \forall\, u \in [0, \bar{x}_{jt}], \quad j = 1, \dots, J, \qquad g^\theta_{jt} = \partial_j \Pi^\theta_{jt}, \tag{A.1}$$

necessary and sufficient. Under (63), each best response $b_j(\mathbf{y}_{-j})$ is unique, and comparing $u = b_j(\mathbf{y}_{-j})$ with $u' = b_j(\mathbf{y}'_{-j})$ in (A.1) gives $(u - u')\{g_j(u, \mathbf{y}_{-j}) - g_j(u', \mathbf{y}'_{-j})\} \ge 0$. Writing the bracket as $[g_j(u, \mathbf{y}_{-j}) - g_j(u', \mathbf{y}_{-j})] + [g_j(u', \mathbf{y}_{-j}) - g_j(u', \mathbf{y}'_{-j})]$ and applying the mean value theorem along x_j and along the segment from $\mathbf{y}'_{-j}$ to $\mathbf{y}_{-j}$ (both inside the convex box $\mathcal{X}_t$, where (63) holds up to its faces), the first bracket contributes at most $-m_j(u - u')^2$ and the second at most $|u - u'| \sum_{k\neq j} L_{jk}|y_k - y'_k|$, hence $m_j|u - u'| \le \sum_{k\neq j} L_{jk}|y_k - y'_k|$. In the weighted maximum norm $\|\mathbf{y}\|_v = \max_j |y_j|/v_j$, we obtain $\|\mathbf{b}(\mathbf{y}) - \mathbf{b}(\mathbf{y}')\|_v \le \gamma_I \|\mathbf{y} - \mathbf{y}'\|_v$ with $\gamma_I < 1$ by (64); since under (63) Nash profiles are exactly the fixed points of the single-valued map $\mathbf{b}$, Banach's fixed-point theorem yields a unique Nash equilibrium and geometric convergence of best responses. Under strong monotonicity instead, summing the J inequalities (A.1) at two equilibria $\mathbf{x}$ and $\mathbf{y}$ (with $u = y_j$ and $u = x_j$, respectively) gives $\sum_j (x_j - y_j)[g_j(\mathbf{y}) - g_j(\mathbf{x})] \le 0$, so $\mu\|\mathbf{x} - \mathbf{y}\|^2 \le 0$ and $\mathbf{x} = \mathbf{y}$ (Vives, 1999). Measurability of the selection in Ω_t follows in the first case because each b_j is measurable in Ω_t (measurable maximum theorem) and Lipschitz in $\mathbf{y}_{-j}$, so the best-response iterates are measurable and the equilibrium is their pointwise limit; in the second because the equilibrium is the unique minimizer over $\mathcal{X}_t$ of the Carathéodory residual $\mathbf{x} \mapsto \|\mathbf{x} - \mathrm{proj}_{\mathcal{X}_t}(\mathbf{x} + \mathbf{g}^\theta_t(\mathbf{x}; \Omega_t))\|$, whose zeros are exactly the solutions of (A.1), and is therefore measurable by the same theorem. [← Back to Proposition 4] □

A.5 Proof of Proposition 5

Proof. Fix a realization ω with positive, absolutely continuous $p_i(\cdot, \omega)$ and absolutely continuous $H_i(\cdot, \omega)$, so that $p_i H_i$ is absolutely continuous and the fundamental theorem of calculus applies to it. Since every term of the ledgers (70)–(72) and of (69) is integrable on $[0, 1]$, the integrals below may be taken term by term. Writing local carrier demand as $D = \sum_i r_i(Y_i + |H_i|)$ and using $Y_i = e_i Y_i$ from (37) and local labor clearing $n = \sum_i e_i L_i + L_T$, summing the local ledgers (70)–(72) gives

$$R^{\mathrm{net}} + \Pi_F + \Pi_T + \Pi_\psi = \sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| + \tau_{j(\ell)}(Y_T - D) - wn - \sum_i e_i \Psi_i - p_M I. \tag{A.2}$$

The land rent R cancels between (70) and (72), so the identity uses neither free entry nor the sorting rule. Because τ_j is constant within each county $\mathcal{C}_j$, county service clearing (51) implies

$$\int_{\mathcal{C}_j} \tau_j Y_T \, d\ell = \tau_j K_j = \tau_j D_j = \int_{\mathcal{C}_j} \tau_j D \, d\ell, \tag{A.3}$$

so integrating (A.2) over $[0, 1]$ yields

$$b_t \bar{L}_t = \int_0^1 (R^{\mathrm{net}} + \Pi_F + \Pi_T + \Pi_\psi) d\ell = \int_0^1 \Big[\sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| - wn - \sum_i e_i \Psi_i - p_M I\Big] d\ell. \tag{A.4}$$

Meanwhile, combining goods conservation (69) with $H_i\partial_\ell p_i = (\kappa p_i + r_i\tau_{j(\ell)})|H_i|$ from spatial no-arbitrage (45) cancels the physical loss terms $\pm\kappa p_i|H_i|$ in $\partial_\ell(p_iH_i) = p_i\partial_\ell H_i + H_i\partial_\ell p_i$, giving $\partial_\ell(p_iH_i) = p_i(Y_i - nc_i - e_i\Psi_i/p_i - I_i) + r_i\tau_{j(\ell)}|H_i|$ ($I_A = 0$, $I_M = I$). Integrating over $[0,1]$ under $H_i(0) = H_i(1) = 0$, summing over $i \in \{A, M\}$, and substituting (A.4) gives $\int_0^1 \sum_i p_i nc_i\, d\ell = \int_0^1 wn\, d\ell + b_t\bar{L}_t$. [← Back to Proposition 5] □

A.6 Proof of Proposition 6

A.6.1 Part 1: Firm attainment, active-research threshold, and pure-site sorting

Fix $\ell \in [0,1]$ and sector $i \in \{A, M\}$ at quotes $p_i(\omega) \geq 0$, $\tau_{j(\ell)}(\omega) \geq 0$ with $\bar{m}_i^V \equiv \mathbb{E}_t[(p_i - r_i\tau_{j(\ell)})_+V_i] < \infty$ (bounded quotes $p_i \leq \bar{p}$ suffice), and $w > 0$, and write $m_i(\phi) = \mathbb{E}_t[(p_i - r_i\tau_{j(\ell)})_+G_i(\phi)]$. Throughout Part 1 the quoted processes $(p_i(\ell,\cdot), \tau_{j(\ell)}(\cdot))$ are evaluated as independent of the own-site success index $U_i = \Phi(X_{it}^s(\ell))$, jointly with the quality draw $V_i = \Phi(X_{it}^g(\ell))^{-1/a_i}$. This is the atomistic entrant's evaluation convention in (28), not a consequence of atomicity: an individual site's draw does not move county aggregates, but under (24) it is correlated with its neighbours' draws, which do.

LEMMA A.1: *Under (22)–(23) with $a_i > 1/(1-\alpha_i) > 1$, $m_i(\phi)$ is continuous on $[0,1]$ and satisfies*

$$\bar{m}_i := \sup_{0\leq\phi\leq 1} m_i(\phi) = m_i(1) = \bar{m}_i^V < \infty \qquad \Big(\leq \bar{p}\,\frac{a_i}{a_i - 1} \text{ at bounded quotes}\Big). \tag{A.5}$$

For each $\phi \in [0,1)$, optimal production labor is unique and given by

$$L_i^*(\phi) = \left[\frac{\alpha_i Z_i m_i(\phi)}{(1+\psi_i)w}\right]^{1/(1-\alpha_i)}, \tag{A.6}$$

and the profiled bid rent

$$R_i(\phi; w) = \Lambda_i w^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi)^{1/(1-\alpha_i)} - \Psi_i(\phi; w), \qquad \Lambda_i = (1-\alpha_i)Z_i\left[\frac{\alpha_i Z_i}{1+\psi_i}\right]^{\alpha_i/(1-\alpha_i)}, \tag{A.7}$$

is finite and attained on $[0,1)$.

Proof. Since $G_i(\phi;\ell,\omega) = 1 + \mathbf{1}_{\{U_i<\phi\}}(V_i - 1)$ with U_i independent of the pair $(V_i, (p_i - r_i\tau_{j(\ell)})_+)$ by the maintained convention (X^s and X^g are independent fields), $m_i(\phi) = m_i(0) + \phi\,\mathbb{E}_t[(p_i - r_i\tau_{j(\ell)})_+(V_i - 1)] = m_i(0)[1 + \kappa_i(\ell)\phi]$ is affine and nondecreasing on $[0,1]$ with $m_i(1) = \bar{m}_i^V < \infty$ and $\mathbb{E}[G_i(\phi)] = 1 + \phi/(a_i - 1)$, giving (A.5). Strict concavity of (29) in L_i yields (A.6)–(A.7). Since $R_i(\cdot; w)$ is continuous on $(0,1)$ with $\lim_{\phi\to 0^+} R_i(\phi; w) = R_i(0; w) - w(f_i(\ell) + \chi_i) < R_i(0; w)$, it is upper semicontinuous on $[0,1)$, while $\Psi_i(\phi; w) \geq w\chi_i/(1-\phi) \to \infty$ as $\phi \uparrow 1$; hence the supremum is attained on $[0,1)$. □

At an interior optimum $\tilde{\phi}_i \in (0,1)$, the envelope theorem gives

$$Z_i L_i^*(\tilde{\phi}_i)^{\alpha_i}\,\partial_\phi \mathbb{E}_t\Big[(p_i - r_i\tau_{j(\ell)})_+G_i(\tilde{\phi}_i)\Big] = \partial_\phi\Psi_i(\tilde{\phi}_i; w) = \frac{w\chi_i}{(1-\tilde{\phi}_i)^2}, \tag{A.8}$$

matching (32). For $w_2 > w_1$ with optimal plans (ϕ_2, L_2) and (ϕ_1, L_1), writing $R_i(\phi; w, \ell)/w = \pi_i^0(w,\ell)[1 + \Gamma_i(\phi;\ell)] - C_i(\phi)$ with $C_i(\phi) = \Psi_i(\phi; w, \ell)/w$ and summing the revealed-preference inequalities gives

$$[\pi_i^0(w_2,\ell) - \pi_i^0(w_1,\ell)]\,[\Gamma_i(\phi_2;\ell) - \Gamma_i(\phi_1;\ell)] \geq 0. \tag{A.9}$$

Since $\pi_i^0(w_2,\ell) < \pi_i^0(w_1,\ell)$ and $\Gamma_i(\cdot;\ell)$ is strictly increasing, $\phi_2 \le \phi_1$, $m_i(\phi_2;\ell) \le m_i(\phi_1;\ell)$, and $L_2 = L_i^*(\phi_2;w_2,\ell) < L_i^*(\phi_1;w_1,\ell) = L_1$: thus $\phi_i^*(w,\ell)$ is weakly decreasing and $L_i^*(\phi_i^*(w,\ell);w,\ell)$ is strictly decreasing in w. Since $\partial^2_{\phi\phi}[m_i(\phi)^{1/(1-\alpha_i)}] = \frac{\alpha_i}{(1-\alpha_i)^2} m_i(\phi)^{(2\alpha_i-1)/(1-\alpha_i)}[\partial_\phi m_i]^2$ with $\partial_\phi m_i = m_i(0;\ell)\kappa_i(\ell)$, strict concavity $\partial^2_{\phi\phi} R_i(\phi;w,\ell) < 0$ on $(0,1)$ holds for $\chi_i > \bar{\chi}_{it} \equiv \frac{\alpha_i}{2(1-\alpha_i)^2}$ $\sup_{\ell\in[0,1],\, u\in\mathcal{U}_t,\, \phi\in(0,1)} (1-\phi)^3[m_i(0;\ell)\kappa_i(\ell)]^2 \Lambda_i(\ell) w(\ell)^{-1/(1-\alpha_i)} m_i(\phi;\ell)^{(2\alpha_i-1)/(1-\alpha_i)}$, a constant on the quote domain of Section III.G (Condition 2 presupposes it is finite), so at most one interior stationary point $\tilde{\phi}_i(w,\ell)$ exists, and (33) uniquely selects $\phi_i^*(w,\ell) \in \{0, \tilde{\phi}_i(w,\ell)\}$. The reservation threshold is well defined: $S_i(\pi) \equiv \sup_{\phi\in(0,1)}[\pi\Gamma_i(\phi;\ell) - f_i(\ell) - \chi_i/(1-\phi)]$ is convex and nondecreasing in π as a supremum of affine functions with nonnegative slopes, $S_i(0) = -f_i(\ell) - \chi_i < 0$, and $S_i(\pi) \to \infty$ when $\kappa_i(\ell) > 0$, so S_i has a unique root $\underline{\pi}_i(\ell) \in (0,\infty)$ and research is active exactly when $\pi_i^0(w,\ell) > \underline{\pi}_i(\ell)$; at equality both 0 and $\tilde{\phi}_i$ are optimal and the convention $\phi_i^* = 0$ applies. If $\kappa_i(\ell) = 0$, research is never active ($\underline{\pi}_i(\ell) = \infty$), and if $m_i(0;\ell) = 0$ the sector is dormant at ℓ with $R_i = L_i^* = \phi_i^* = 0$. Under Condition 2, $R_A(\ell) - R_M(\ell)$ changes sign exactly once, at $b_{ct} \in (0,1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing. With e_{it} and ϕ_{it}^* fixed at every ℓ, (38) determines $A_{it}^+(\ell,\omega)$ pointwise, and Lemma S.C.1 (Online Appendix) ensures $F_{it}(\ell,\omega)$ in (39) is finite a.s. with finite mean everywhere on $[0,1]$.

A.6.2 Part 2: Existence and uniqueness on the quote domain

Proof. Steps 1–4 establish the operator claims of Part 1 and Step 5 proves Part 2. Fix $\mathbf{x}_t \in \mathcal{X}_t$ and $u = (w,p,\tau) \in \mathcal{U}_t$, and let $(e_i, \phi_i^*, L_i^*, L_T^*)$ be the plans that u induces through (30)–(34) and (50).

Step 1 (Labor clearing pins $\bar{u}_t$). By Part 1, $L_i^*(\phi_i^*;w,\ell)$ and $L_T^*(w,\ell)$ are strictly decreasing in w, with $L_i^*(0;w,\ell) \le L_i^*(\phi_i^*;w,\ell) \le [\alpha_i Z_i \bar{m}_i/((1+\psi_i)w)]^{1/(1-\alpha_i)}$ by $G_i \ge 1$ and (A.5). Hence, for every $(p,\tau) \in \mathcal{P}_t \times \mathcal{M}_t$, aggregate labor demand at a uniform wage exceeds $\bar{L}_t$ at $\underline{w}_t$ and falls below $\bar{L}_t$ at $\overline{W}_t$ (Condition 1). Since $\widetilde{w}(\ell)$ in (74) is increasing in $\bar{u}_t$, with $\mathbb{E}_t[1/\mathcal{P}]$ and $\mathbb{E}_t[(b_t - \gamma p_A)/\mathcal{P}]$ bounded on $\mathcal{U}_t$, the intermediate value theorem gives a unique $\bar{u}_t(u)$ at which the third line of (74) holds, continuous in u; the invariance clause of Condition 1 places $\widetilde{w}$ in $\mathcal{W}_t$.

Step 2 (Spot dispatch at the input quotes). With weights $\vartheta(\ell,\omega) = w(\ell) + b_t(\omega;u) - \gamma p_A(\ell,\omega) > 0$ (Condition 1) and research demand $e_i\Psi_i/p_i$ evaluated at u, consider for each ω the program $\max \int_0^1 \vartheta n[\eta \log(c_A - \gamma) + (1-\eta)\log c_M]\, d\ell$ over (c_i, Y_i, H_i) with $0 \le Y_i \le e_i Z_i G_i (L_i^*)^{\alpha_i}$, H_i absolutely continuous with $H_i(0) = H_i(1) = 0$, the free-disposal form $\partial_\ell H_i \le Y_i - n c_i - e_i\Psi_i/p_i - I_i - \kappa|H_i|$ of (69), and $D_j(\omega) \le K_j$. The program is concave with a Slater point (Condition 1), so it has a solution supported by multipliers $\widetilde{p}_i(\cdot,\omega) \ge 0$, the absolutely continuous adjoint of the flow constraint, and $\widetilde{\tau}_j(\omega) \ge 0$. Stationarity in c_i gives $\widetilde{p}_A = \eta\vartheta/(c_A - \gamma)$ and $\widetilde{p}_M = (1-\eta)\vartheta/c_M$, so prices are positive, the flow constraint binds, and consumption is Stone–Geary demand (16) at income ϑ; stationarity in Y_i gives full utilization where $\widetilde{p}_i > r_i \widetilde{\tau}_{j(\ell)}$ and none where $\widetilde{p}_i < r_i \widetilde{\tau}_{j(\ell)}$, as in (27); integrating the H_i-terms by parts gives (45) a.e.; and complementary slackness is (51). Consumption is unique by strict concavity, so $\widetilde{p}_i$ is unique at every (occupied) site, and $\widetilde{\tau}_j$ is unique whenever the binding county contains a trading or partially utilized site, where (45) or $\widetilde{p}_i = r_i\widetilde{\tau}_j$ pins it; Condition 1 takes the multipliers unique, so $\mathcal{T}_t^p$ is single valued and measurable in ω. Dividing the state-ω multipliers by the agricultural multiplier at $\ell = 1$ imposes the numeraire. At a fixed point, where $\widetilde{p}_A = p_A$, the divisor $\lambda(\omega)$ equals one: the stationarity conditions give $\int_0^1 \sum_i \widetilde{p}_i n c_i\, d\ell = \lambda^{-1}\int_0^1 \vartheta n\, d\ell + \gamma \int_0^1 \widetilde{p}_A n\, d\ell$ for the normalized prices, while Proposition 5 at the normalized quotes gives $\int_0^1 \sum_i \widetilde{p}_i n c_i\, d\ell = \int_0^1 (w + b_t) n\, d\ell$, and subtracting yields $(\lambda^{-1} - 1)\int_0^1 \vartheta n\, d\ell = 0$ with $\int_0^1 \vartheta n\, d\ell > 0$.

Step 3 (Continuity). The sorting margin $R_A - R_M$ and the activation margins $\pi_i^0(w(\ell),\ell) - \underline{\pi}_i(\ell)$ are continuous in u uniformly in ℓ, since bid rents depend on u through $w(\ell)$, $m_i(0;\ell)$, and $\mathbb{E}_t[\tau_{j(\ell)}]$,

and they vanish on Lebesgue-null sets of sites (Condition 2); hence the measure of sites whose sector or research regime differs between nearby quotes vanishes, and, each regime's L_i^* and L_T^* being continuous, the plan profile is continuous in u in $L^1([0,1])$. The data of the dispatch program are therefore continuous in u for almost every ω (G_i is almost surely continuous in ϕ by (24)), its solution and multipliers are unique, and the maximum theorem makes $(\widetilde{p},\widetilde{\tau})(\omega;\cdot)$ continuous in u; the expectations entering $\mathcal{T}_t^w$ and b_t are then continuous by dominated convergence, quotes and $\mathbb{E}_t[G_i]$ being bounded on $\mathcal{U}_t$. Thus $\mathcal{T}_t$ is a continuous self-map of $\mathcal{U}_t$, the self-map property being the invariance clause of Condition 1, with $|\partial_\ell \widetilde{p}_i| \le \kappa_0 \overline{p}_i + r_i \overline{\tau}_{jt}$ a.e., so its prices are Lipschitz as Definition 1 requires.

Step 4 (Fixed points are equilibria, and conversely). A fixed point $u^* = \mathcal{T}_t(u^*)$ satisfies Definition 1: plans are optimal at the quotes and land is sorted by construction; Step 2 delivers utilization, Stone–Geary consumption at income $w + b_t - \gamma p_A$, spatial no-arbitrage, goods balance, and county clearing; Step 1 delivers mobility (17) and national labor clearing; carriers hire by (50) at $\mathbb{E}_t[\tau_j]$; and b_t is the ledger dividend (73). Conversely, at a conditional dated equilibrium with quotes in $\mathcal{U}_t$, the equilibrium allocation satisfies the first-order conditions of the concave dispatch program at weights $w + b_t - \gamma p_A$ with multipliers (p,τ), so $(p,\tau) = \mathcal{T}_t^p(u)$ by uniqueness of the multipliers, and the wage satisfies (18) at the unique clearing $\bar{u}_t$, so $w = \mathcal{T}_t^w(u)$.

Step 5 (Existence and uniqueness). $\mathcal{U}_t$ is a closed subset of a Banach space (continuous wages, essentially bounded measurable families of continuous price profiles, and essentially bounded tolls) under $\|\cdot\|_{\mathcal{U}_t}$, so under Condition 3 Banach's fixed-point theorem yields exactly one fixed point $u^* \in \mathcal{U}_t$; by Step 4 it is a conditional dated equilibrium, and any other equilibrium with quotes in $\mathcal{U}_t$ would be a second fixed point. □

A.6.3 Part 3: Small-friction sufficient condition

Proof. This part is a regular-perturbation argument around the frictionless benchmark; it proves the existence of a friction threshold $\bar{\varepsilon}_t$ without computing it.

Step 1 (Frictionless benchmark). Consider the limiting economy in which goods move without loss, require no county services, and construction demand is absent ($\kappa_0 = m_I = 0$ and $r_A = r_M = 0$; since $r_M = 1$ normalizes service units, this is a benchmark rather than a parameter choice), so that $p_{At} \equiv 1$, $p_{Mt} \equiv p_{Mt}^0 > 0$, $\tau_{jt} \equiv 0$, and $w_t \equiv w_t^0$. Suppose $\chi_i > \bar{f}_i \equiv \sup_{\ell,\,\phi\in(0,1)}(1-\phi)\pi_i^0(w_t^0,\ell)\Gamma_i(\phi;\ell)$. Then $\pi_i^0\Gamma_i(\phi) - f_i(\ell) - \chi_i/(1-\phi) \le (\bar{f}_i - \chi_i)/(1-\phi) \le \bar{f}_i - \chi_i < 0$ for every $\phi \in (0,1)$, so $S_i(\pi_i^0(w_t^0,\ell)) < 0$ and research is dormant ($\phi_{it}^* \equiv 0$) at every site: even the most favorable discovery cannot repay the research bill. With no discoveries the dated economy is deterministic. At uniform quotes and zero research, $\log R_{it} = \log\Lambda_i(\ell) + \frac{1}{1-\alpha_i}\log m_i(0) - \frac{\alpha_i}{1-\alpha_i}\log w_t^0$ with $m_A(0) = 1$ and $m_M(0) = p_{Mt}^0$, so $\log(R_{At}(\ell)/R_{Mt}(\ell))$ is a constant minus the comparative-advantage profile of Condition 2 and is strictly decreasing in ℓ when that profile is strictly increasing; with $R_{At}(0) > R_{Mt}(0)$ and $R_{At}(1) < R_{Mt}(1)$ the sign changes exactly once, at $b_{ct}^0 \in (0,1)$. The triple $(w_t^0, p_{Mt}^0, b_{ct}^0)$ solves sorting, labor clearing, and manufacturing-goods clearing under Stone–Geary expenditure shares,

$$R_{At}(b_{ct}^0) = R_{Mt}(b_{ct}^0), \qquad \int_0^{b_{ct}^0} L_A^*(0;w_t^0,\ell)\,d\ell + \int_{b_{ct}^0}^1 L_M^*(0;w_t^0,\ell)\,d\ell = \bar{L}_t,$$
$$(1-\eta)\left[Y_{At}^0 + p_{Mt}^0 Y_{Mt}^0 - \gamma\bar{L}_t\right] = p_{Mt}^0 Y_{Mt}^0,$$

with $Y_{it}^0 = \int_0^1 e_i Z_i L_i^*(0;w_t^0,\ell)^{\alpha_i}\,d\ell$ and aggregate income $(w_t^0 + b_t^0)\bar{L}_t = Y_{At}^0 + p_{Mt}^0 Y_{Mt}^0$; the strict resource inequality $Y_{At}^0 > \gamma\bar{L}_t$ then gives nominal subsistence slack $w_t^0 + b_t^0 - \gamma > 0$.

Step 2 (Uniform bounds by regular perturbation). At date t the admissible stocks $T_t = T_0 + s_{jt}$ with $0 \le s_{jt} \le \bar{s}_{jt}$ form a compact set, and the frictions enter plans, bid rents, and dispatch continuously through the route loss $\kappa_0 e^{-T_t} \le \kappa_0 e^{-\underline{T}}$, the effective charges $r_i\tau_j \le r_i\overline{\tau}_{jt}$, the carrier payroll wL_T^*, which vanishes with the toll ceiling by (50), and m_I. Take $\mathcal{U}_t$ to be the quotes within a fixed distance $\delta > 0$ of the benchmark quotes $(w_t^0, 1, p_{Mt}^0, 0)$ in $\|\cdot\|_{\mathcal{U}_t}$, with collar $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}] = [w_t^0 - \delta, w_t^0 + \delta]$, price bounds $1 \pm \delta$ and $p_{Mt}^0 \pm \delta$, and toll ceilings $\overline{\tau}_{jt} \le \delta$. For δ small, the strict benchmark inequalities persist on $\mathcal{U}_t$: $S_i(\pi_i^0) \le \bar{f}_i - \chi_i + o(1) < 0$ keeps research dormant, $R_A - R_M$ keeps a single strict sign change near b_{ct}^0 with $R_A(0) > R_M(0)$ and $R_A(1) < R_M(1)$ (Condition 2), and the resource inequalities and nominal slack $w + b_t - \gamma p_A \ge w_t^0 + b_t^0 - \gamma - o(1) > 0$ hold (Condition 1). The wage and price outputs of $\mathcal{T}_t$ are then within $o(1)$ of the benchmark quotes as the frictions vanish, uniformly over the stock set: the agricultural price obeys $p_{At}(\ell, \omega) \in [e^{-\bar{\kappa}} - r_A\overline{\tau}_t,\ e^{\bar{\kappa}}(1 + r_A\overline{\tau}_t)]$ with $\bar{\kappa} = \kappa_0 e^{-\underline{T}}$ and $\overline{\tau}_t = \max_j \overline{\tau}_{jt}$, by Grönwall's inequality applied to (45) leftward from $p_{At}(1, \omega) = 1$; the manufacturing price, whose level (45) leaves free (every constant satisfies it), is pinned by manufacturing goods clearing, $\int_0^1 [(1-\eta)n(w + b_t - \gamma p_A)/p_M + I + \kappa|H_M|]\, d\ell = \int_0^1 Y_M\, d\ell$, whose left side is strictly decreasing in the level of p_M and converges to its benchmark form, so the level converges to p_{Mt}^0 and the profile lies within $O(\bar{\kappa}\,\overline{p}_M + \overline{\tau}_t)$ of it; and the mobility wage $\widetilde{w}$ converges to w_t^0, the price index being nearly uniform and $\bar{u}_t$ clearing labor. Hence there is $\bar{\varepsilon}_t > 0$ such that, for frictions below $\bar{\varepsilon}_t$, the wage and price coordinates of $\mathcal{T}_t$ map $\mathcal{U}_t$ into $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}] \times \prod_i [\underline{p}_i, \overline{p}_i]$ for every admissible stock path. The expected-toll bounds $[\underline{\tau}_{jt}, \overline{\tau}_{jt}]$ are not obtained by perturbation from $\tau \equiv 0$: the benchmark has $K_j = D_j = 0$ and lies outside $\mathcal{M}_t$, and the toll coordinate remains part of the invariance clause of Condition 1.

Step 3 (Condition 3 is not a small-friction property). The toll block of $\mathcal{T}_t$ returns the dispatch multiplier $\widetilde{\tau}_j(\omega)$ at the predetermined capacity

$$K_j = Z_T^{1/(1-\alpha_T)} (\alpha_T \mathbb{E}_t[\tau_j])^{\alpha_T/(1-\alpha_T)} \int_{\mathcal{C}_j} w^{-\alpha_T/(1-\alpha_T)} T_t\, d\ell$$

implied by (50), whereas county service demand D_j responds to the realized toll only through the effective charge $r_i\widetilde{\tau}_j$ in (45) and (27). Where capacity binds, $D_j(\widetilde{\tau}_j) = K_j(\mathbb{E}_t[\tau_j])$ gives $\partial \log \widetilde{\tau}_j / \partial \log \mathbb{E}_t[\tau_j] = [\alpha_T/(1-\alpha_T)]/\varepsilon_{D_j}$, with $\varepsilon_{D_j} = \partial \log D_j / \partial \log \widetilde{\tau}_j < 0$ the toll elasticity of service demand. This diagonal entry of the toll block is invariant to reweighting the toll coordinate and grows without bound as the toll share $r_i\widetilde{\tau}_j/p_i$ vanishes, since ε_{D_j} is of the order of that share; at the benchmark itself ($K_j = D_j = 0$, $\tau_j \equiv 0 \notin \mathcal{M}_t$) the toll component of $\mathcal{T}_t$ is undefined. Condition 3 is therefore not obtained by perturbation from the benchmark and is maintained as a restriction on $\mathcal{T}_t$ at the actual frictions. [← Back to Proposition 6] □

A.7 Proof of Proposition 7

A.7.1 Part 1: Recursive county contraction and state transitions

Forecasting kernel, state domain, and weight.— County j evaluates its continuation on the state $\Omega = (A_A, A_M, T, \mathbf{z}, \overline{\mathbf{z}}, t)$ through the one-period forecasting kernel $K_{\mathbf{x}}^{\theta}$: given $\mathbf{x} \in \mathcal{X}_t$, the route stock moves deterministically by (54), the productivity gaps are drawn from the diagnostic law (12) with the perceived reference (13), and the knowledge profiles $(A_{A,t+1}, A_{M,t+1})$ are integrated under the conditional law of the knowledge trajectory that the county takes as given (Definition 2(ii)). That law is indexed by the date and the current knowledge profile and is held fixed when the county varies its stock and action arguments, so it responds neither to $\mathbf{x}$ nor to $\overline{\mathbf{T}}$; in a dynamic equilibrium it coincides with the law generated by $\mathbf{x}_t^{\theta}$ through (38)–(40). The stage value is $Q_j(\Omega, \mathbf{x}; V) =$

$R_j(\mathbf{x};\Omega)+\beta(K^\theta_{\mathbf{x}}V_j)(\Omega)$, and because $\overline{\mathbf{T}}$ enters the kernel only through the deterministic shift $\overline{\mathbf{T}}+\mathbf{d}_t\mathbf{x}$,

$$\partial_{\overline{T}_k}(K^\theta_{\mathbf{x}}f)(\Omega) = (K^\theta_{\mathbf{x}}\,\partial_{\overline{T}_k}f)(\Omega) \tag{A.10}$$

for every f differentiable in $\overline{\mathbf{T}}$ with $K^\theta_{\mathbf{x}}|\partial_{\overline{T}_k}f|<\infty$. A state domain $\mathcal{D}$ is forecast-invariant if every feasible transition—the physical law of Definition 2(iii) and the perceived update (13), for every $\mathbf{x}\in\mathcal{X}_t$—maps it into itself with probability one. For a measurable weight $W\geq 1$ on $\mathcal{D}$, let $\mathcal{V}_W$ be the Banach space of county-value vectors $V=(V_j)_{j=1}^J$ on $\mathcal{D}$ that are C^1 in $\overline{\mathbf{T}}$ with finite weighted C^1 route-stock norm

$$\|V\|_{W,1} = \max_j \sup_{\Omega\in\mathcal{D}} W(\Omega)^{-1}\Big(|V_j(\Omega)| + \sum_{k=1}^J \overline{T}_k\,|\partial_{\overline{T}_k}V_j(\Omega)|\Big),$$

and write $\|\mathbf{q}\|_v=\max_j|q_j|/v_j$ for the weighted maximum norm of the proof of Proposition 4 with a fixed positive weight vector v ($v_j\equiv 1$ under strong monotonicity). The stock derivative is controlled because county policies respond to marginal-q, the stock derivative of the continuation value (62). The weight

$$W(\Omega_t) = 1 + \sum_{i\in\{A,M\}}\|A_{it}\|_\infty^{1/(1-\alpha_i)} + \sum_{j=1}^J \overline{T}_{jt}^{\,1-\alpha_T}$$

carries the exponents at which county rents grow in knowledge ($\Lambda_i\propto Z_i^{1/(1-\alpha_i)}$ in (31)) and carrier capacity grows in the stock ($Y_T\propto T^{1-\alpha_T}$). Under any knowledge law generated by (38)–(40) its one-step drift is $(K^\theta_{\mathbf{x}}W)(\Omega_t)\leq\kappa_W W(\Omega_t)$ with $\kappa_W=\max\{\kappa_A,\kappa_T\}$: pathwise $A_{i,t+1}\leq F_{it}$, $F_{it}\leq\|A_{it}\|_\infty\sup_r G_{it}(r)$, and $G_{it}(r)\leq\Phi(X^g_{it}(r))^{-1/a_i}$ whatever the plans (proof of Lemma S.C.1, Online Appendix), so

$$\mathbb{E}_t\|A_{i,t+1}\|_\infty^{1/(1-\alpha_i)}\leq\kappa_A\|A_{it}\|_\infty^{1/(1-\alpha_i)},\qquad \kappa_A=\max_{i\in\{A,M\}}\mathbb{E}\Big[\sup_{r\in[0,1]}\Phi\big(X^g_{it}(r)\big)^{-1/(a_i(1-\alpha_i))}\Big]<\infty,$$

finite because $1/(1-\alpha_i)<a_i$ (Condition 1), the Borell–TIS argument of that proof applying with $b=1/(1-\alpha_i)$, and $\kappa_A=1$ wherever research is inactive ($G_{it}\equiv 1$). The stock term drifts at $\kappa_T=\sup_{\mathcal{D}}\max_j(1+d_{jt}\bar{x}_{jt}/\overline{T}_{jt})^{1-\alpha_T}$, finite on domains on which one-period increments are bounded relative to stocks. The contraction condition of Theorem A.1 therefore restricts $\beta\kappa_A$ and the maximal one-period stock growth factor on $\mathcal{D}$; in the no-research regime of Part 2 rents are bounded and the theorem is applied with $W\equiv 1$, $\kappa_W=1$.

THEOREM A.1: *On a forecast-invariant state domain $\mathcal{D}$ where each $\mathbf{x}\in\mathcal{X}_t$ has a unique measurable conditional dated equilibrium, let $W\geq 1$ be a measurable weight with $(K^\theta_{\mathbf{x}}W)(\Omega)\leq\kappa_W W(\Omega)$ for every $\mathbf{x}\in\mathcal{X}_t$, and let $\mathcal{C}\subseteq\mathcal{V}_W$ be closed with $\mathbf{0}\in\mathcal{C}$. Suppose that for every $U,V\in\mathcal{C}$ and every county j: (i) $|R_j(\mathbf{x};\Omega)|+\sum_k\overline{T}_k|\partial_{\overline{T}_k}R_j(\mathbf{x};\Omega)|\leq C_R W(\Omega)$, and the level and weighted stock-derivative changes of $Q_j(\Omega,\cdot\,;V)$ across profiles $\mathbf{x},\mathbf{x}'\in\mathcal{X}_t$ sum to at most $BW(\Omega)\|\mathbf{x}-\mathbf{x}'\|_v$; (ii) the stage game has a unique measurable Nash profile $a_V(\Omega)$, differentiable in $\overline{\mathbf{T}}$ outside the switching states of (62) (where one-sided derivatives are used and suprema are essential), with $\|a_V(\Omega)-a_U(\Omega)\|_v\leq L_a\|V-U\|_{W,1}$, $\sum_k\overline{T}_k\|\partial_{\overline{T}_k}a_V(\Omega)\|_v\leq A_1$, and $\sum_k\overline{T}_k\|\partial_{\overline{T}_k}a_V(\Omega)-\partial_{\overline{T}_k}a_U(\Omega)\|_v\leq L_a'\|V-U\|_{W,1}$; (iii) the rival-action gradient along the equilibrium, $\mathbf{y}\mapsto\sum_{h\neq j}\partial_{x_h}Q_j(\Omega,a_V(\Omega);V)\,y_h$, is Lipschitz in V with modulus $B_2W(\Omega)\|\mathbf{y}\|_v$; and that the Nash Bellman operator $(\mathcal{B}^\theta V)_j(\Omega)=\max_{x_j}Q_j(\Omega,x_j,a_{V,-j}(\Omega);V)=Q_j(\Omega,a_V(\Omega);V)$ maps $\mathcal{C}$ into itself with*

$$\rho\equiv\beta\kappa_W+B\,(L_a+L_a')+A_1B_2<1.$$

Then $\mathcal{B}^\theta$ has a unique fixed point $V^\theta\in\mathcal{C}$, the finite-horizon iterates $(\mathcal{B}^\theta)^H\mathbf{0}$ converge to it geometrically at rate ρ, and a_{V^θ} is the unique Markov perfect county policy with values in $\mathcal{C}$.

Proof. Fix $U, V \in \mathcal{C}$ and j, and abbreviate $a_V = a_V(\Omega)$. Since $a_{V,j}$ is county j's best response to $a_{V,-j}$, $(\mathcal{B}^\theta V)_j = Q_j(a_V; V)$, and

$$(\mathcal{B}^\theta V)_j - (\mathcal{B}^\theta U)_j = [Q_j(a_V; V) - Q_j(a_V; U)] + [Q_j(a_V; U) - Q_j(a_U; U)].$$

Levels. The first bracket equals $\beta K^\theta_{a_V}(V_j - U_j)$ at the fixed profile a_V, so positivity of the kernel and the drift bound give $|K^\theta_{a_V}(V_j - U_j)| \le K^\theta_{a_V}[W]\,\|V - U\|_{W,1} \le \kappa_W W \|V - U\|_{W,1}$; the second is at most $BW\|a_V - a_U\|_v \le BWL_a\|V - U\|_{W,1}$ by (i)–(ii). *Stock derivatives.* Outside the switching states, the chain rule gives $\partial_{\overline{T}_k}(\mathcal{B}^\theta V)_j = \partial_{\overline{T}_k} Q_j(a_V; V) + \sum_{h \ne j} \partial_{x_h} Q_j(a_V; V)\, \partial_{\overline{T}_k} a_{V,h}$, the own-action term vanishing by the first-order condition at an interior best response and by local constancy of $a_{V,j}$ at a corner. The first term obeys the same two-bracket decomposition. At the fixed profile a_V, (A.10) and $\overline{T}_k \le \overline{T}'_k$ along every transition give $\sum_k \overline{T}_k |\partial_{\overline{T}_k} K^\theta_{a_V}(V_j - U_j)| \le K^\theta_{a_V}[\sum_k \overline{T}'_k |\partial_{\overline{T}_k}(V_j - U_j)|]$, which together with the level term is at most $K^\theta_{a_V}[W']\,\|V - U\|_{W,1} \le \kappa_W W \|V - U\|_{W,1}$; shifting the profile from a_V to a_U at the continuation U costs, levels and weighted derivatives together, at most $BW\|a_V - a_U\|_v \le BWL_a\|V - U\|_{W,1}$ by (i)–(ii). In the chain-rule term, the difference between V and U is $\sum_{h \ne j} \partial_{x_h} Q_j(a_V; V)[\partial_{\overline{T}_k} a_{V,h} - \partial_{\overline{T}_k} a_{U,h}] + \sum_{h \ne j}[\partial_{x_h} Q_j(a_V; V) - \partial_{x_h} Q_j(a_U; U)]\partial_{\overline{T}_k} a_{U,h}$; weighting by $\overline{T}_k$ and summing over k, the first sum is at most $BW \sum_k \overline{T}_k \|\partial_{\overline{T}_k} a_V - \partial_{\overline{T}_k} a_U\|_v \le BWL'_a\|V - U\|_{W,1}$, the rival-action gradient being bounded by BW in the dual of $\|\cdot\|_v$ by (i), and the second at most $B_2 W\|V - U\|_{W,1} \sum_k \overline{T}_k \|\partial_{\overline{T}_k} a_U\|_v \le A_1 B_2 W \|V - U\|_{W,1}$ by (ii)–(iii). Collecting terms and dividing by W, $\|\mathcal{B}^\theta V - \mathcal{B}^\theta U\|_{W,1} \le \rho \|V - U\|_{W,1}$, so Banach's fixed-point theorem on the closed subset $\mathcal{C}$ of the Banach space $\mathcal{V}_W$ yields a unique fixed point $V^\theta \in \mathcal{C}$ and $\|(\mathcal{B}^\theta)^H \mathbf{0} - V^\theta\|_{W,1} \le \rho^H \|V^\theta\|_{W,1}$. The stage Nash profile a_{V^θ} satisfies (58) under the values (57) and is therefore the county policy of Definition 2(ii); conversely, the value vector of any Markov perfect equilibrium with values in $\mathcal{C}$ is a fixed point of $\mathcal{B}^\theta$, hence equals V^θ, and its policy is the unique stage profile a_{V^θ}. Hypothesis (ii) on the stock derivatives of the profile is not implied by C^1 bounds on $\mathcal{C}$, because the profile responds to second stock derivatives of V through the implicit function theorem; it must be verified on the class at hand, for instance on a finite-dimensional approximation class on which stock derivatives are controlled by levels. Part 2 verifies the remaining hypotheses in the small-friction regime. □

COROLLARY A.1: *Under Theorem A.1 and for the Markov perfect equilibrium in $\mathcal{C}$, every realization of the physical shock and discovery laws from Ω_0 generates a unique path of allocations, knowledge frontiers, productivity gaps, and route stocks.*

Proof. Composing the unique measurable conditional dated equilibrium map with the unique measurable county policy a_{V^θ} fixes the allocation and investment profile at each state, and iterating (10), (38)–(40), and (54) determines the realized path by induction. □

A.7.2 Part 2: Small-friction argument on the rescaled stock corridor

Proof. Fix J counties $\mathcal{C}_j = [c_{j-1}, c_j]$ (ten equal counties $[(j-1)/10, j/10]$ in the application) in the small-friction regime of Proposition 6(3): research is dormant ($\phi^*_{it} \equiv 0$), so the dated economy is deterministic and knowledge evolves by diffusion alone with $\|A_{i,t+1}\|_\infty \le \|A_{it}\|_\infty$; wages and prices lie in the stock-independent collar of Appendix A.6.3, Step 2; and expected tolls are positive, $\mathbb{E}_t[\tau_j] \ge \underline{\tau}_{jt} > 0$, on the quote domain (74) (tolls vanish only at slack capacity). Throughout, $m_I = c\bar{\varepsilon}$ and $\kappa_0 e^{-\underline{T}} \le c'\bar{\varepsilon}$ with $c, c' > 0$ fixed and $\bar{\varepsilon} \to 0$. At date t with $n \ge 1$ periods remaining in horizon H, county j maximizes $\Pi^{(n),\theta}_{jt}(\mathbf{x}_t; \Omega_t) = R_{jt}(\mathbf{x}_t; \Omega_t) + \beta H^{(n-1)}_{jt}(\overline{\mathbf{T}}_{t+1}; \mathbf{z}_t, \overline{\mathbf{z}}_t)$, where $H^{(n-1)}_{jt} = \mathbb{E}^\theta_t[V^{(n-1),\theta}_{j,t+1}]$ ($V^{(0),\theta}_H \equiv 0$) and $R_{jt} = \widetilde{R}_{jt} - X_{jt} D^I_{jt}$ with $X_{jt} = m_I(x_{jt} + \frac{1}{2}\chi_I x^2_{jt})$ and

$D^I_{jt} \geq D_{\min} > 0$ for every county with an open project (61); counties with $\bar{x}_{jt} = 0$ set $x_{jt} = 0$ and drop out of the stage game.

Step 1 (Stage-game strong monotonicity and attenuated stock sensitivity). Since $\mathbf{x}_t$ enters the dated equilibrium only through construction demand $I_t(\ell; \mathbf{x}_t) = \sum_r e_{rt} k_{rt}(\ell) X_{rt}$ and inherited stocks enter it through tolls and route loss, assume that the dated-equilibrium map $(\mathbf{x}, \overline{\mathbf{T}}_t) \mapsto (\widetilde{R}_{jt}, D^I_{jt})$ is C^2 and satisfies, uniformly on $\mathcal{X}_t$ and on the stock set of Step 3, the small-friction general-equilibrium bounds

$$\begin{aligned} |\partial_{X_j} \widetilde{R}_{jt}| \leq \tfrac{1}{4} D_{\min}, \qquad & |\partial^2_{X_i X_j} \widetilde{R}_{jt}| + 2|\partial_{X_i} D^I_{jt}| + X_{jt} |\partial^2_{X_i X_j} D^I_{jt}| \leq c_0, \\ |\partial^2_{x_j \overline{T}_i} R_{jt}| \leq c_1 m_I (1+\chi_I)\, \varepsilon\, \overline{T}_{it}^{-1/\alpha_T}, \qquad & \partial^2_{\overline{T}_j \overline{T}_j} R_{jt} \leq -c_R\, \varepsilon\, \overline{T}_{jt}^{-(1+\alpha_T)/\alpha_T} < 0, \end{aligned} \tag{A.11}$$

for all i, j. Local construction demand moves p_M only through the no-arbitrage wedge $O(r_M \tau + \kappa_0 e^{-T})$ of (45), which is why the first two bounds hold for small frictions; the stock exponents follow from the toll elasticity $\partial_{\overline{T}_j} \log \mathbb{E}_t[\tau_j] = -(1-\alpha_T)/(\alpha_T \overline{T}_{jt})$ of (50) and (51), since rents depend on $\overline{T}_{jt}$ through $\mathbb{E}_t[\tau_j] \propto \varepsilon \overline{T}_{jt}^{-(1-\alpha_T)/\alpha_T}$, whose first and second stock derivatives are of order $\varepsilon \overline{T}_{jt}^{-1/\alpha_T}$ and $\varepsilon \overline{T}_{jt}^{-(1+\alpha_T)/\alpha_T}$, with the route-loss channel $\kappa_0 e^{-T}$ dominated by the toll channel under $\kappa_0 e^{-\underline{T}} \leq c' \bar{\varepsilon}$ and the last bound using the positive expected toll. Differentiating R_{jt} with $X'_{jt} = m_I(1 + \chi_I x_{jt})$ and $X''_{jt} = m_I \chi_I$,

$$\begin{aligned} \partial^2_{x_j x_j} R_{jt} = m_I \chi_I [\partial_{X_j} \widetilde{R}_{jt} - D^I_{jt} - X_{jt} \partial_{X_j} D^I_{jt}] + (X'_{jt})^2 [\partial^2_{X_j X_j} \widetilde{R}_{jt} - 2\partial_{X_j} D^I_{jt} - X_{jt} \partial^2_{X_j X_j} D^I_{jt}] \\ \leq -\tfrac{1}{2} m_I \chi_I D_{\min} + c_0 m_I^2 (1+\chi_I)^2, \end{aligned}$$

using $X_{jt} |\partial_{X_j} D^I_{jt}| \leq c_0 m_I (1+\chi_I) \leq \frac{1}{4} D_{\min}$ under the window (A.13) below, and $|\partial^2_{x_i x_j} R_{jt}| \leq c_0 m_I^2 (1+\chi_I)^2$ for $i \neq j$. Suppose the continuation $H^{(n-1)}_{jt}$ is concave in $\overline{T}_{j,t+1}$ with

$$|\partial^2_{\overline{T}_j \overline{T}_j} H^{(n-1)}_{jt}| \leq C_1 \varepsilon\, \overline{T}_{j,t+1}^{-(1+\alpha_T)/\alpha_T}, \qquad |\partial^2_{\overline{T}_i \overline{T}_j} H^{(n-1)}_{jt}| \leq C_2 \varepsilon^2 (\overline{T}_{i,t+1} \overline{T}_{j,t+1})^{-1/\alpha_T} \quad (i \neq j), \tag{A.12}$$

the orders of the own and cross stock curvature of toll-driven rents. The Jacobian of the stage pseudogradient $\mathbf{g}^{(n),\theta}_t = (\partial_{x_j} \Pi^{(n),\theta}_{jt})_j$ then has diagonal and off-diagonal entries

$$\begin{aligned} \partial^2_{x_j x_j} R_{jt} + \beta d^2_{jt} \partial^2_{\overline{T}_j \overline{T}_j} H^{(n-1)}_{jt} \leq -\tfrac{1}{2} m_I \chi_I D_{\min} + c_0 m_I^2 (1+\chi_I)^2, \\ |\partial^2_{x_i x_j} \Pi^{(n),\theta}_{jt}| \leq c_0 m_I^2 (1+\chi_I)^2 + \beta d_{it} d_{jt} C_2 \varepsilon^2 (\overline{T}_{i,t+1} \overline{T}_{j,t+1})^{-1/\alpha_T} \quad (i \neq j). \end{aligned}$$

Under the adjustment-cost window and the continuation cross-stock bound

$$J c_0 m_I (1+\chi_I)^2 \leq \tfrac{3}{16} \chi_I D_{\min}, \qquad (J-1)\beta \max_{i \neq j} d_{it} d_{jt} C_2 \varepsilon^2 (\overline{T}_{i,t+1} \overline{T}_{j,t+1})^{-1/\alpha_T} \leq \tfrac{3}{16} m_I \chi_I D_{\min}, \tag{A.13}$$

Gershgorin's theorem then bounds every eigenvalue of the symmetric part of $-D_{\mathbf{x}} \mathbf{g}^{(n),\theta}_t$ below by $\frac{1}{8} m_I \chi_I D_{\min}$, so the pseudogradient is strongly monotone with modulus $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$ and the stage Nash profile $\mathbf{x}^{(n),\theta}_t$ is unique (Proposition 4). On each active interior set $\mathcal{A} = \{r : 0 < x^{(n),\theta}_{rt} < \bar{x}_{rt}\}$, locally constant outside the switching states of (62), the implicit function theorem gives $\partial_{\overline{T}_{it}} \mathbf{x}_{\mathcal{A}} = -[\partial_{\mathbf{x}_{\mathcal{A}}} \mathbf{g}_{\mathcal{A}}]^{-1} \partial_{\overline{T}_{it}} \mathbf{g}_{\mathcal{A}}$ with $\|[\partial_{\mathbf{x}_{\mathcal{A}}} \mathbf{g}_{\mathcal{A}}]^{-1}\| \leq 1/\mu$ and, at fixed actions, $\partial_{\overline{T}_{it}} g_{jt} = \partial^2_{x_j \overline{T}_i} R_{jt} + \beta d_{jt} \partial^2_{\overline{T}_j \overline{T}_i} H^{(n-1)}_{jt}$, so by (A.11)–(A.12) and $\overline{T}_{j,t+1} \geq \overline{T}_{jt}$,

$$|\partial_{\overline{T}_{it}} g_{jt}| \leq \varepsilon\, \overline{T}_{it}^{-1/\alpha_T} \Big[c_1 m_I (1+\chi_I) + \beta d_{jt} \Big(\mathbf{1}_{\{i=j\}} \frac{C_1}{\overline{T}_{jt}} + \mathbf{1}_{\{i \neq j\}} C_2 \varepsilon \overline{T}_{jt}^{-1/\alpha_T} \Big) \Big],$$

which is (75) with $C_u = \sqrt{J}\,[c_1 m_I(1+\chi_I) + \beta \max_j d_{jt}(C_1/\overline{T}_{jt} + C_2\varepsilon \overline{T}_{jt}^{-1/\alpha_T})]$, uniform on stock sets on which $d_{jt}/\overline{T}_{jt}$ and $d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T}$ are bounded. The first term of C_u is the toll dependence of D^I_{jt} and of marginal rents, the second the continuation curvature; since $m_I = O(\bar{\varepsilon})$, the attenuation $C_u\varepsilon/\mu$ is of order $1/(\chi_I D_{\min})$.

Step 2 (Backward induction on continuation curvature; sketch). Differentiating $V^{(n),\theta}_{jt}(\Omega_t) = \Pi^{(n),\theta}_{jt}(\mathbf{x}^{(n),\theta}_t(\Omega_t);\Omega_t)$ in $\overline{T}_{jt}$ outside the switching states, the own-action term $\partial_{x_j}\Pi^{(n),\theta}_{jt}\,\partial_{\overline{T}_j}x^{(n),\theta}_{jt}$ vanishes (first-order condition at an interior best response, local constancy at a corner), leaving $\partial_{\overline{T}_j}R_{jt} + \beta\partial_{\overline{T}_j}H^{(n-1)}_{jt}$ plus the rival feedback $\sum_{h\neq j}\partial_{x_h}\Pi^{(n),\theta}_{jt}\partial_{\overline{T}_j}x^{(n),\theta}_{ht}$. Differentiating once more on the interior branch, where $\partial_{\overline{T}_j}x_{jt} = -\partial^2_{x_j\overline{T}_j}\Pi_{jt}/\partial^2_{x_jx_j}\Pi_{jt}$ up to rival terms, and writing R_{xx}, R_{xT}, R_{TT} for the own-action, cross, and own-stock second derivatives of R_{jt} and H'' for $\partial^2_{\overline{T}_j\overline{T}_j}H^{(n-1)}_{jt} \le 0$,

$$\partial^2_{\overline{T}_j\overline{T}_j}V^{(n),\theta}_{jt} = R_{TT} + \beta H''\,\frac{|R_{xx}|}{|R_{xx}| + \beta d^2_{jt}|H''|} + \frac{R^2_{xT} + 2\beta d_{jt}R_{xT}H''}{|R_{xx}| + \beta d^2_{jt}|H''|} + \text{rival terms.}$$

Continuation concavity passes through with a factor in $(0,1]$, and own-stock concavity is preserved whenever $|R_{xT}|(|R_{xT}|+2\beta d_{jt}|H''|)/\mu$ and the rival terms are dominated by $|R_{TT}|$; by (A.11)–(A.12) and $|R_{xx}| \ge \mu$, the first requirement holds when

$$32c_1^2 m_I(1+\chi_I)^2\varepsilon \le c_R\chi_I D_{\min}\,T_{\text{lo}}^{(1-\alpha_T)/\alpha_T}, \qquad 64\beta c_1 C_1(1+\chi_I)\max_j d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T} \le c_R\chi_I D_{\min}, \tag{A.14}$$

with T_{lo} the lower edge of the stock set; both hold in the regime of Step 1, the first because $m_I\chi_I\varepsilon \to 0$ and the second because the gross marginal value of a package, $\beta d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T}$ up to constants, is of the order of the marginal construction cost $m_I\chi_I D^I_{jt}$ on the interior branch and $m_I\chi_I \le \bar{m}$ (Step 3). The cross-stock derivatives obey the same decomposition, $\partial^2_{\overline{T}_i\overline{T}_j}V^{(n),\theta}_{jt} = \partial^2_{\overline{T}_i\overline{T}_j}R_{jt} + \beta\partial^2_{\overline{T}_i\overline{T}_j}H^{(n-1)}_{jt} + (R_{xT} + \beta d_{jt}H'')\,\partial_{\overline{T}_i}x_{jt} + \text{rival terms}$, so that with (75) the constant in (A.12) satisfies the recursion $C_2^{(n)} \le a + bC_2^{(n-1)}$ with $b = \beta\big[1 + \sqrt{J}\big(c_1 m_I(1+\chi_I) + \beta\max_j d_{jt}C_1/\overline{T}_{jt}\big)\max_j d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T}/\mu\big]$, and $C_2^{(n)} \le a/(1-b)$ uniformly in n whenever $b<1$, which requires $d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T}$ and $d_{jt}/\overline{T}_{jt}$ to be small relative to primitive constants; C_1 is bounded by the same argument. Policies are only piecewise C^1 in the stock because of the clipping in (62), so $V^{(n),\theta}_{jt}$ is piecewise C^2; for each stock vector the set of $\mathbf{z}_{t+1}$ at which some county sits exactly at a clipping boundary has probability zero under the truncated-Gaussian law, and when the boundaries move with $\mathbf{z}_{t+1}$ with nonvanishing derivative (transversality), integration against its density smooths the $O(1/\chi_I)$ jumps of $\partial_{\overline{T}_j}x_{ht}$ into a contribution of order ε/χ_I times the boundary density, so that $H^{(n)}_{j,t-1} = \mathbb{E}^\theta_{t-1}[V^{(n),\theta}_{jt}]$ carries (A.12) in the semiconcave sense. The recursion for the constants and the transversality step are argued rather than computed, which is why this step is a sketch.

Step 3 (Rescaled stock set and infinite-horizon convergence; sketch). For $t \ge 1881$ the schedule is stationary up to package growth: $o_{jt} = o_{j,1881}$ and $d_{jt} = g^{t-1881}\bar{d}_j$ with $\bar{d}_j = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j}\iota_{j,1881}(\ell)\,d\ell$, and only counties with $o_{j,1881} = 1$ build; the others keep constant stocks, enter rivals' payoffs through bounded tolls, and are excluded from the stock set. Rescale stocks by the toll-balanced factor g^{α_T}, $\widetilde{T}_{jt} = g^{-\alpha_T(t-1881)}\overline{T}_{jt}$, so that $\widetilde{T}_{j,t+1} = g^{-\alpha_T}(\widetilde{T}_{jt} + \bar{d}_j g^{(1-\alpha_T)(t-1881)}x_{jt})$. By the toll formula preceding Proposition 7, the gross marginal value of a package, $\widetilde{q}^\theta_{jt} = \beta d_{jt}\mathbb{E}^\theta_t[\partial_{\overline{T}_j}V^\theta_{j,t+1}]$, and the continuation cross terms $\beta d_{it}d_{jt}\partial^2_{\overline{T}_i\overline{T}_j}H_{jt}$ scale as $d_{jt}\varepsilon\overline{T}_{jt}^{-1/\alpha_T} = \bar{d}_j\varepsilon\widetilde{T}_{jt}^{-1/\alpha_T}$ and $\beta\bar{d}_i\bar{d}_j\varepsilon^2(\widetilde{T}_{it}\widetilde{T}_{jt})^{-1/\alpha_T}$, which depend on t only through $\widetilde{\mathbf{T}}_t$, whereas the marginal construction cost $m_I(1+\chi_I x_{jt})D^I_{jt} \ge$

$m_I D_{\min}$ is bounded below. The clipped rule (62) thus balances a stock-stationary marginal value against a bounded marginal cost: along the balanced path $\overline{T}_{jt} \propto g^{\alpha_T t}$ and $x_{jt} \propto g^{-(1-\alpha_T)t} \to 0^+$, so intensities are not bounded away from zero, and no interior corridor in the g^{-t}-rescaled stock exists (there $d_{jt}\varepsilon \overline{T}_{jt}^{-1/\alpha_T} \propto g^{-(1-\alpha_T)t}$ would vanish against the bounded marginal cost and force $x_{jt} = 0$). Suppose instead that at the lower edge the rule builds enough to keep pace with the rescaling: for $\widetilde{T}_{jt} \in [L_j, g^{\alpha_T} L_j]$ and every $t \geq 1881$, $\bar{d}_j g^{(1-\alpha_T)(t-1881)} x_{jt}^{\theta} \geq g^{\alpha_T} L_j - \widetilde{T}_{jt}$, a primitive inequality on $\beta \bar{d}_j \varepsilon L_j^{-1/\alpha_T}$ relative to $m_I(1+\chi_I) D_{jt}^I$ to be verified from the toll formula (its right side is nonincreasing in t on the interior branch). Then the lower stock set $\mathcal{K}_t = \prod_j [L_j g^{\alpha_T (t-1881)}, \infty)$ is forward invariant, and on it $d_{jt}\varepsilon \overline{T}_{jt}^{-1/\alpha_T} \leq \bar{d}_j \varepsilon L_j^{-1/\alpha_T}$, so the constants c_0, c_1, C_1, C_2, μ and the inequalities (A.13) and (A.14) of Steps 1–2 are uniform in t, research remains dormant, and rents are bounded (knowledge nonincreasing in supremum norm, wages and prices in the stock-independent collar). Theorem A.1 then applies with $W \equiv 1$ and $\kappa_W = 1$: $|R_{jt}| \leq C_R$ and $\overline{T}_k |\partial_{\overline{T}_k} R_{jt}| = O(\varepsilon \overline{T}_k^{1-1/\alpha_T})$ are bounded; the implicit function theorem gives $L_a \leq \beta \max_j (d_{jt}/\overline{T}_{j,t+1})/\mu$; the rival-action dependence of R_{jt} and of the continuation gives $B = O(c_0 m_I(1+\chi_I) + \beta \max_j d_{jt}\varepsilon \overline{T}_{jt}^{-1/\alpha_T})$; and (75) gives $A_1 \leq J C_u \varepsilon \max_k \overline{T}_{kt}^{1-1/\alpha_T}/\mu$, so that $BL_a = O(1/(\chi_I D_{\min}))$ once $m_I = O(\bar{\varepsilon})$ and $d_{jt}/\overline{T}_{jt}$ is bounded. The contraction condition then reads $\beta + O(1/(\chi_I D_{\min})) + BL_a' + A_1 B_2 < 1$, which holds for $\chi_I \geq \underline{\chi}$ given the policy-derivative clause (ii) of Theorem A.1 (L_a' and B_2), which the perturbation argument does not supply; together with the window (A.13), which for $\chi_I \geq 1$ is implied by $m_I \chi_I \leq \bar{m} \equiv \frac{3}{16} D_{\min}/(J c_0) - 3 m_I$, this is the two-sided range $\underline{\chi} \leq \chi_I \leq \bar{m}/m_I$, nonempty for small m_I (for instance $\chi_I \asymp \bar{\varepsilon}^{-1/2}$). Two uniformity questions remain open. Along the toll-balanced path $d_{jt}/\overline{T}_{jt} = \bar{d}_j g^{(1-\alpha_T)(t-1881)}/\widetilde{T}_{jt}$ grows without bound, so L_a and the own-continuation part of C_u are uniform in t only on date ranges on which this ratio is bounded; restoring uniformity requires the own continuation curvature $\beta d_{jt}^2 |\partial^2_{\overline{T}_j \overline{T}_j} H_{jt}|$, which grows at the same rate and has the concave sign, to be carried in the modulus. And the constant recursion of Step 2 is argued rather than computed. The infinite-horizon conclusion of Proposition 7(2) is conditional on these steps. The pre-1881 segment, on which the opportunity masks o_{jt} vary, is a finite backward induction from the post-1881 fixed point by Steps 1–2. [← Back to Proposition 7] □

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